

(12) **United States Patent**
Wu et al.

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(45) **Date of Patent:** **Sep. 16, 2025**

(54) **THREE-DIMENSIONAL MEMORY ARRAY
WITH LOCAL LINE SELECTOR**

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patent is extended or adjusted under 35
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(65) **Prior Publication Data**
US 2024/0324230 A1 Sep. 26, 2024

Related U.S. Application Data
(60) Continuation of application No. 18/308,031, filed on
Apr. 27, 2023, now Pat. No. 12,035,534, which is a
(Continued)

(51) **Int. Cl.**
H10B 43/35 (2023.01)
H01L 23/522 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10B 43/35** (2023.02); **H01L 23/5226**
(2013.01); **H10B 41/27** (2023.02); **H10B**
41/35 (2023.02); **H10B 43/27** (2023.02)

(58) **Field of Classification Search**
None
See application file for complete search history.

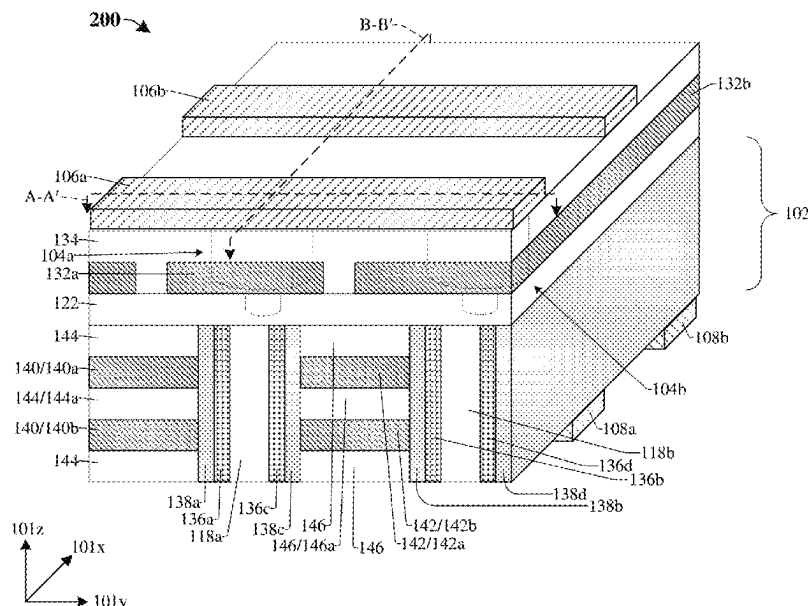
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Primary Examiner — Sarah K Salerno
(74) *Attorney, Agent, or Firm* — Eschweiler & Potashnik,
LLC

(57) **ABSTRACT**
The present disclosure relates to an integrated chip including
a three-dimensional memory array. The three-dimensional
memory array includes a first local line and a second local
line that are elongated vertically, a first memory cell extend-
ing between the first local line and the second local line, and
a second memory cell directly under the first memory cell,
extending between the first local line and the second local
line, and coupled in parallel with the first memory cell. The
first memory cell is coupled to a first word line and the
second memory cell is coupled to a second word line. The
first word line and the second word line are elongated
horizontally. A first global line is disposed at a first height
and is elongated horizontally. A first selector extends verti-
cally from the first local line to the first global line.

20 Claims, 41 Drawing Sheets



Related U.S. Application Data

division of application No. 17/168,317, filed on Feb. 5, 2021, now Pat. No. 11,672,123.

(51) Int. Cl.

H10B 41/27 (2023.01)

H10B 41/35 (2023.01)

H10B 43/27 (2023.01)

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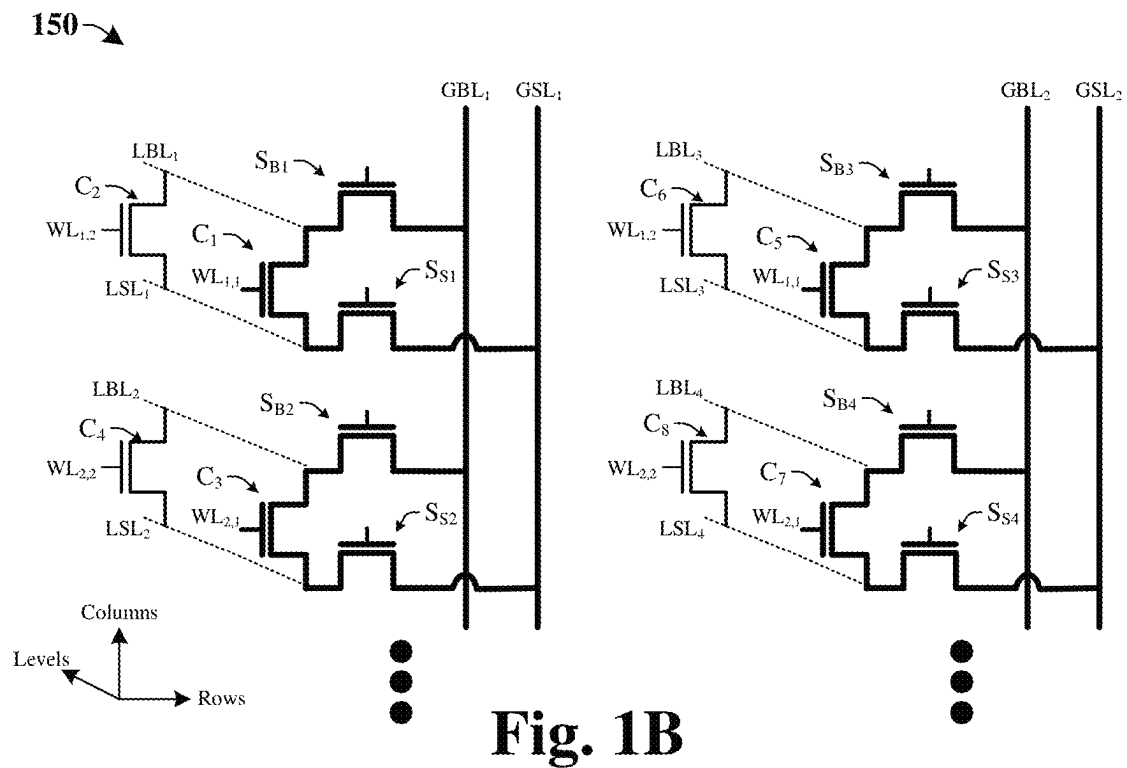
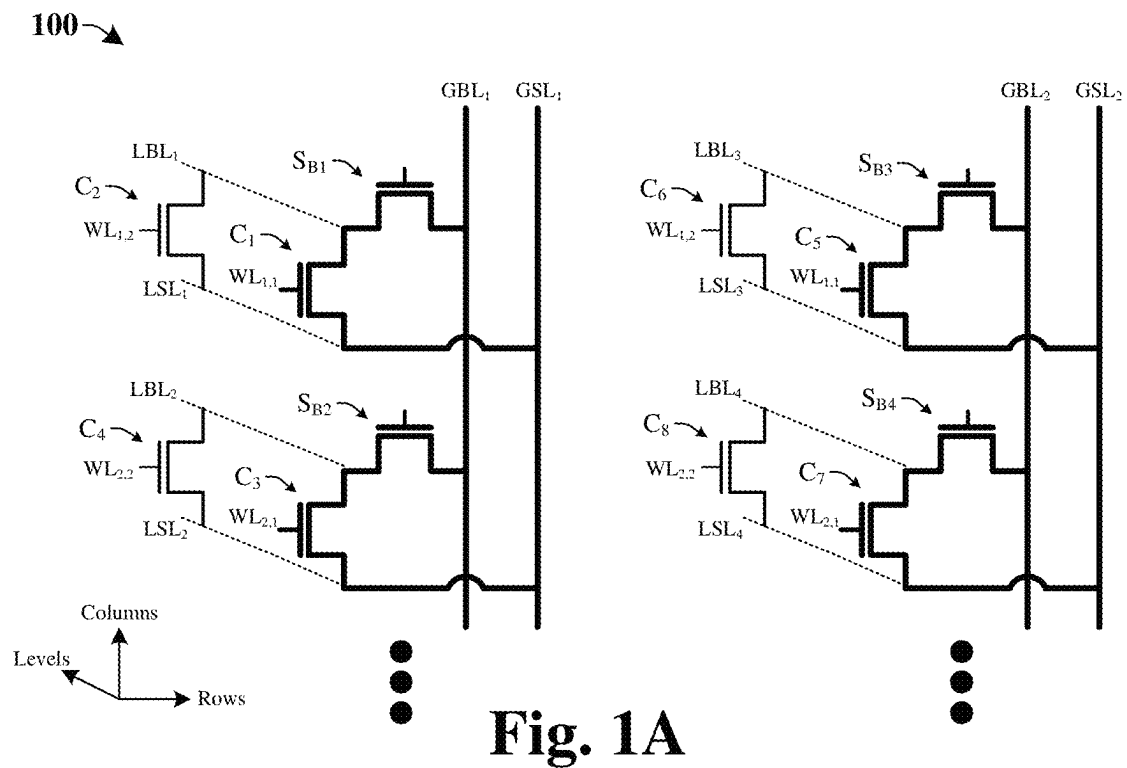
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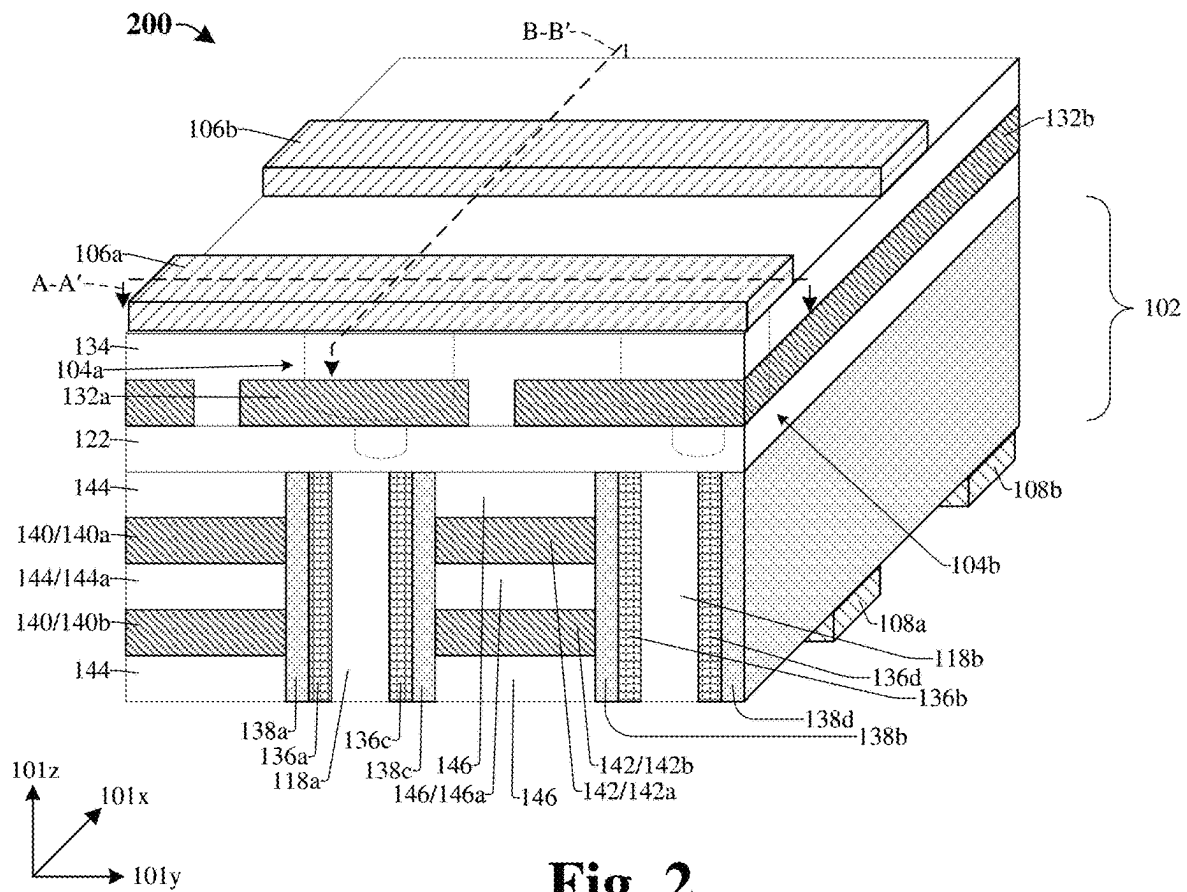


Fig. 2

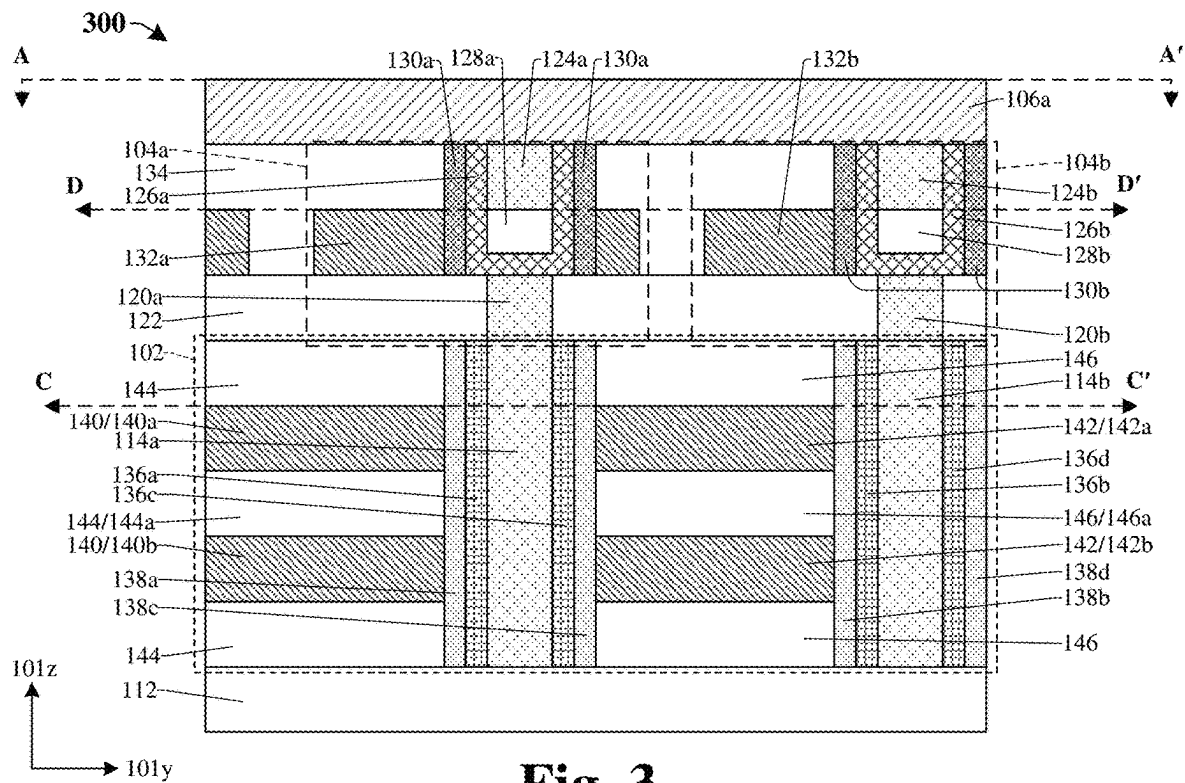


Fig. 3

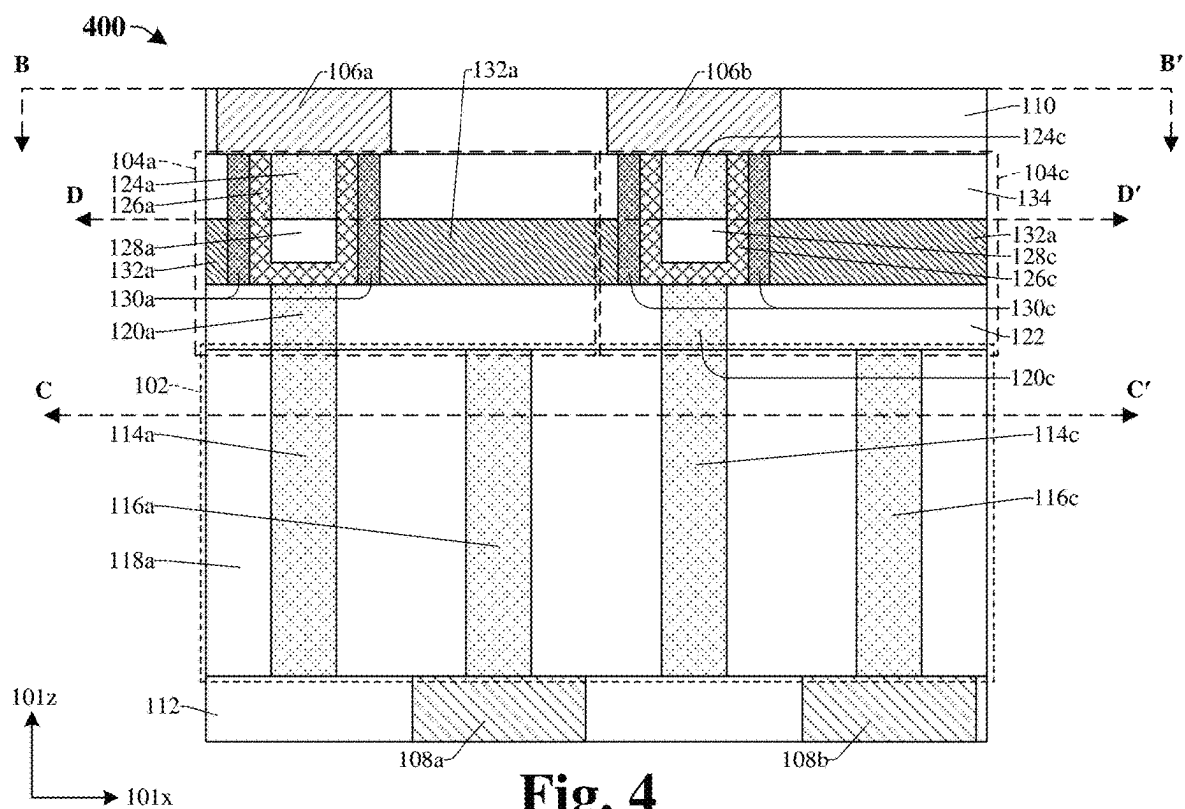
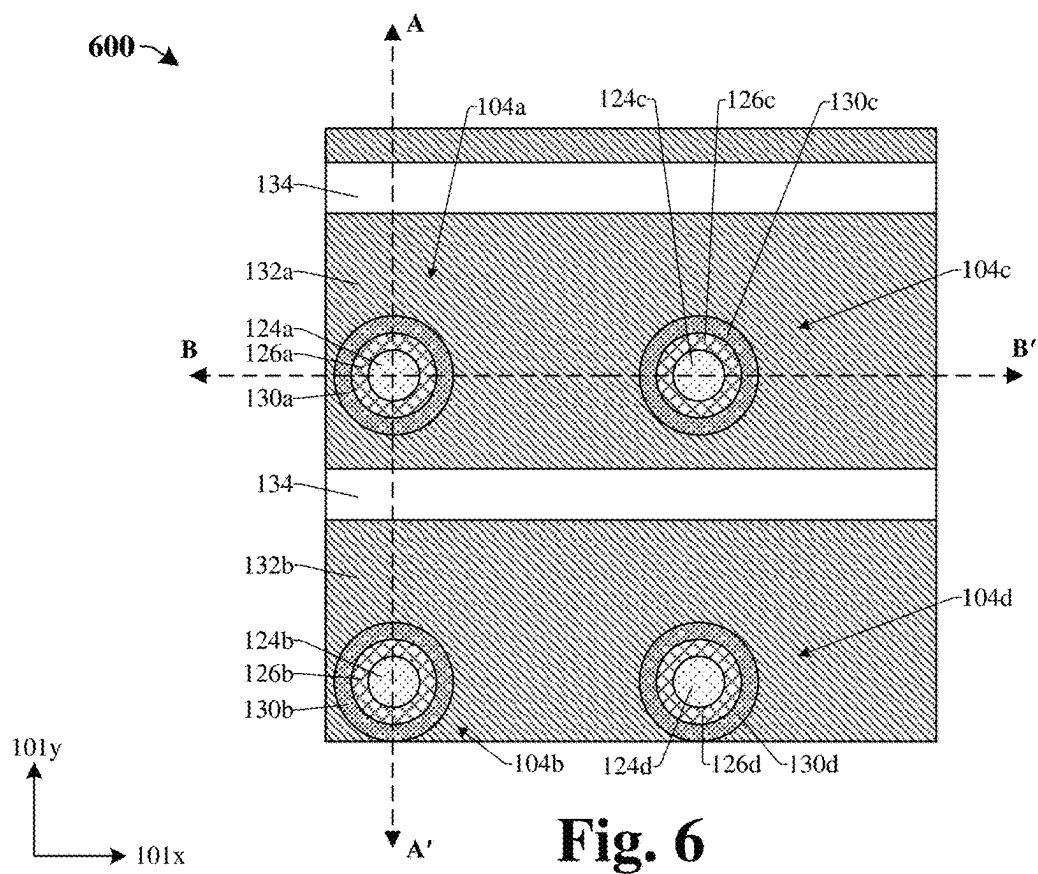
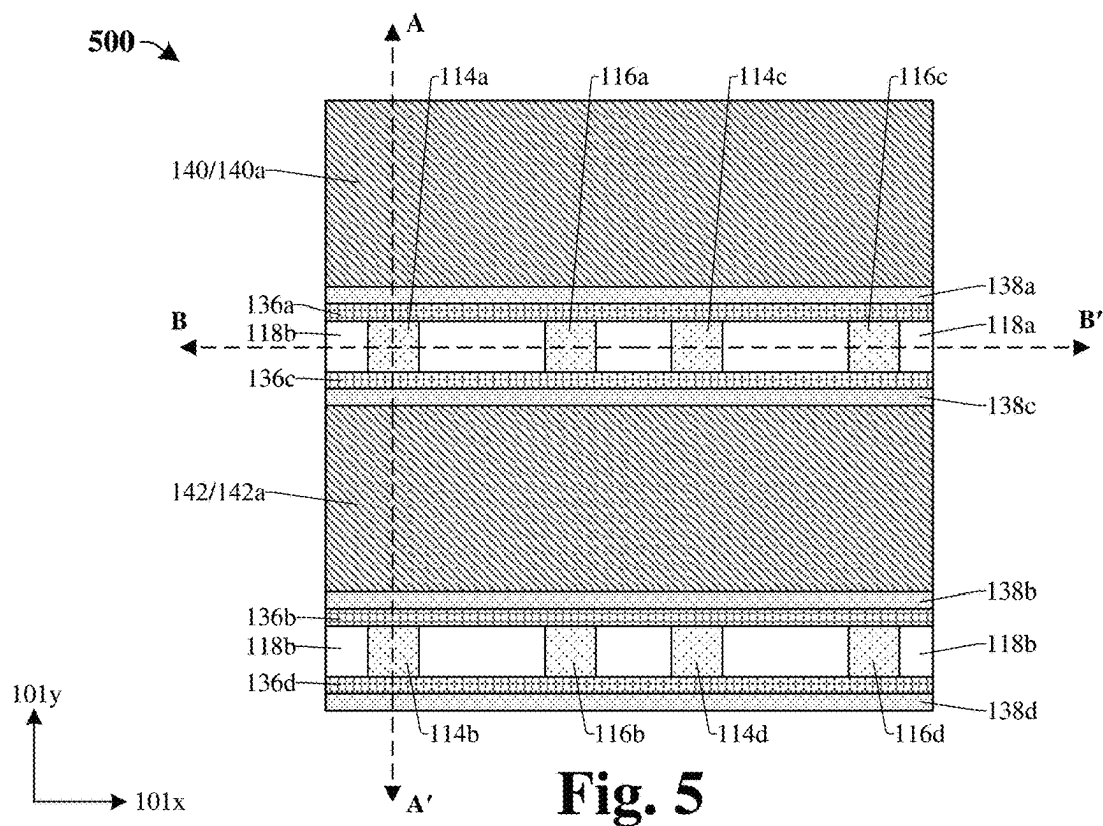
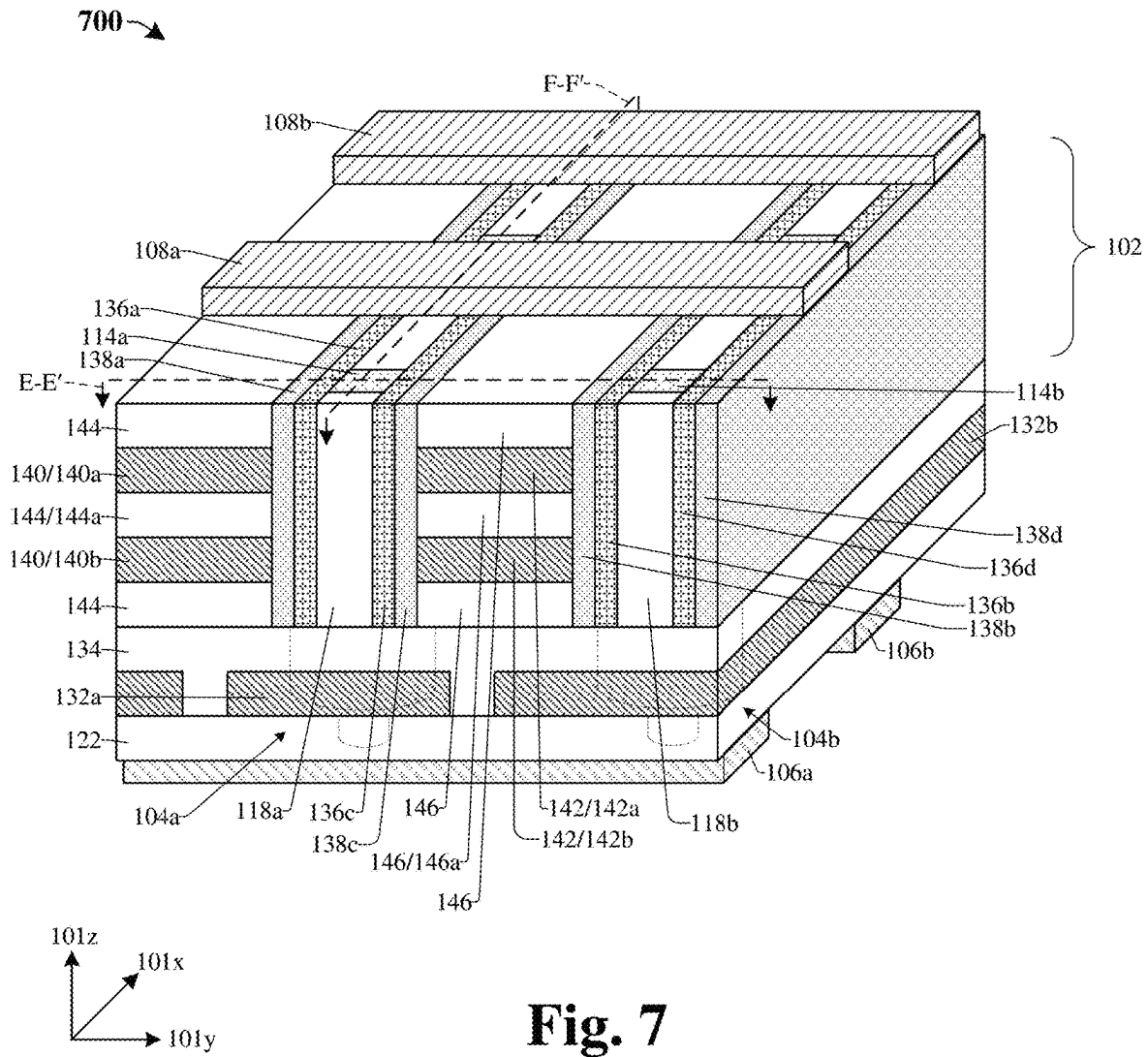


Fig. 4





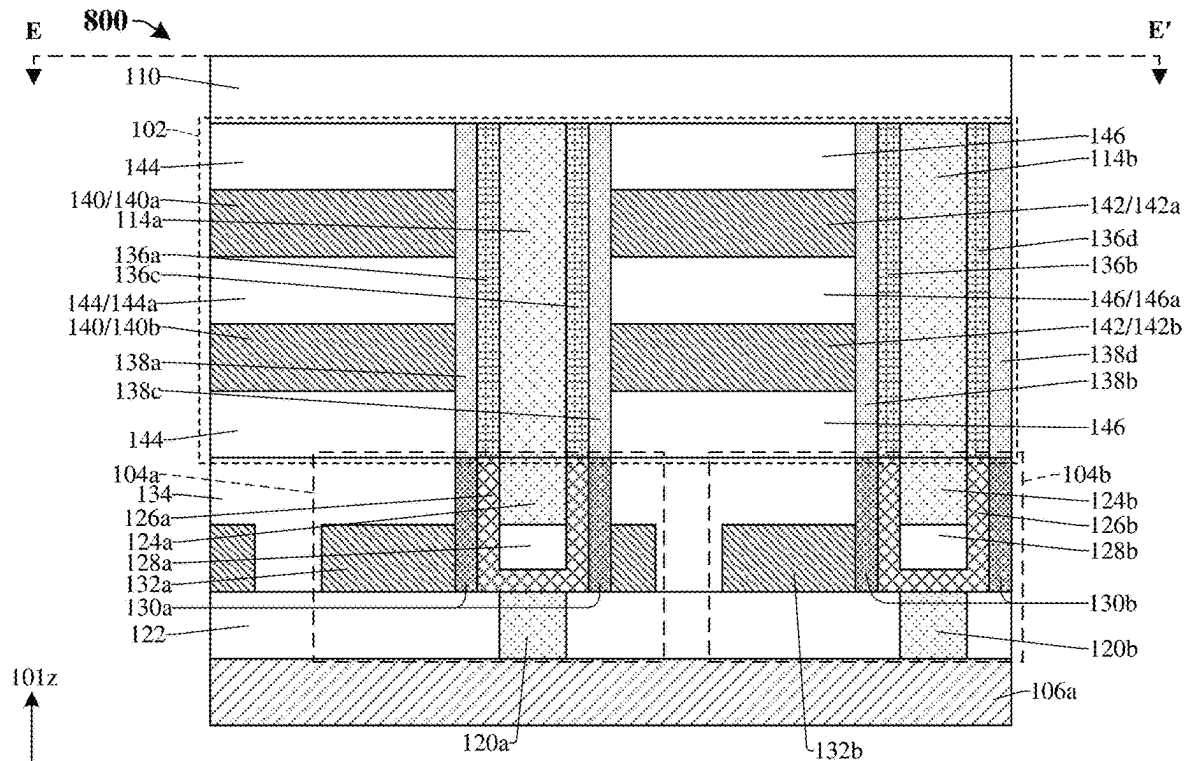


Fig. 8

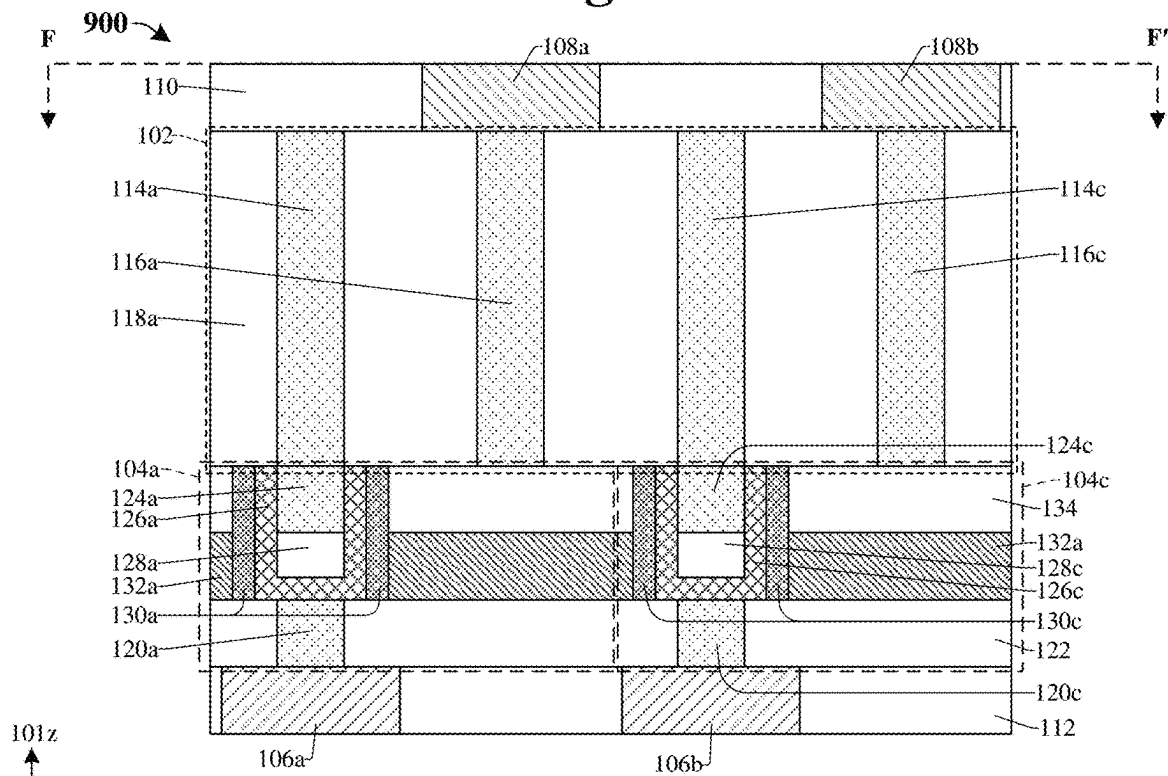
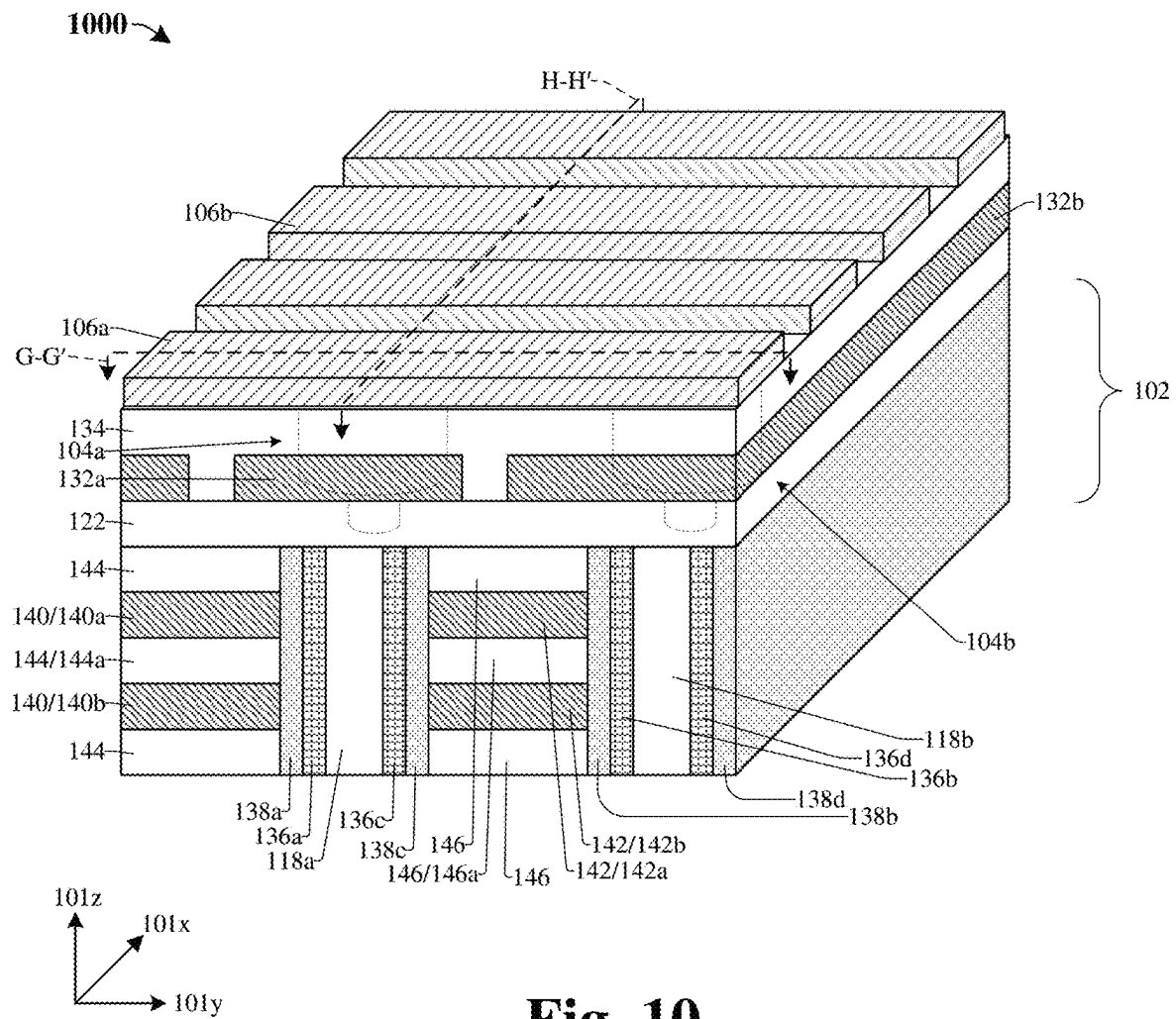
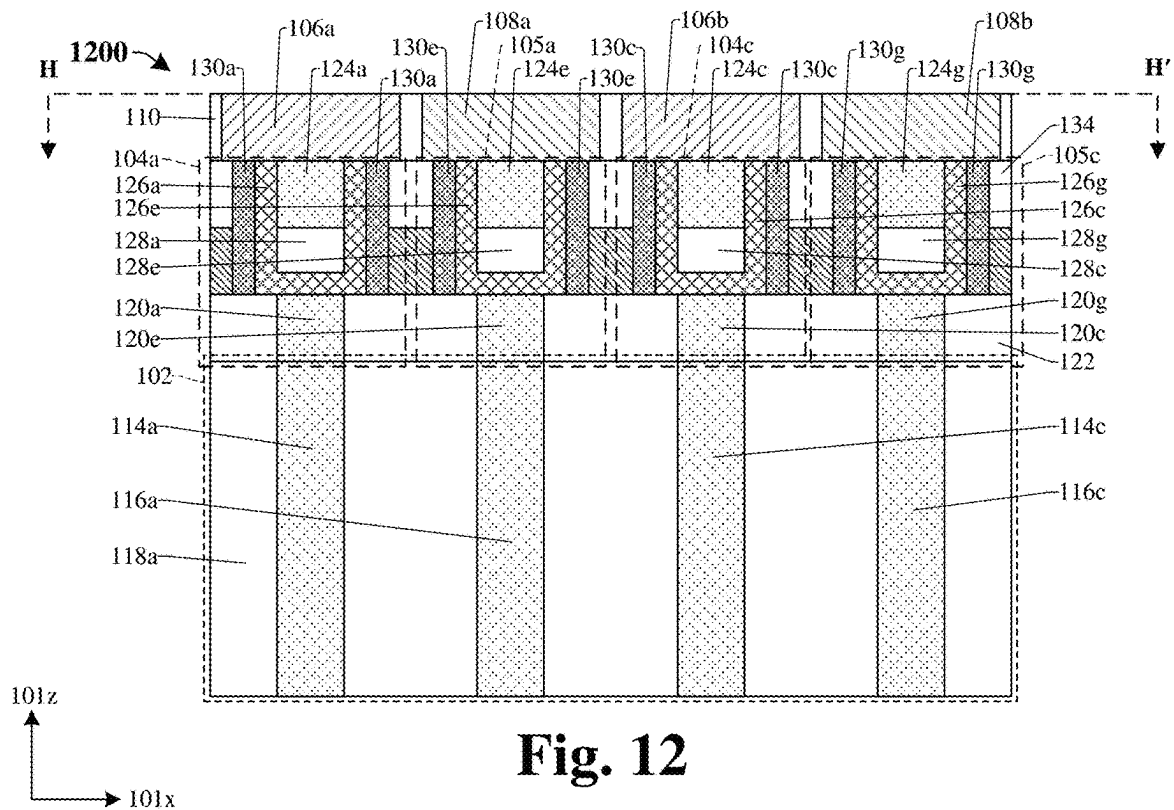
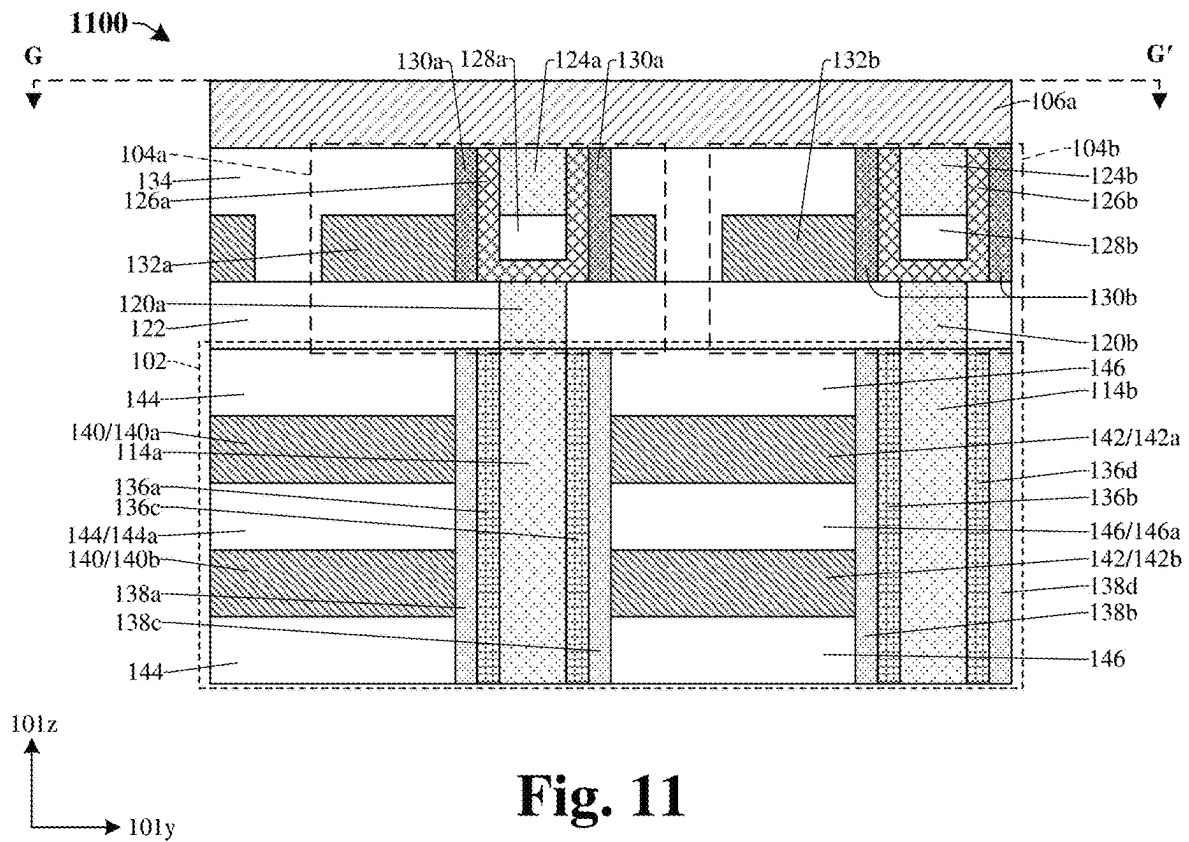
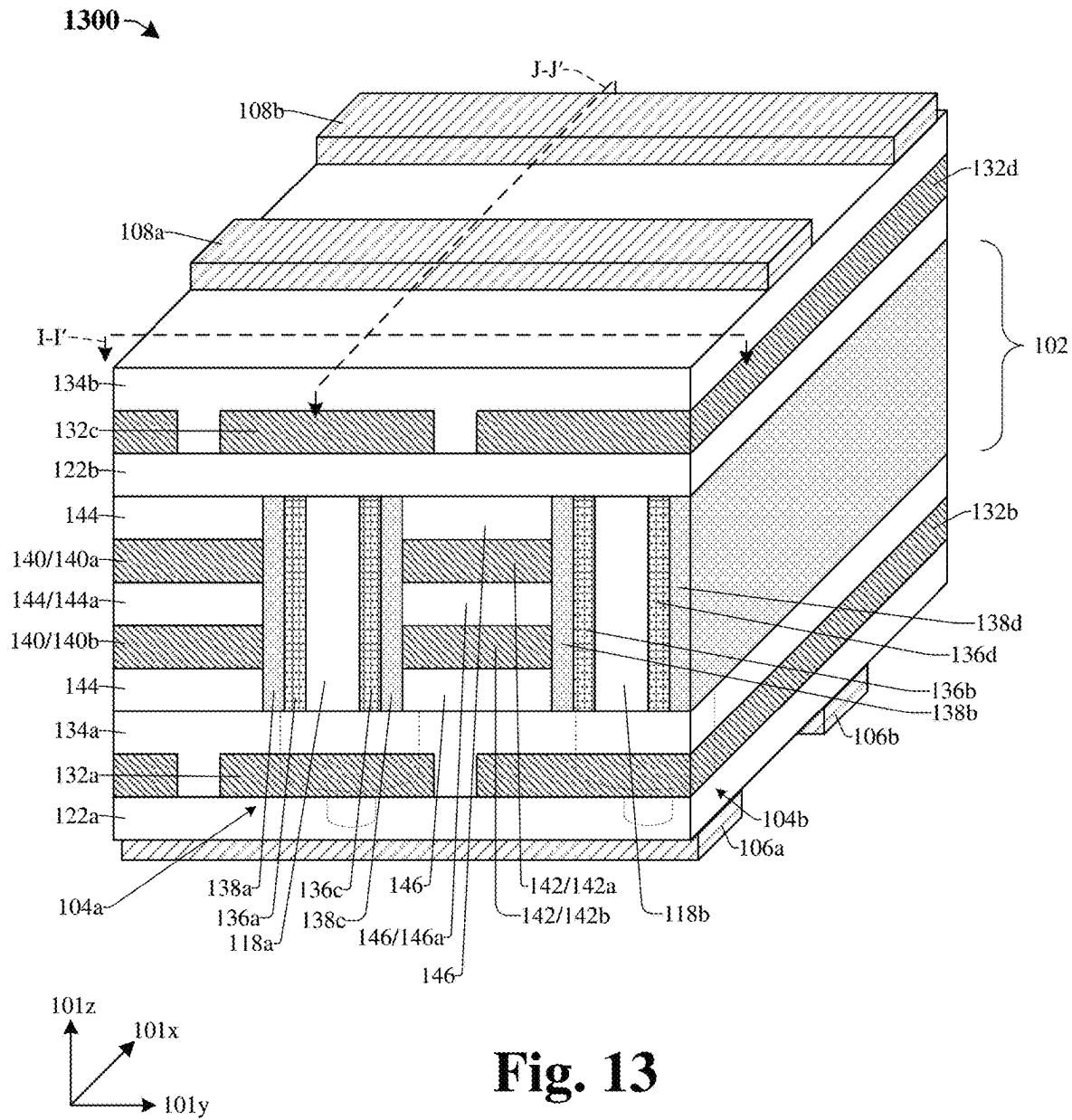


Fig. 9







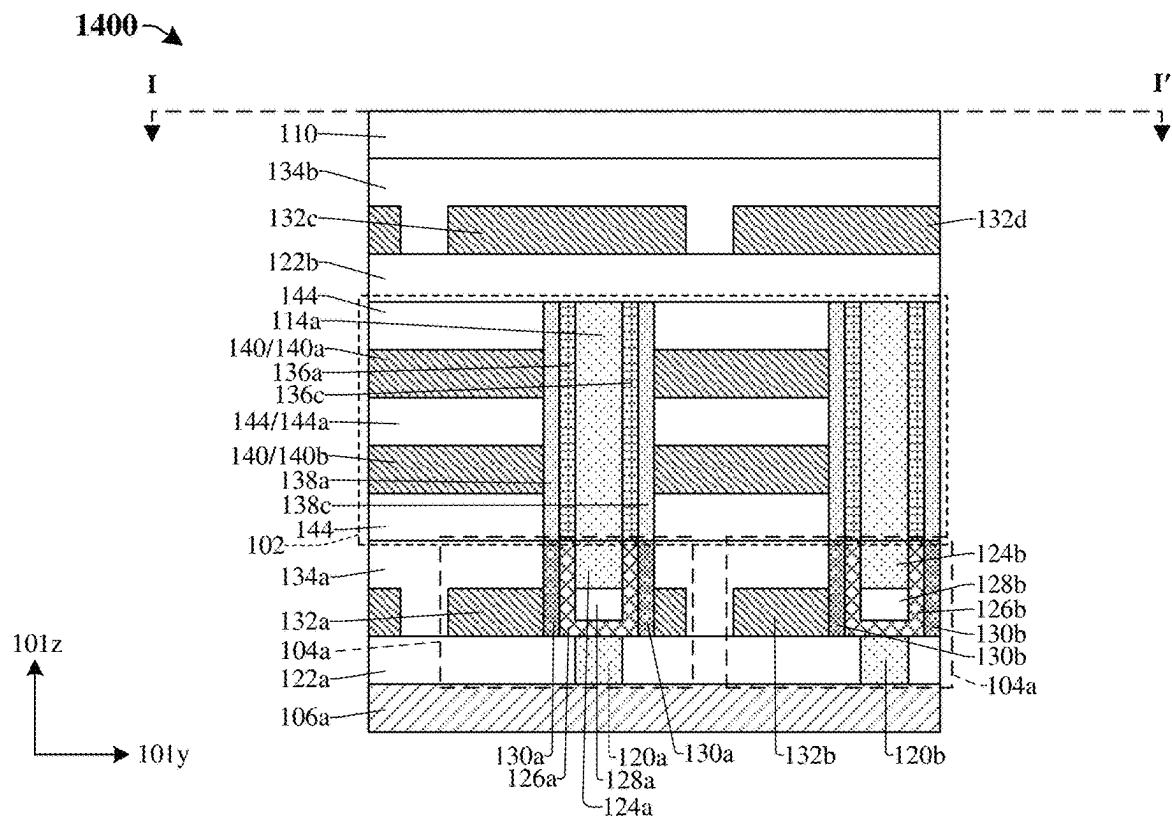


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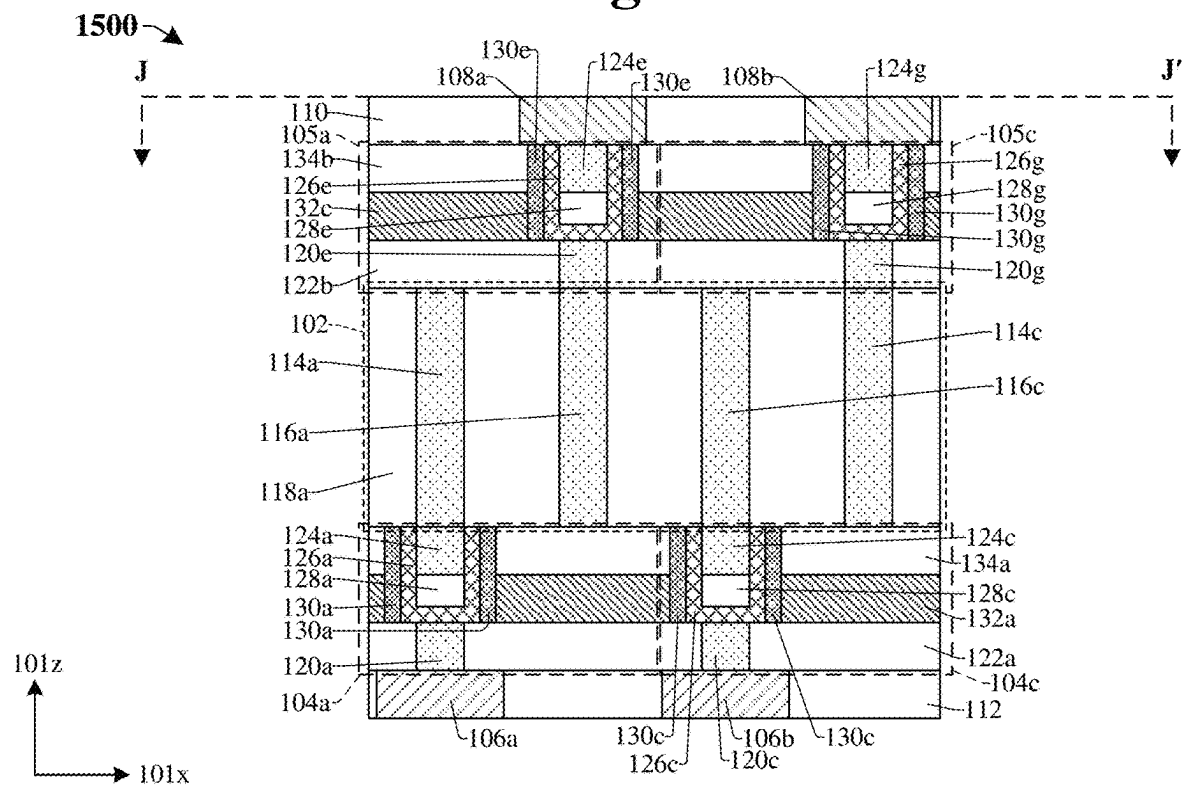


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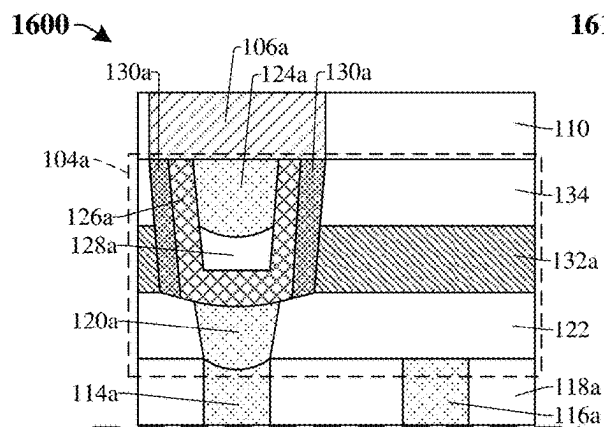


Fig. 16A

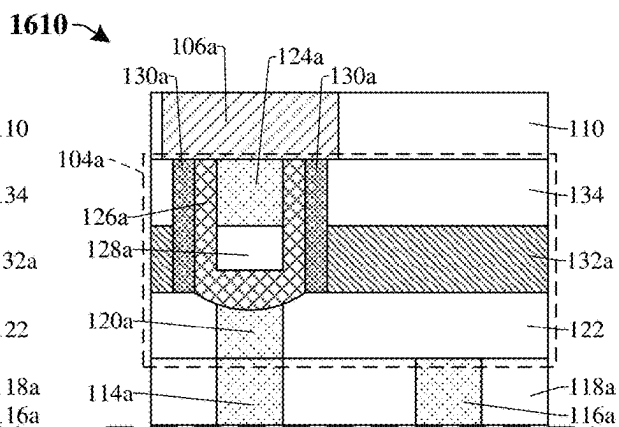


Fig. 16B

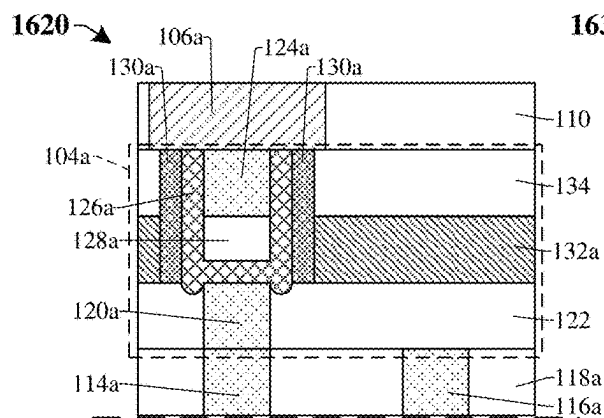
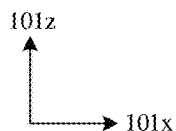
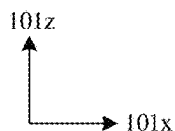


Fig. 16C

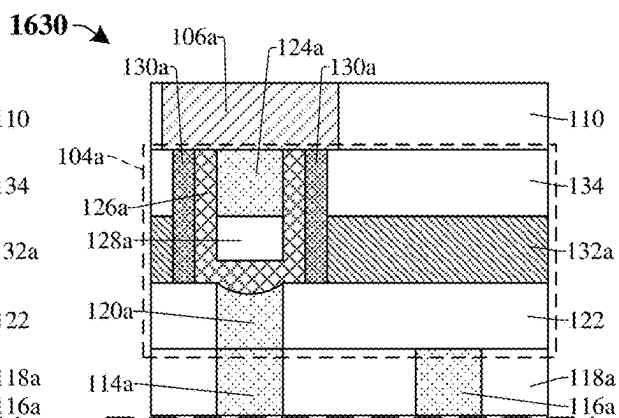
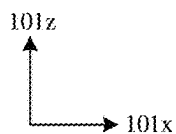
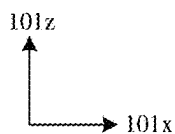
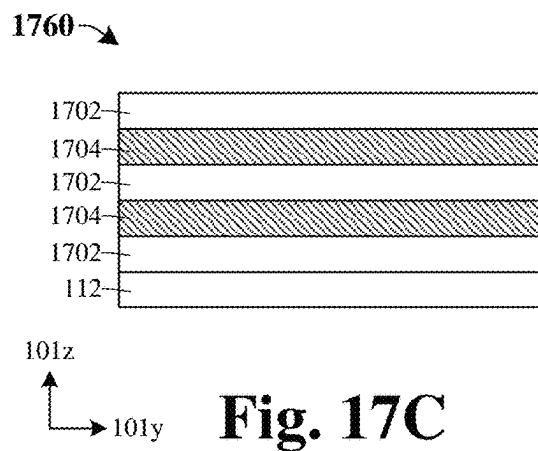
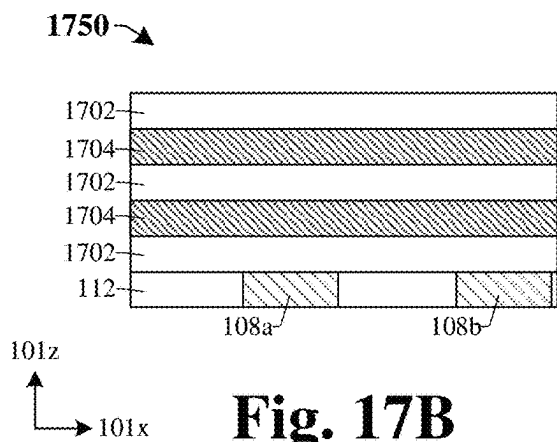
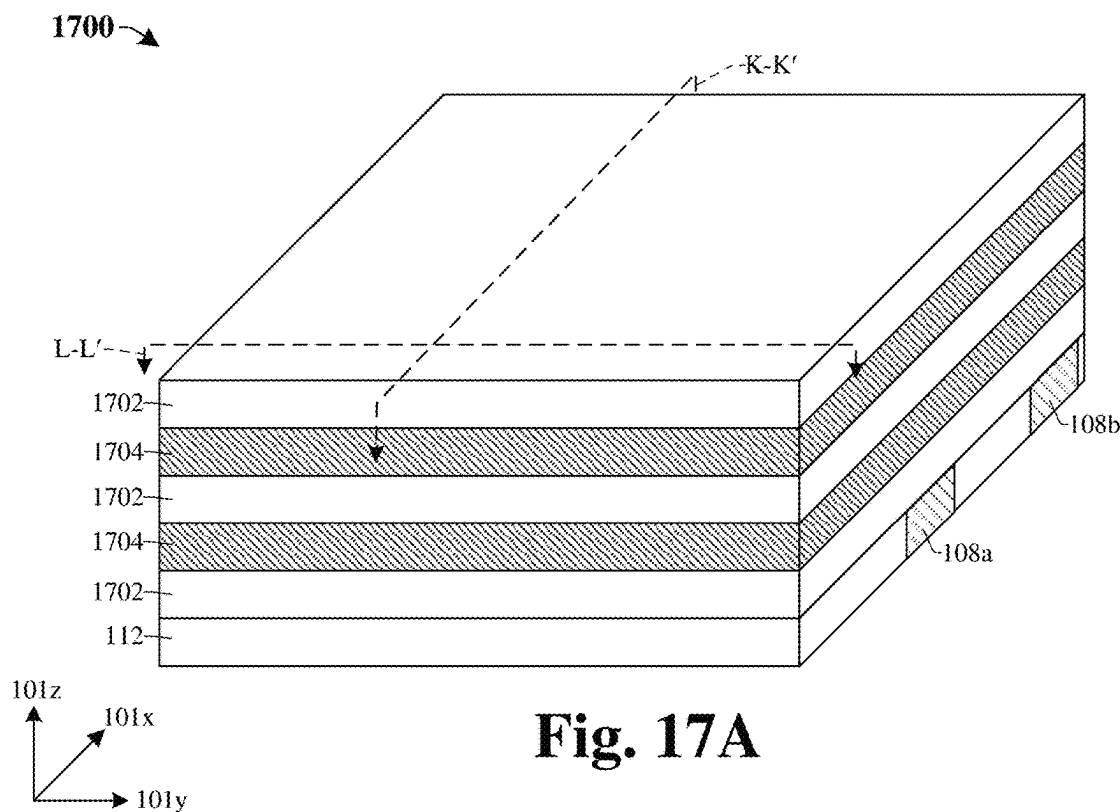
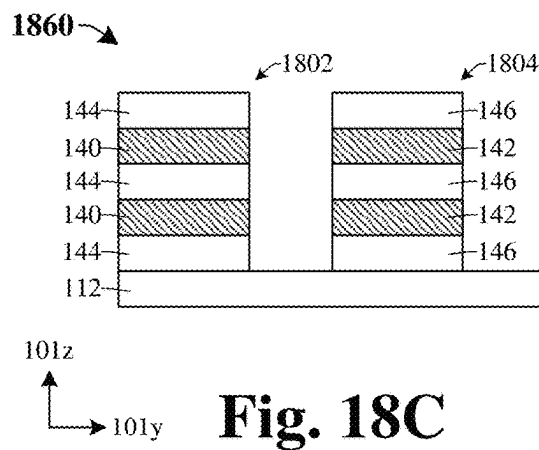
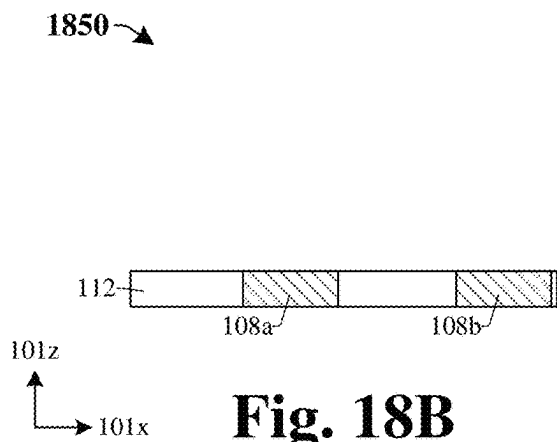
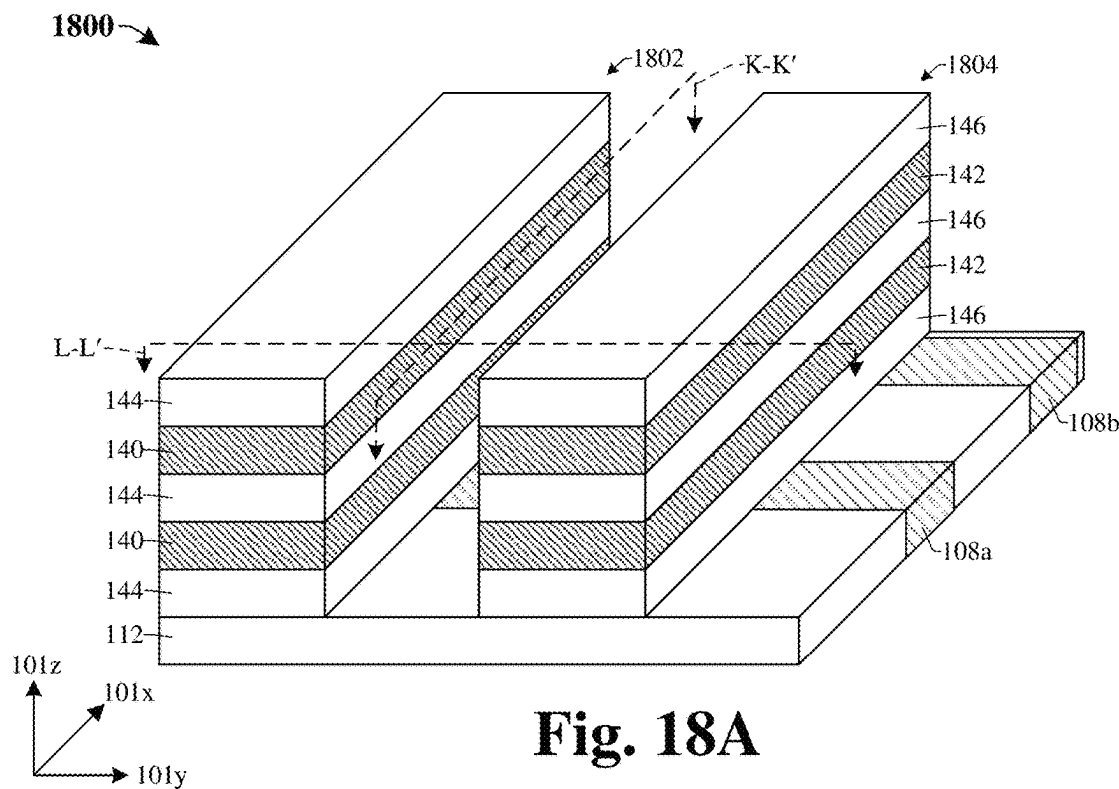


Fig. 16D







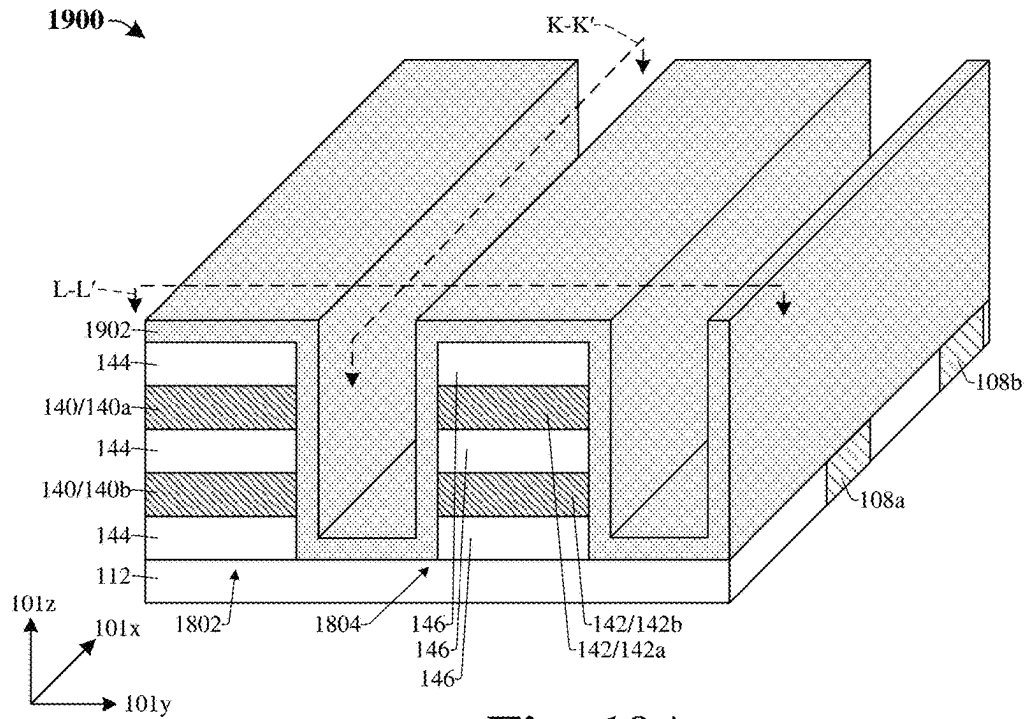


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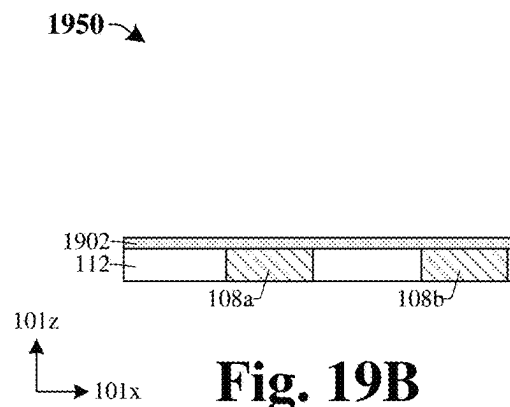


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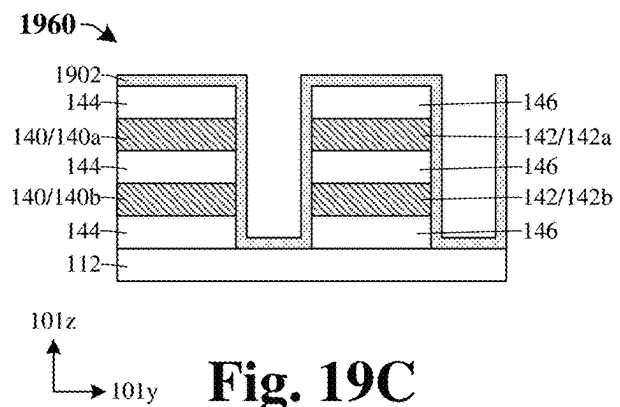
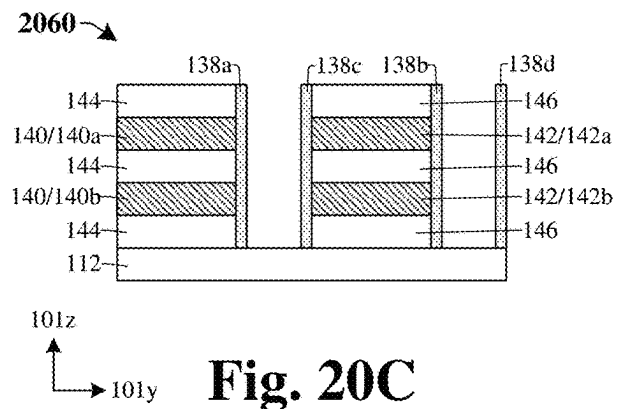
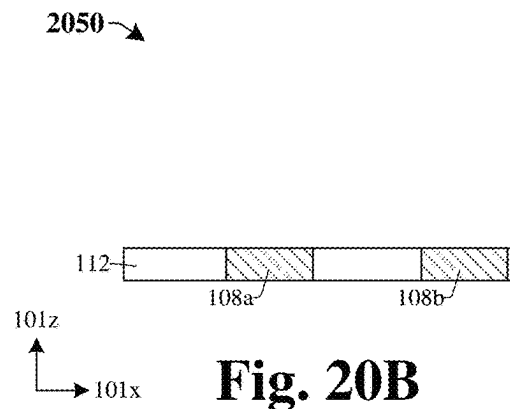
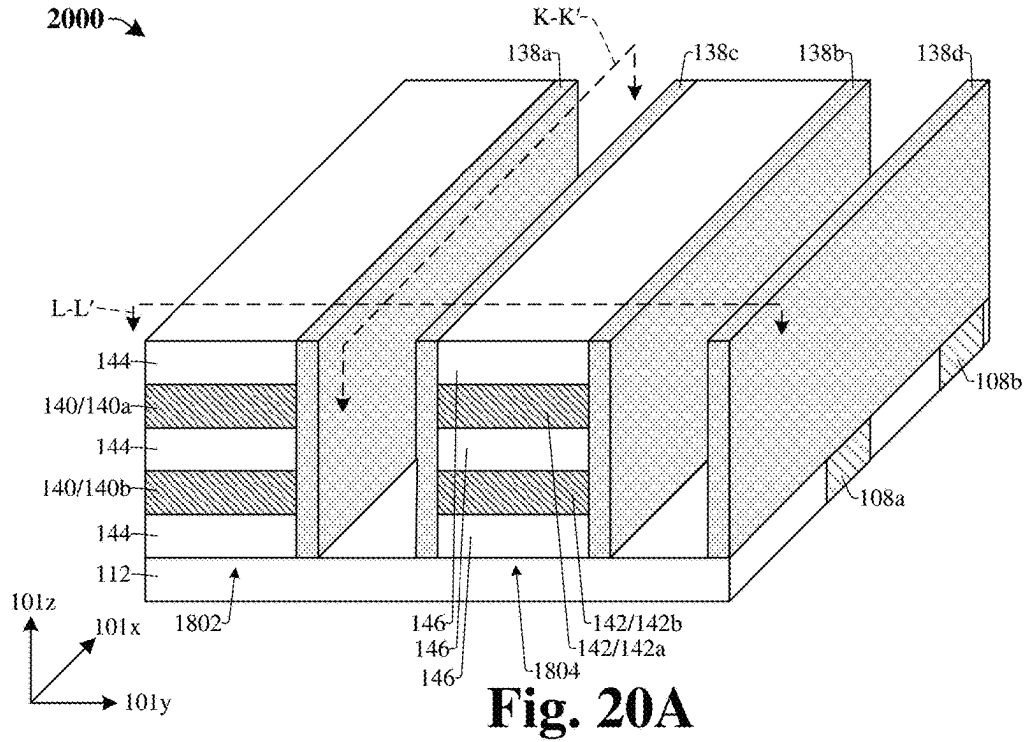


Fig. 19C



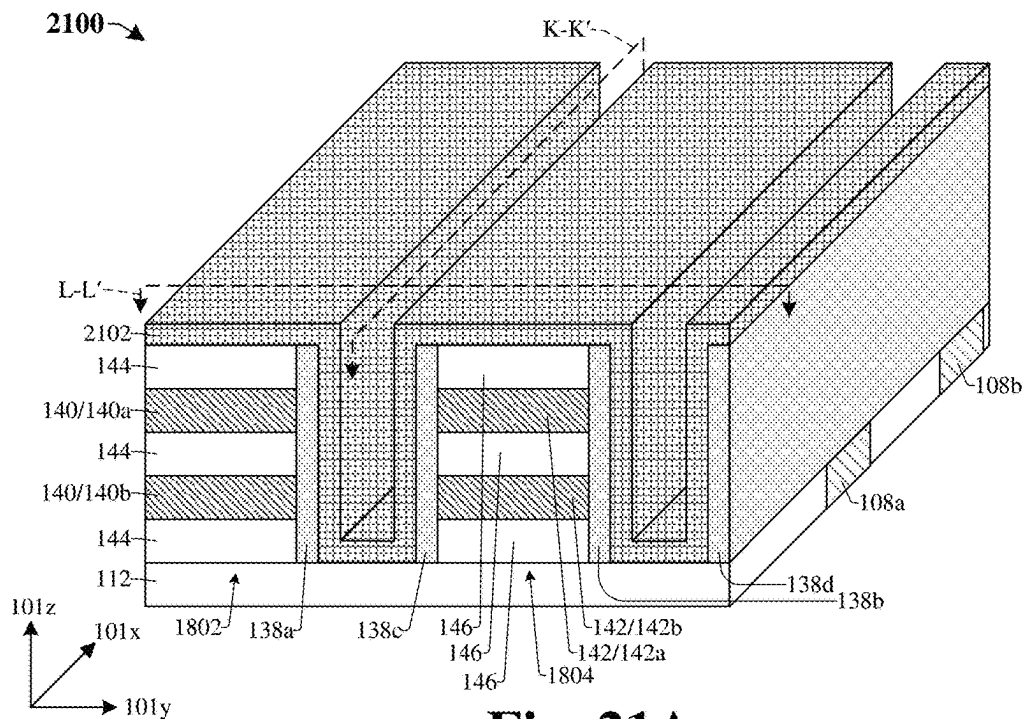


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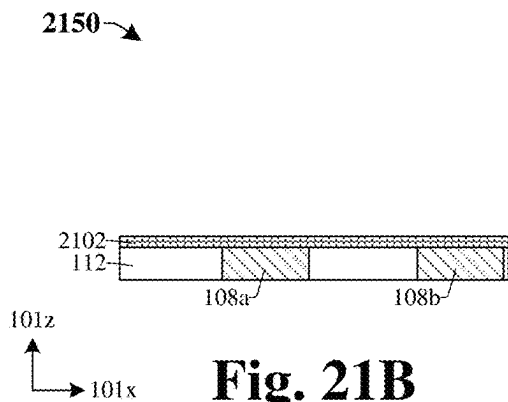


Fig. 21B

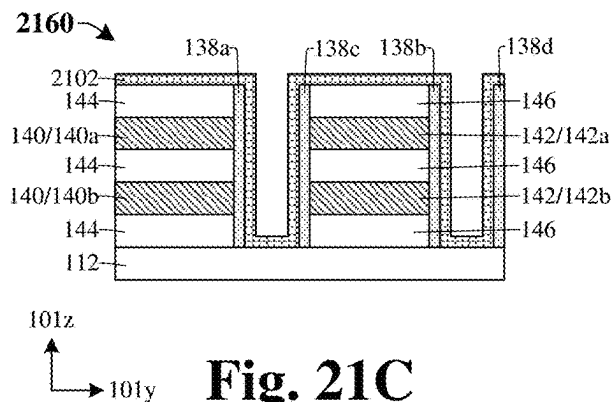


Fig. 21C

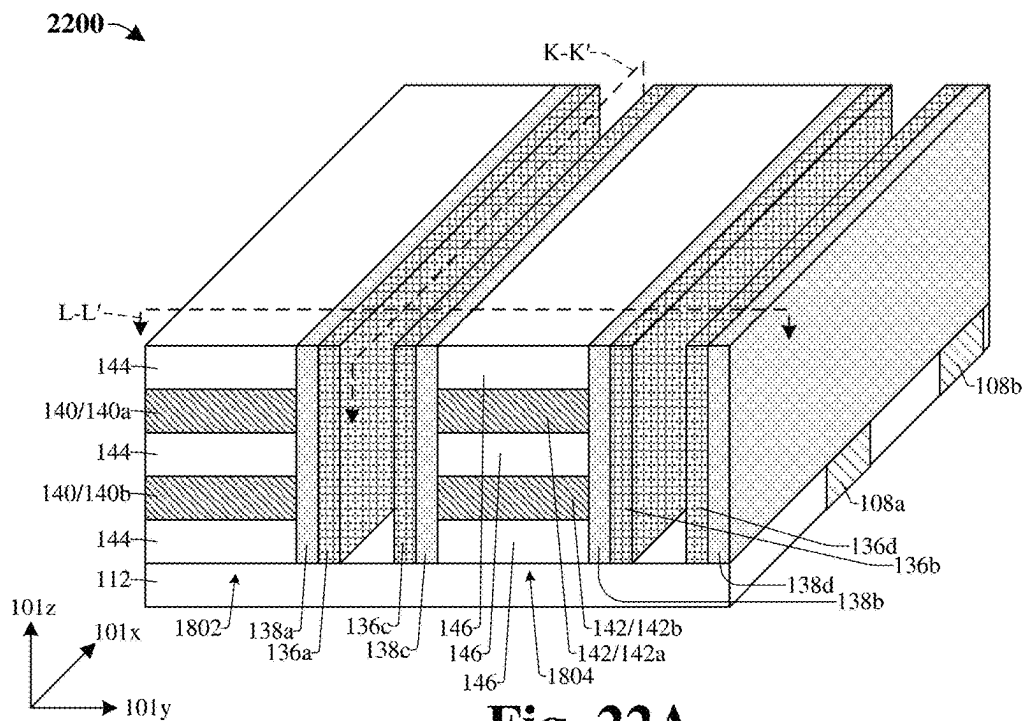


Fig. 22A

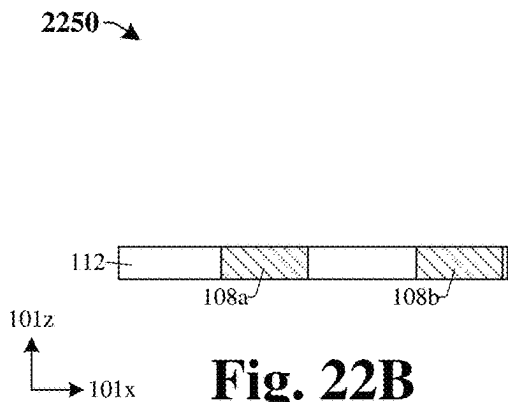


Fig. 22B

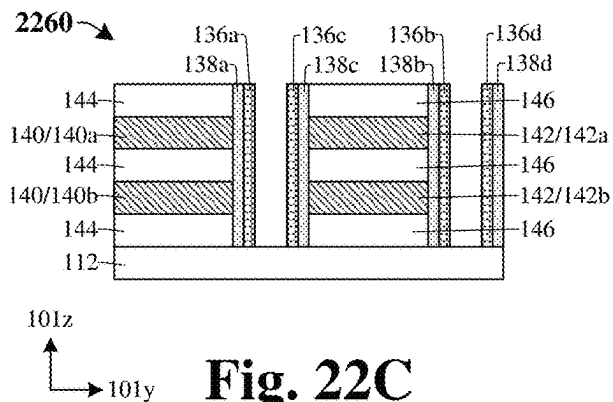
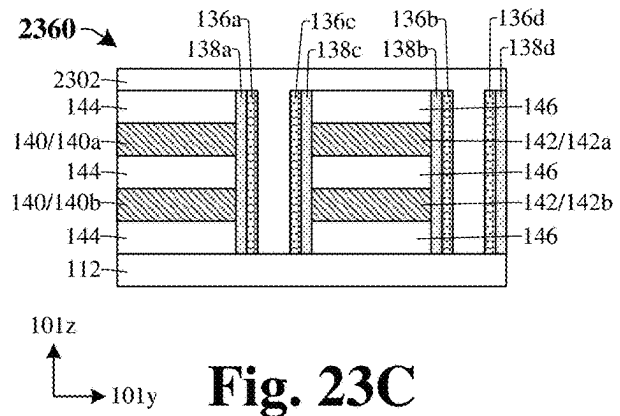
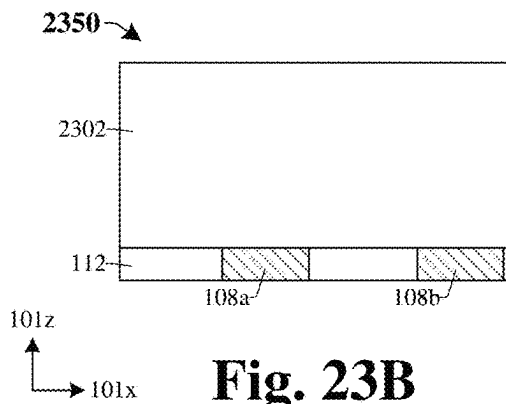
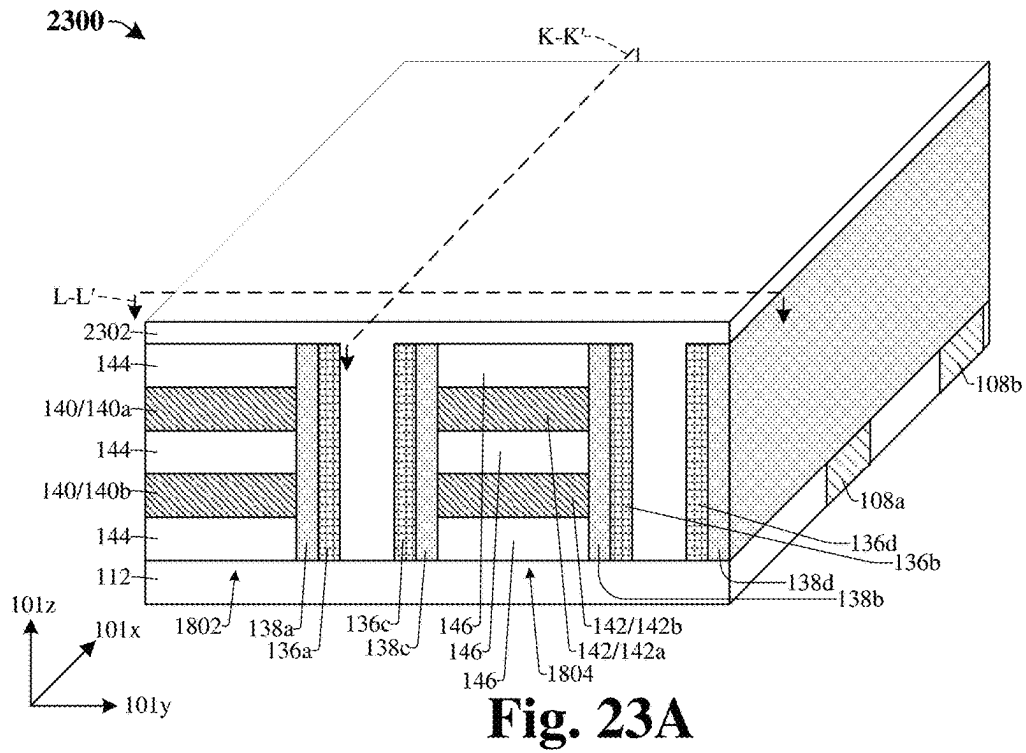


Fig. 22C



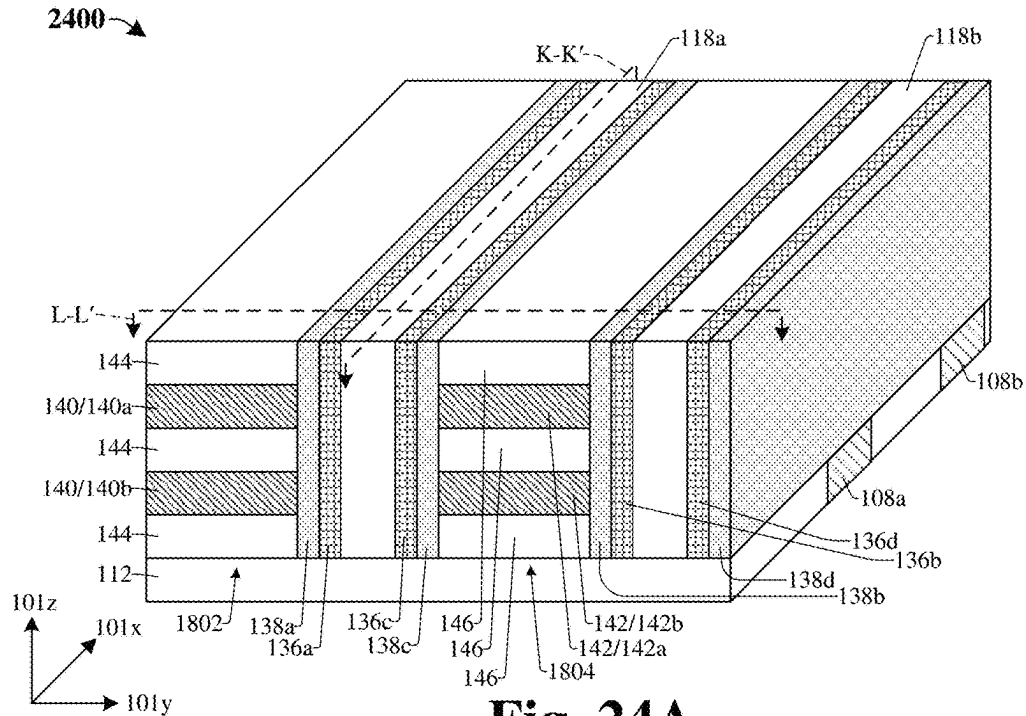


Fig. 24A

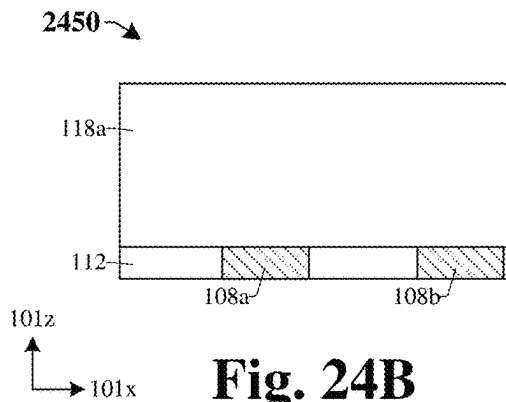


Fig. 24B

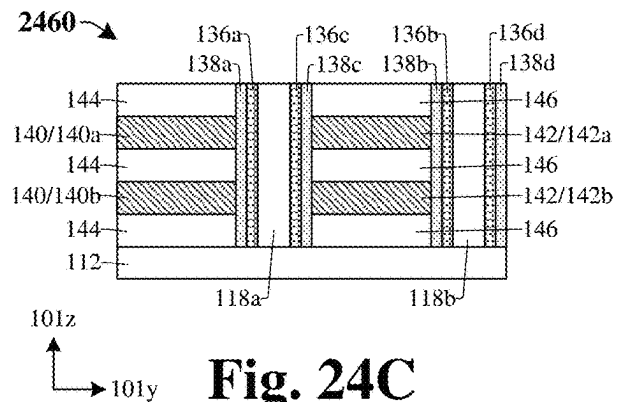


Fig. 24C

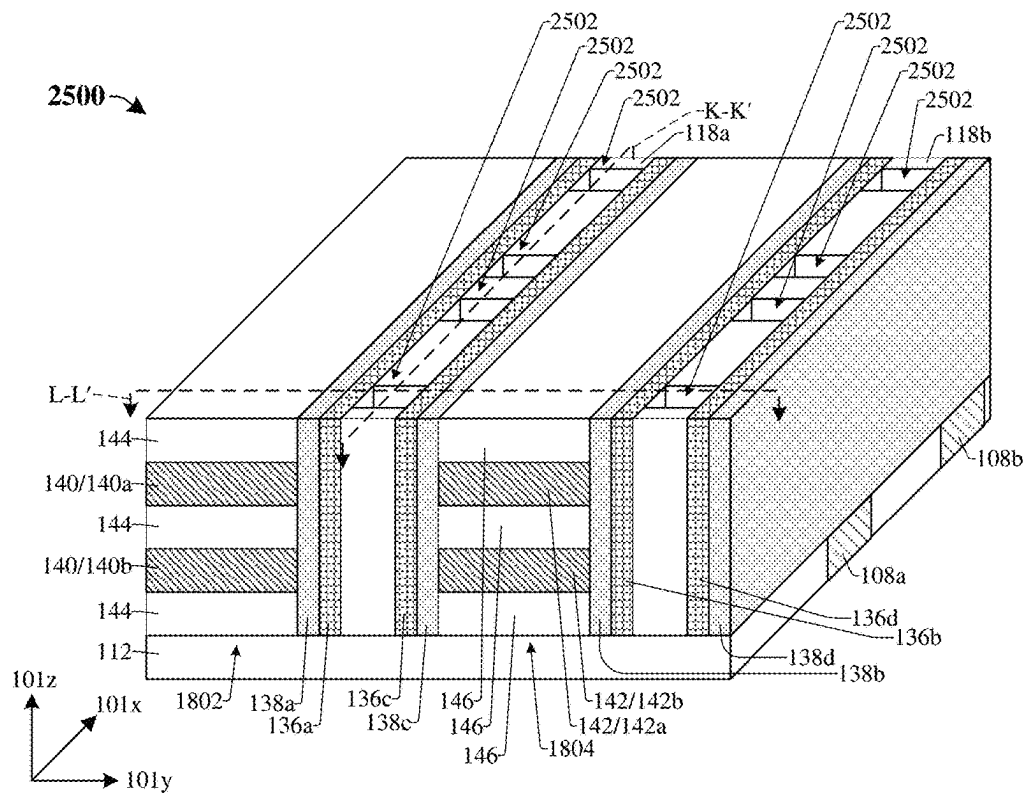


Fig. 25A

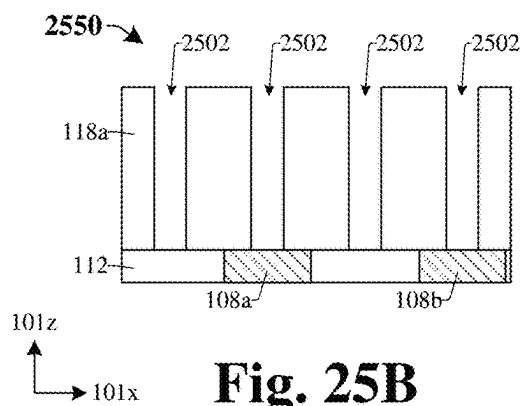


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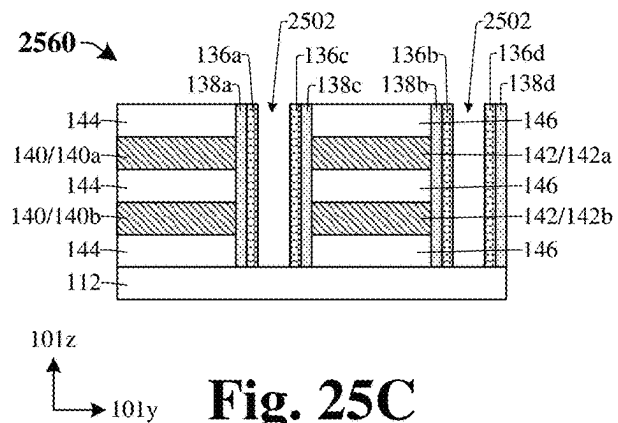


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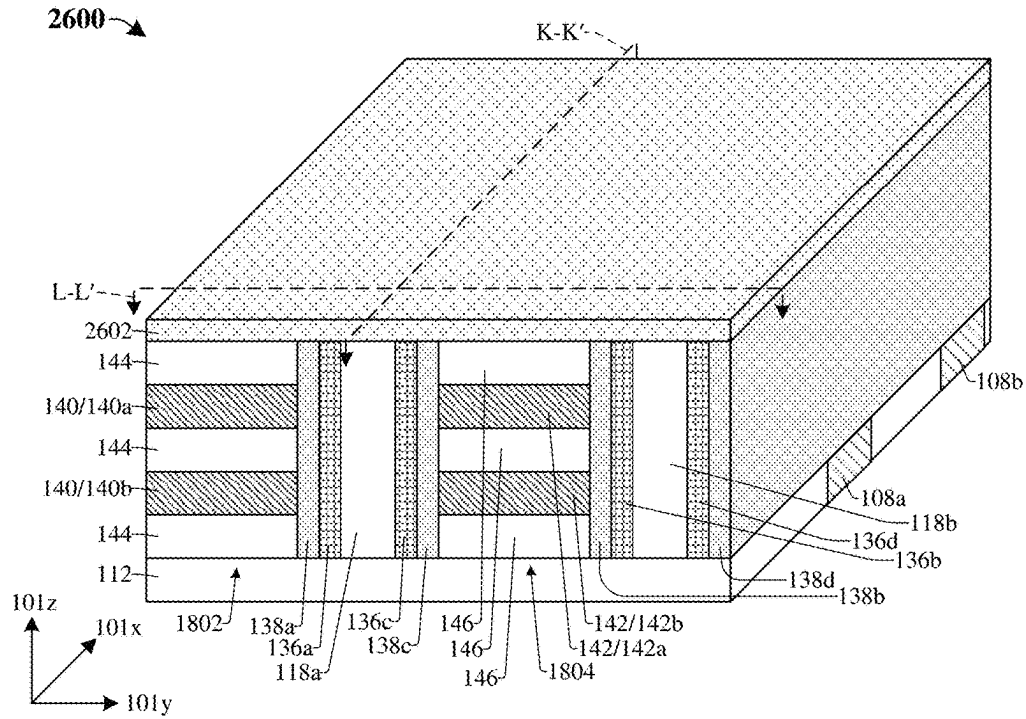


Fig. 26A

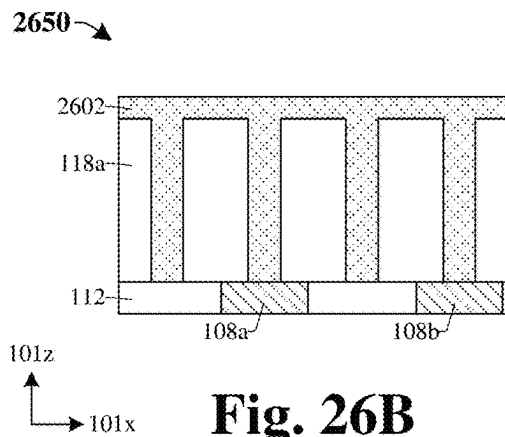


Fig. 26B

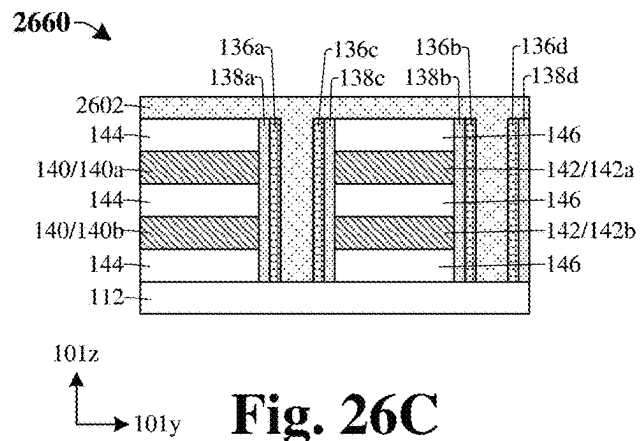


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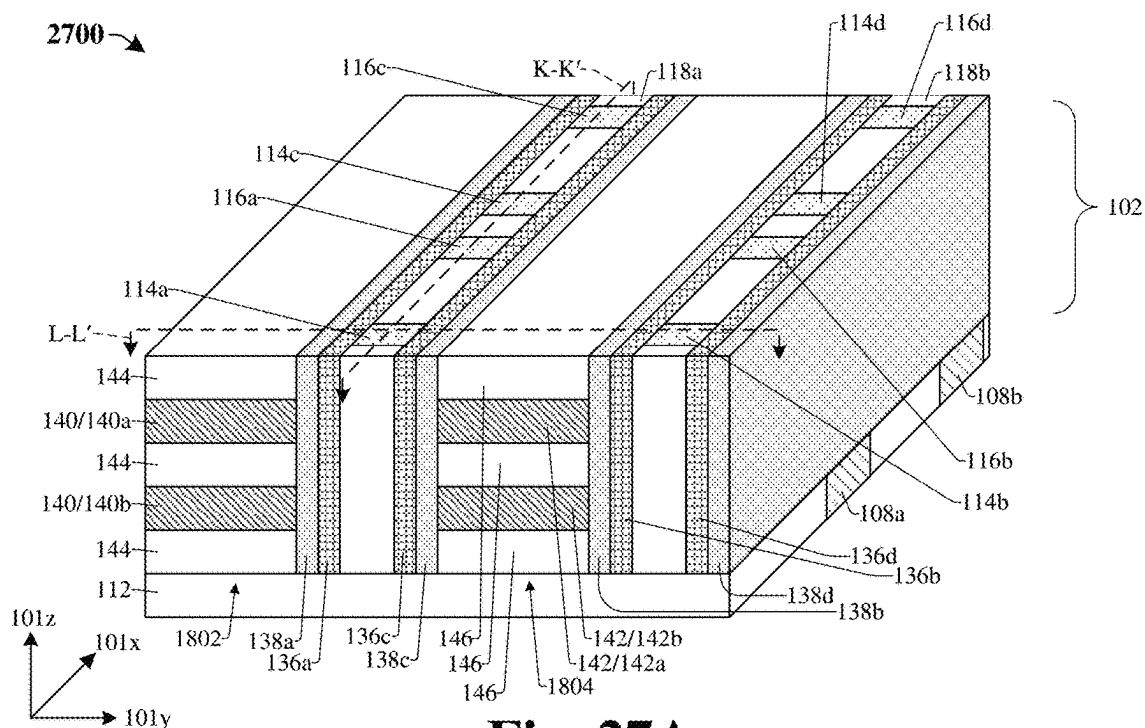


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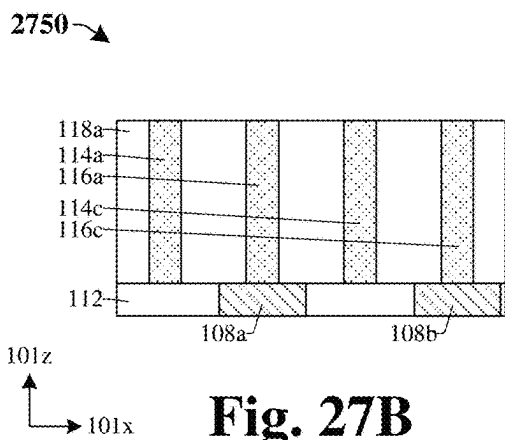


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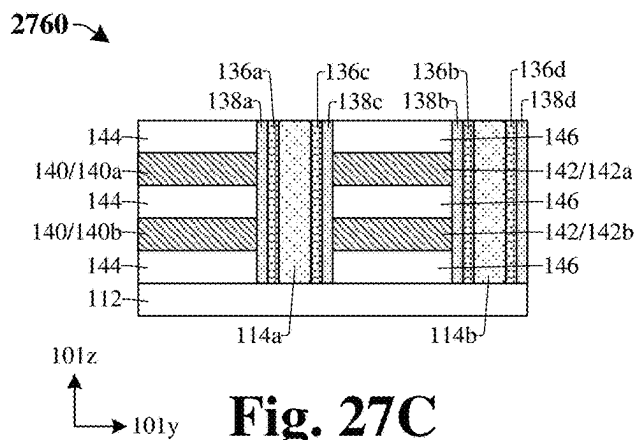


Fig. 27C

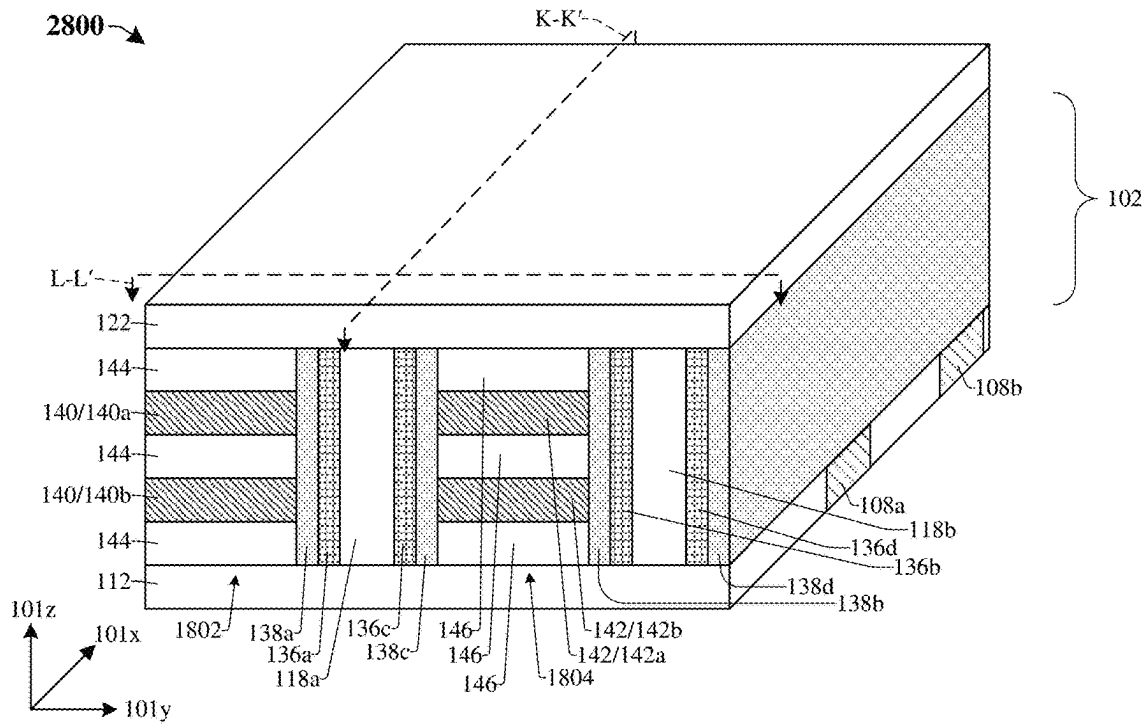


Fig. 28A

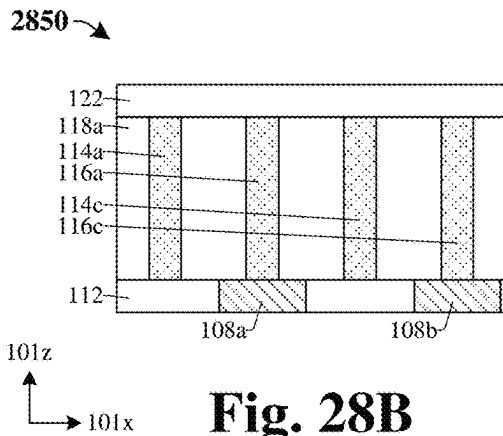


Fig. 28B

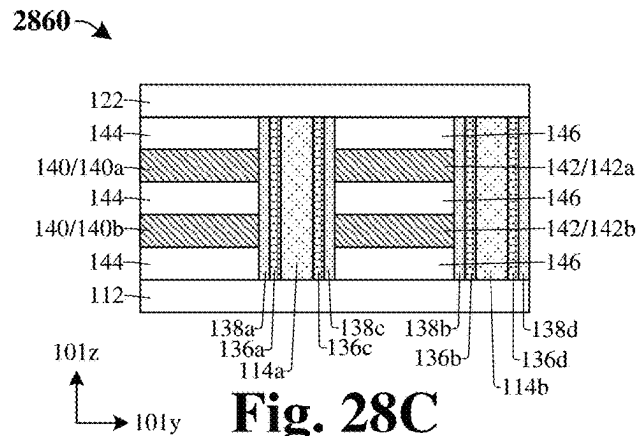


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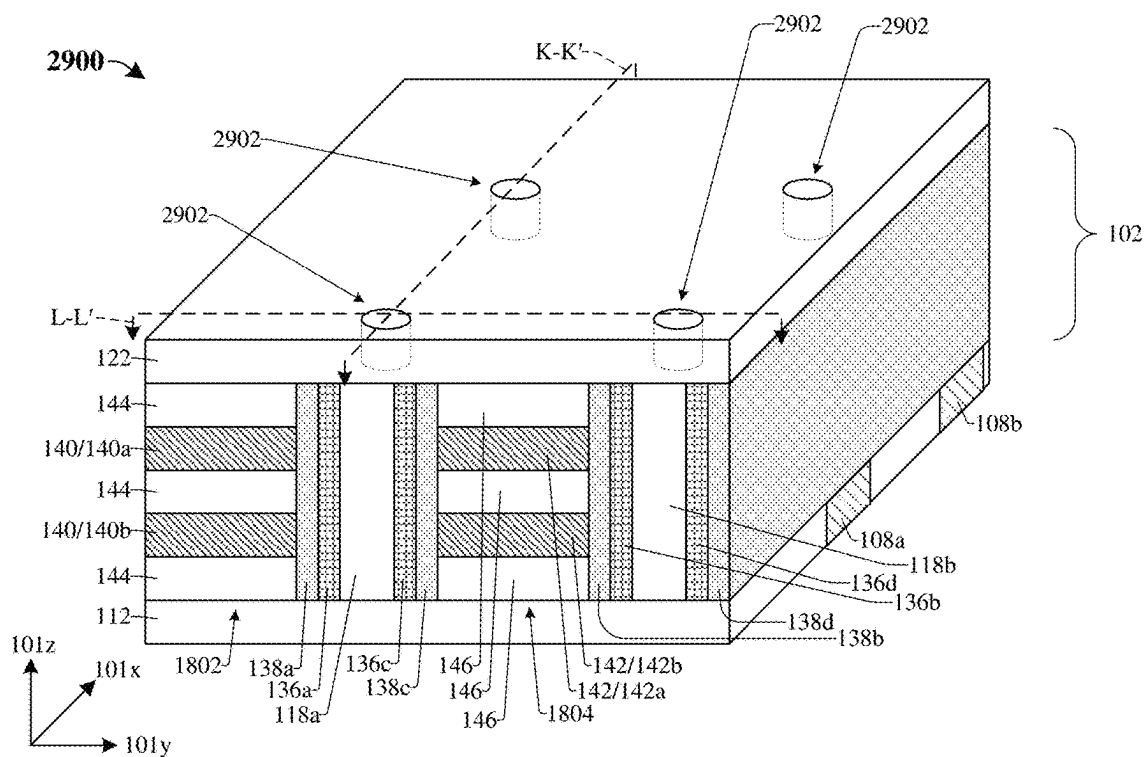


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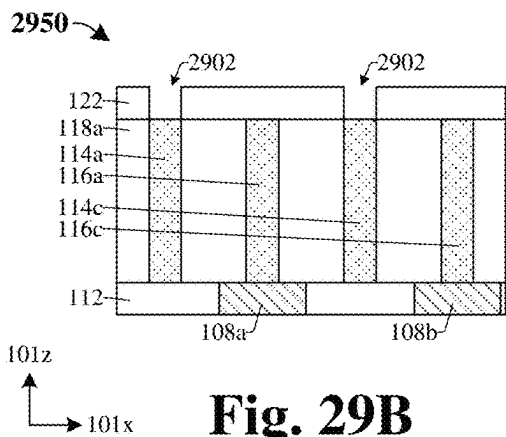


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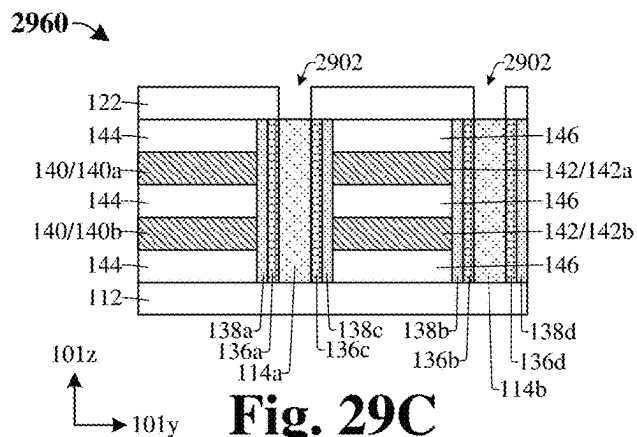


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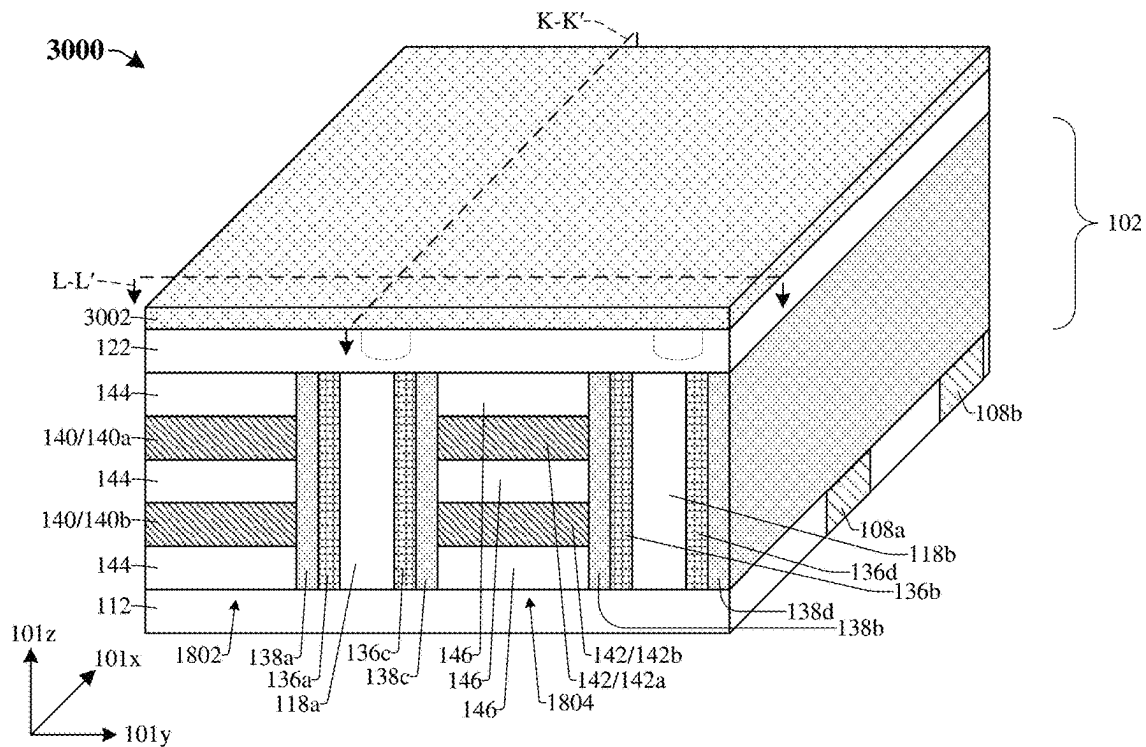


Fig. 30A

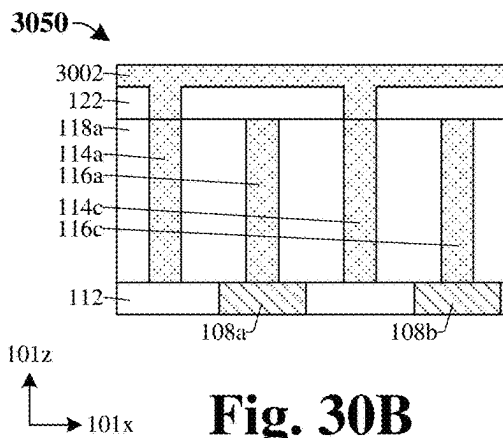


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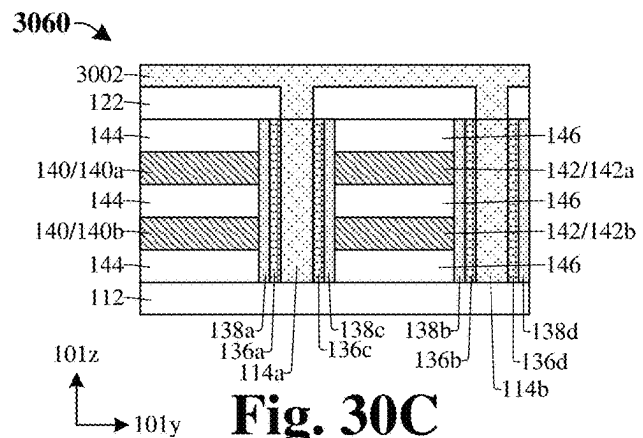


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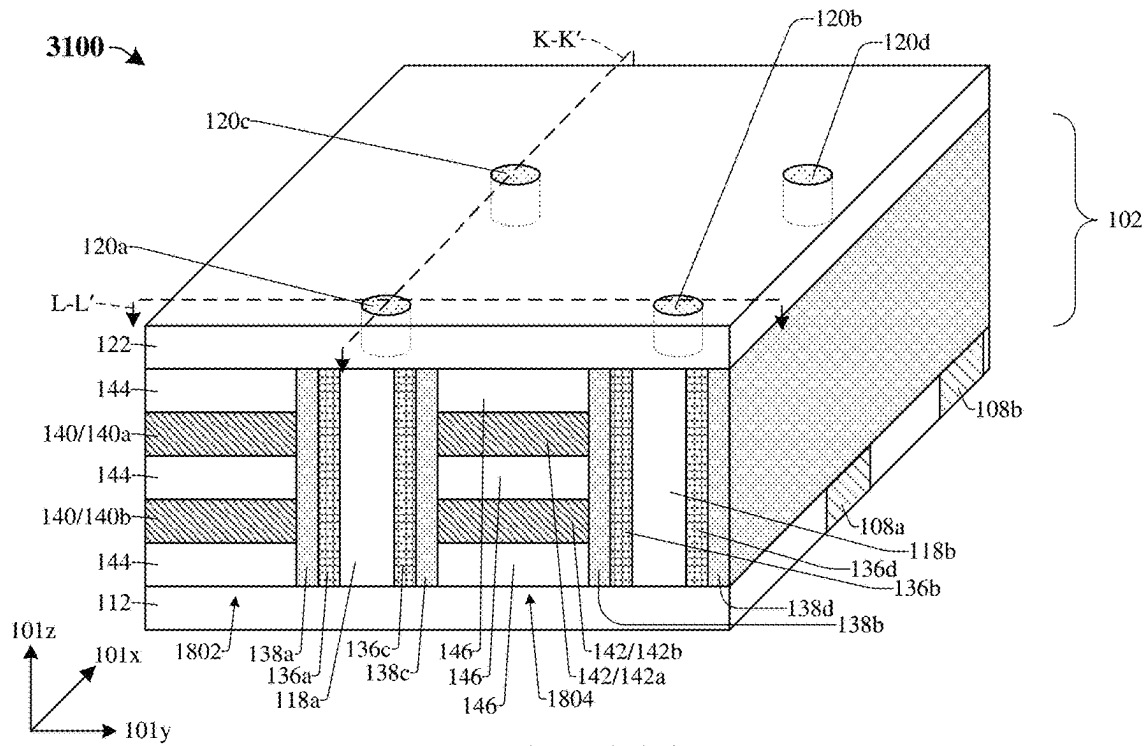


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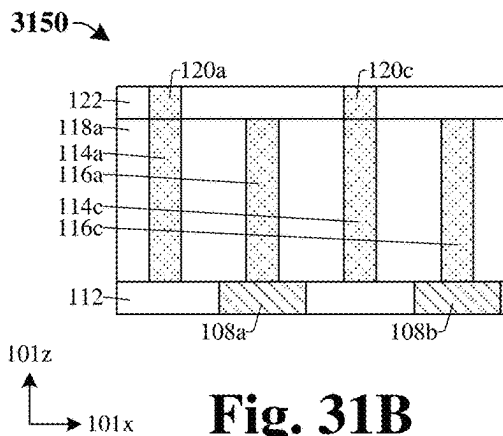


Fig. 31B

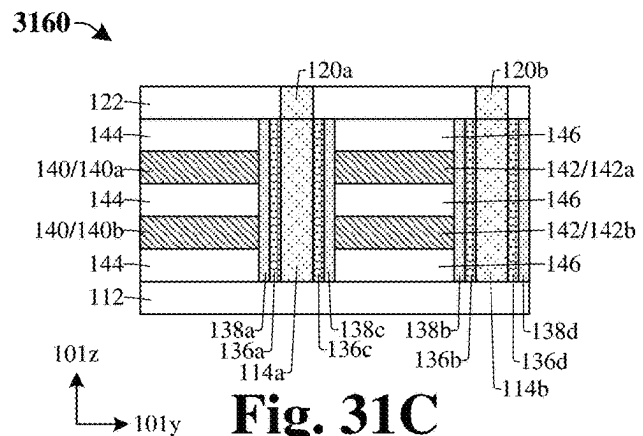


Fig. 31C

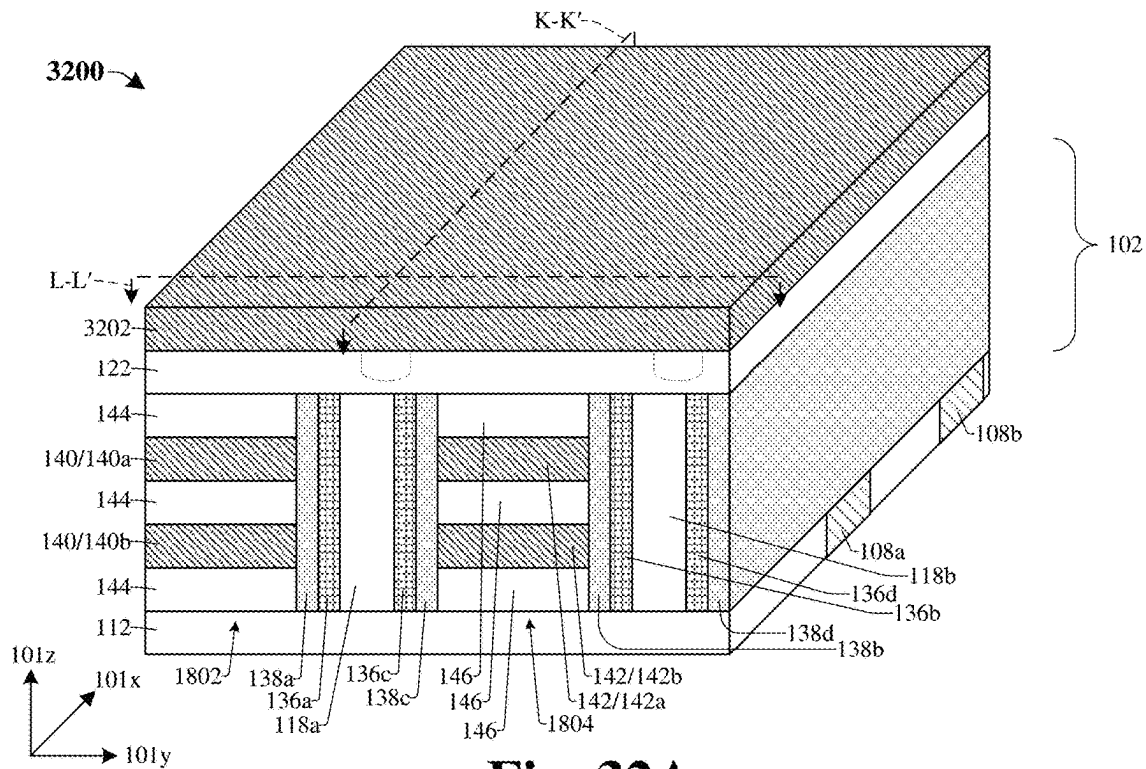


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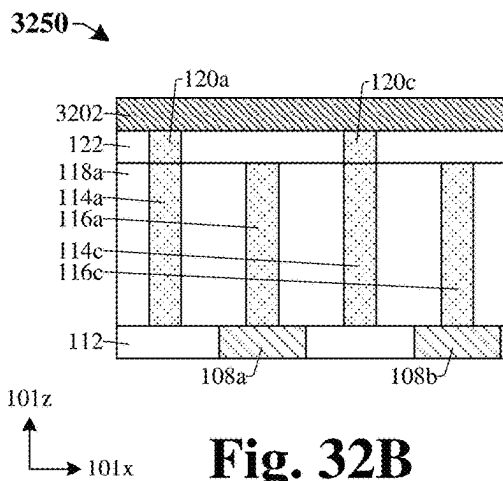


Fig. 32B

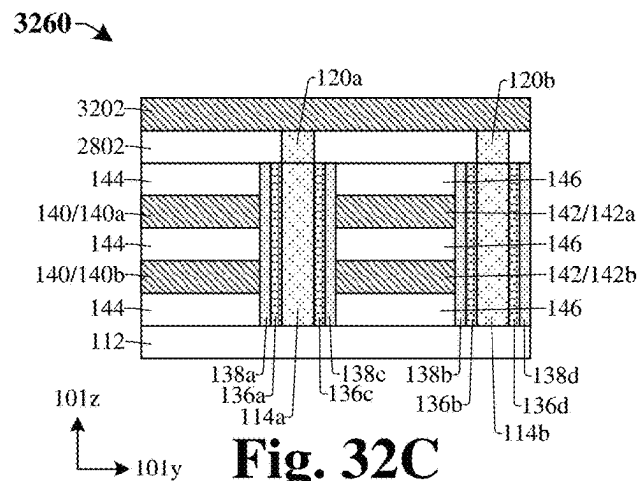


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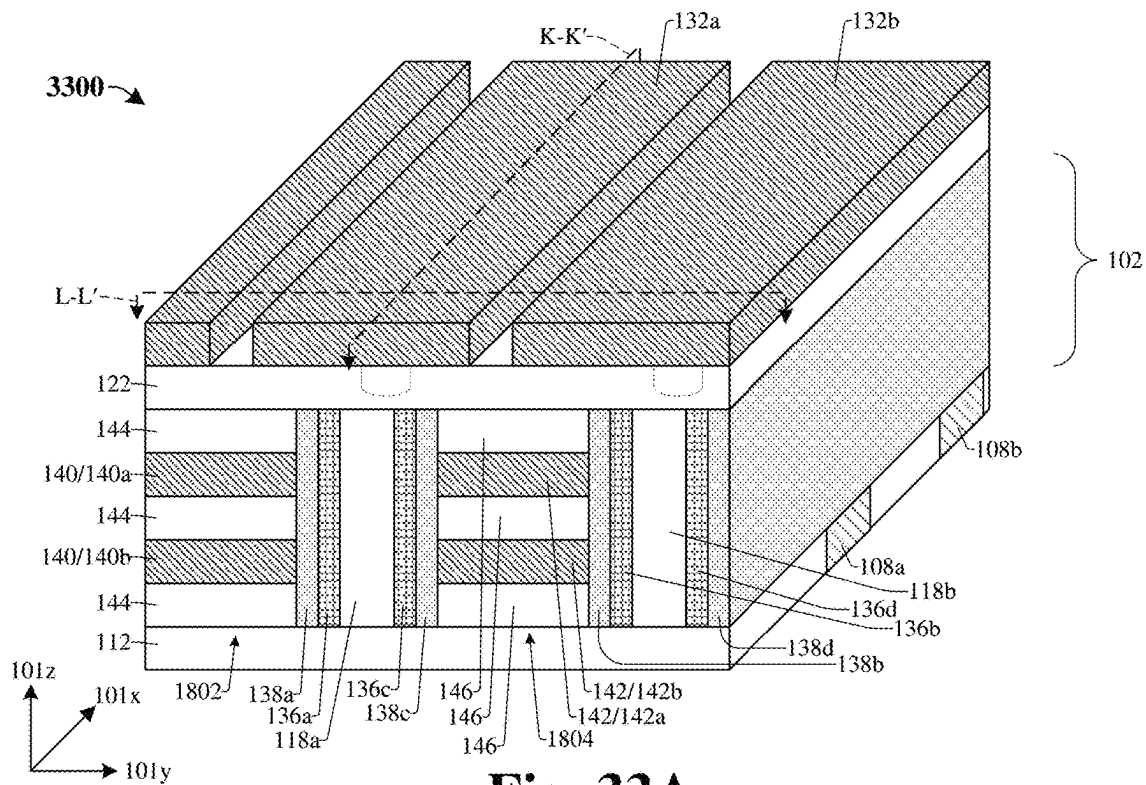


Fig. 33A

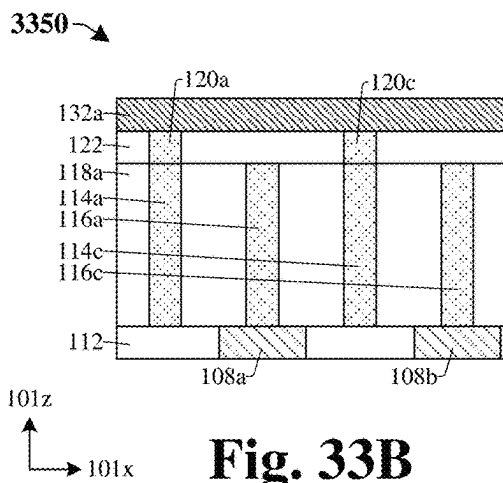


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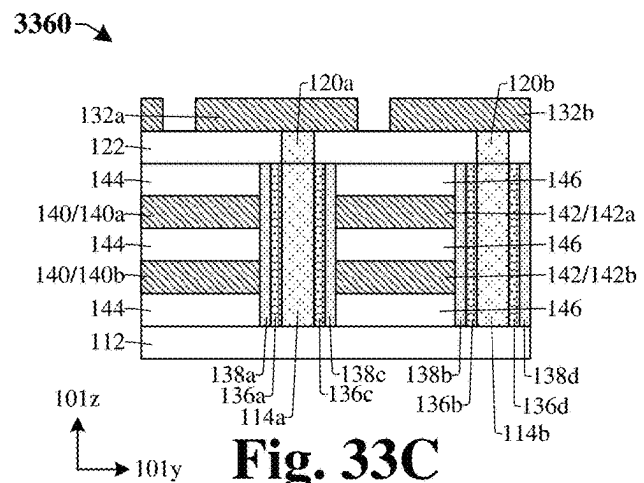
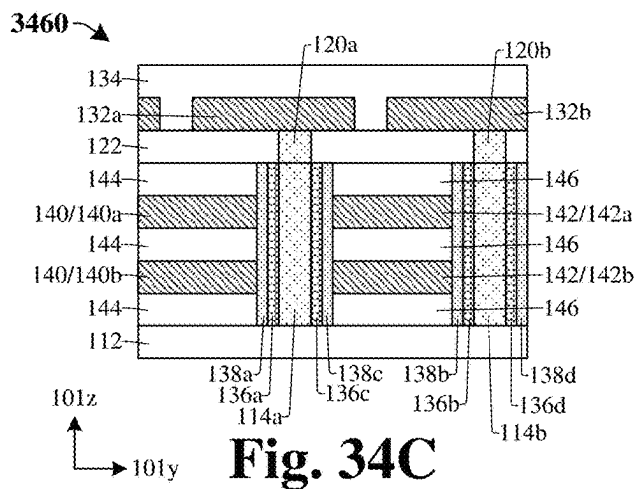
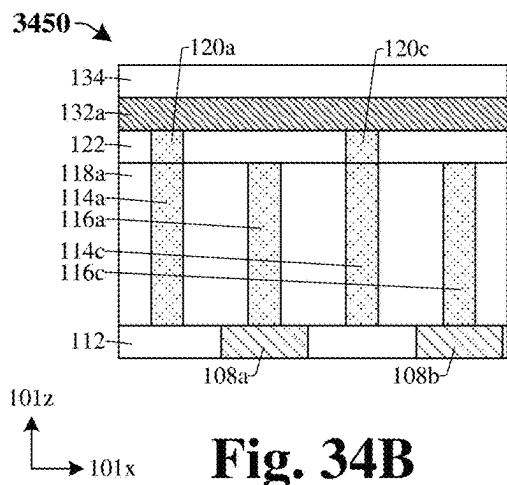
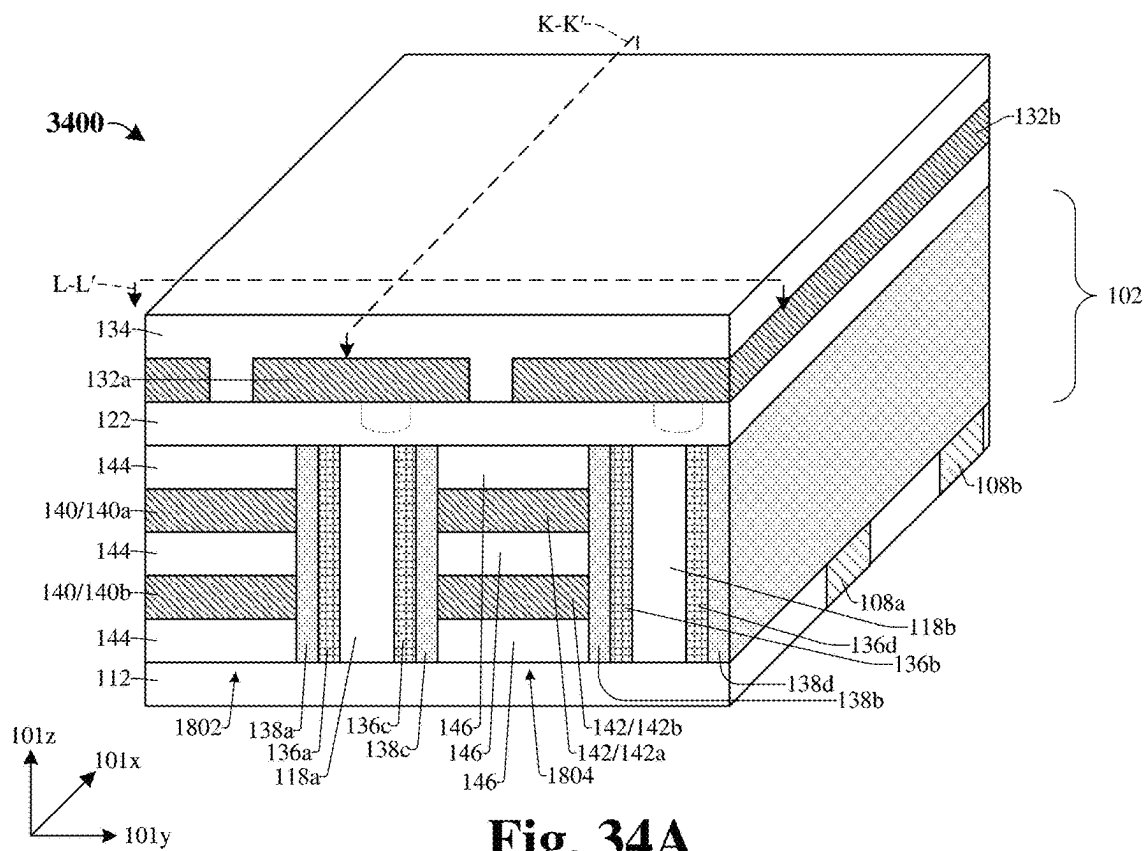
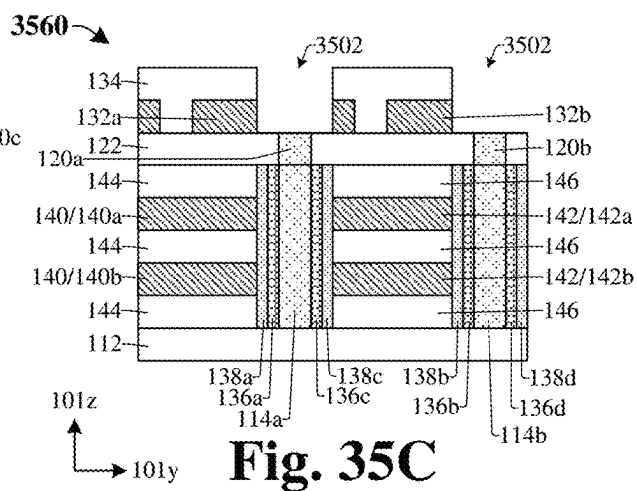
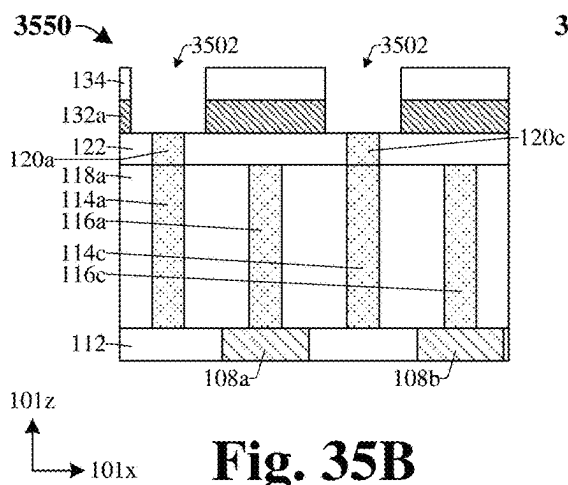
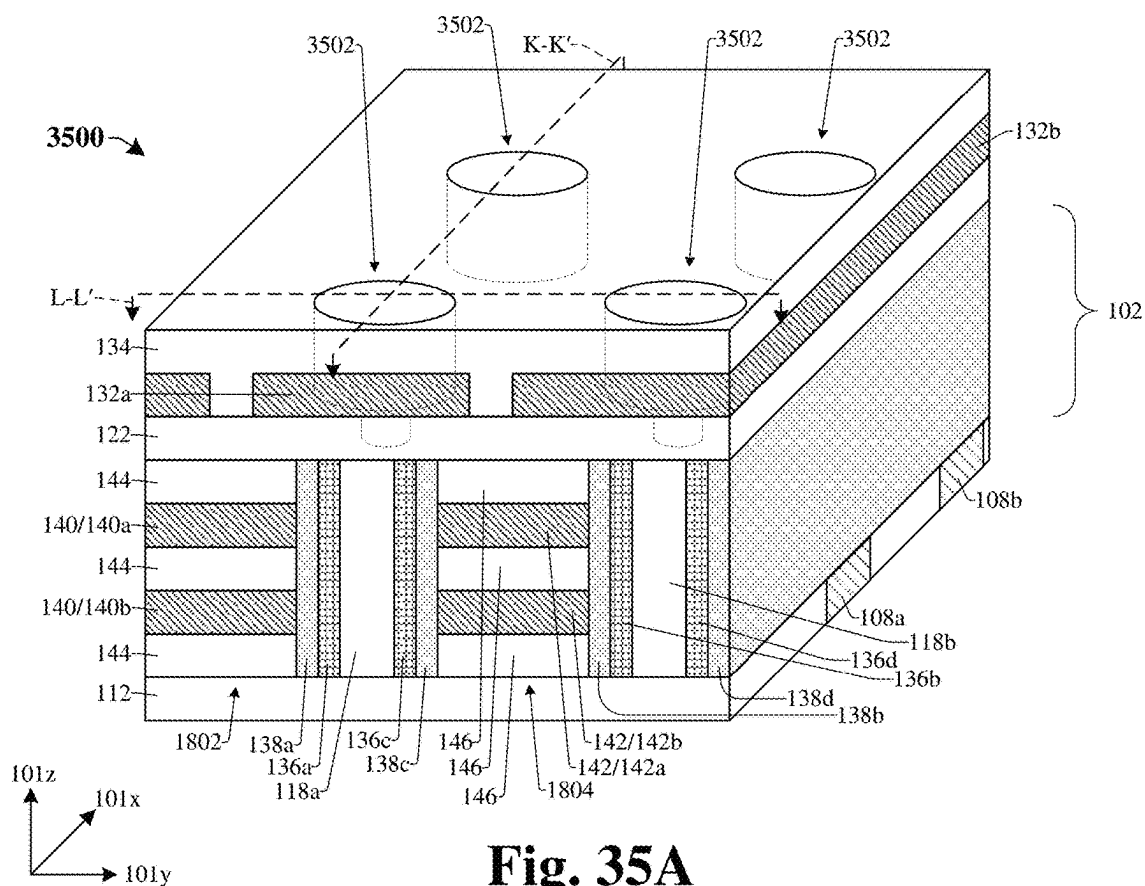
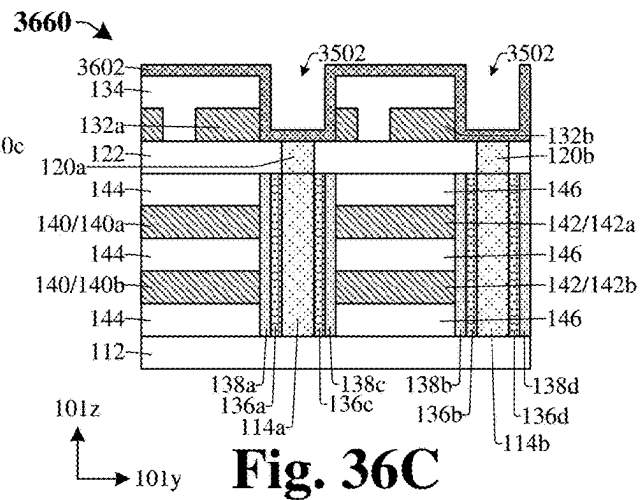
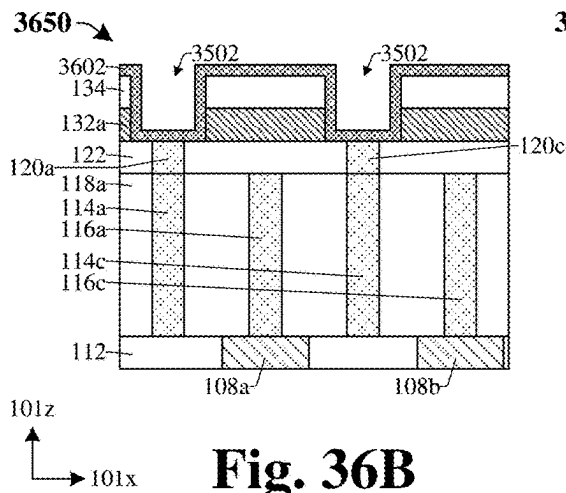
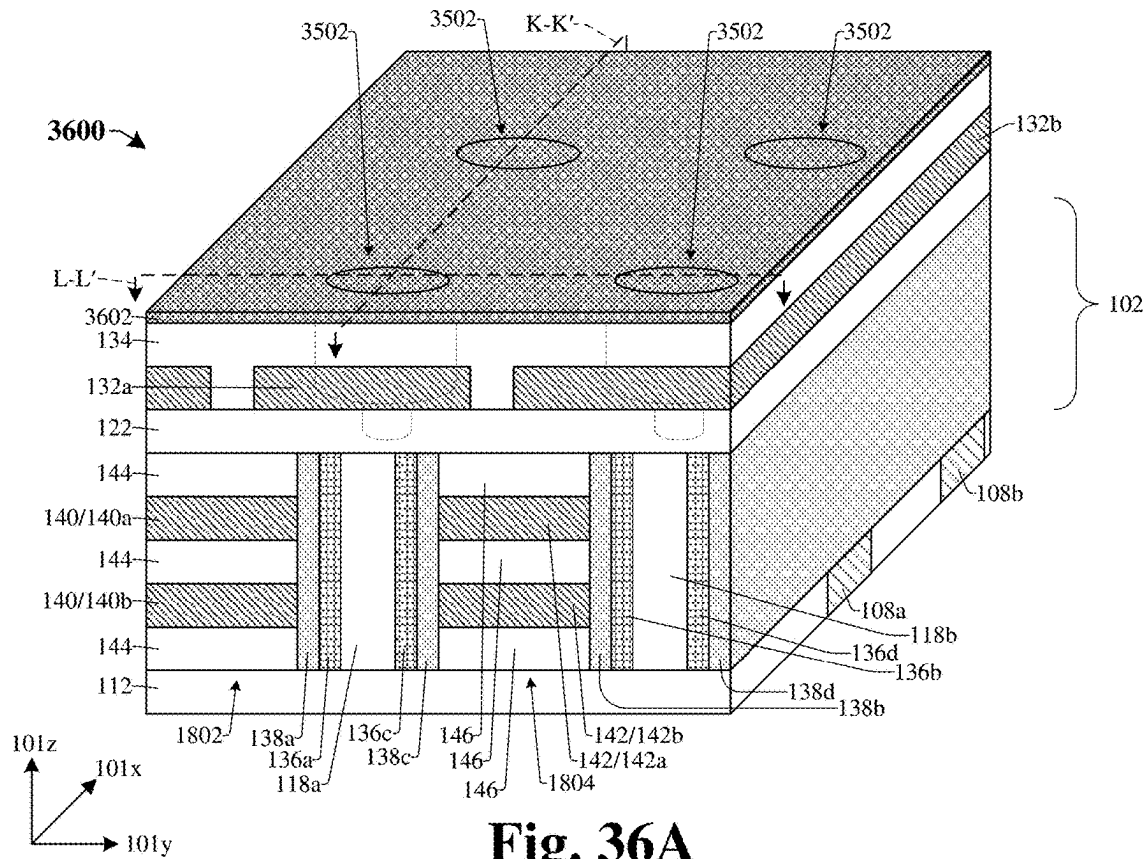


Fig. 33C







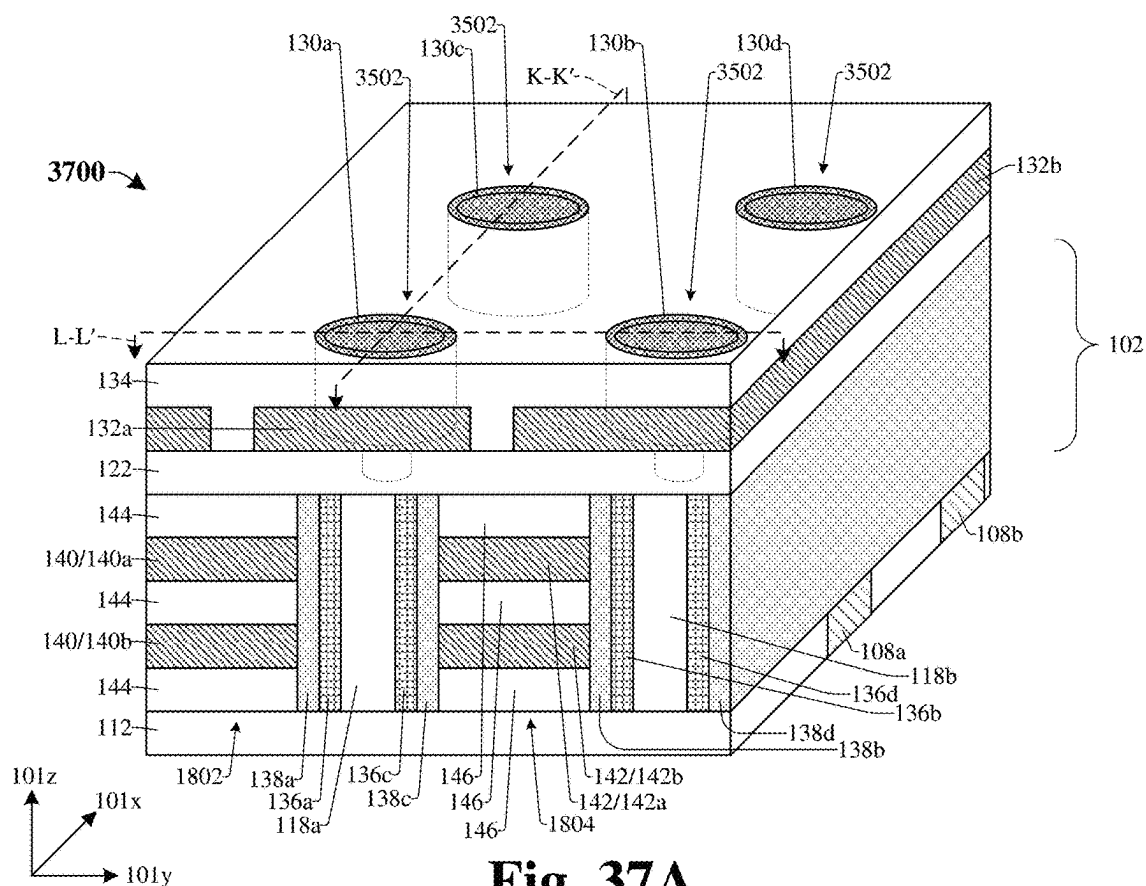


Fig. 37A

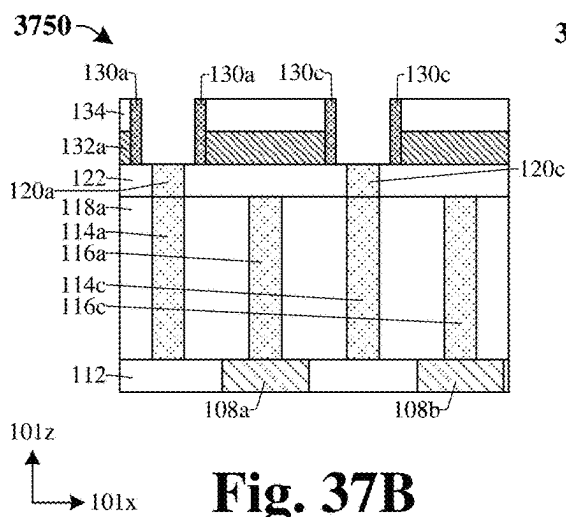


Fig. 37B

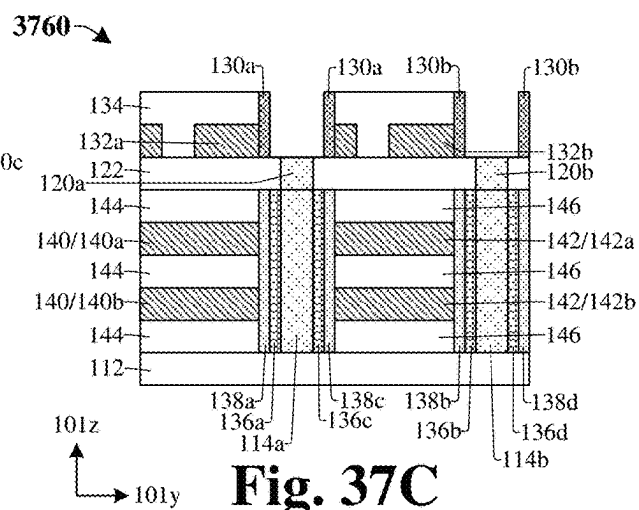
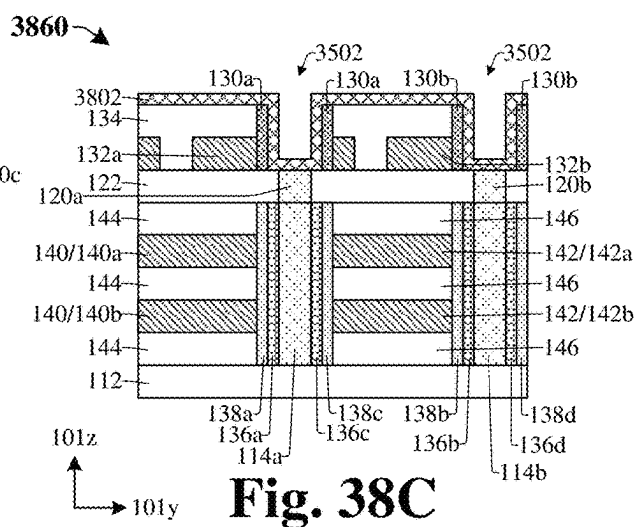
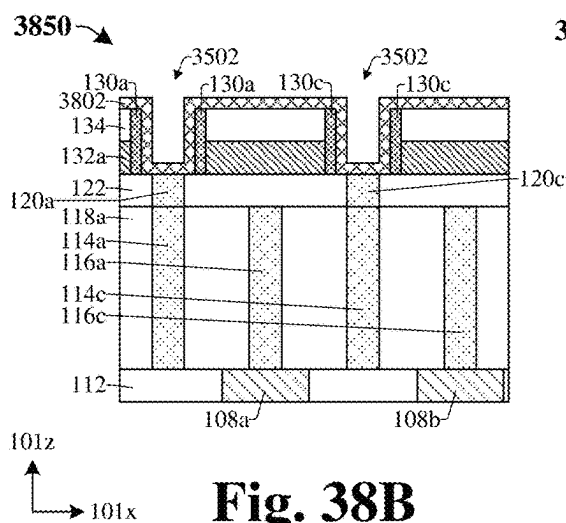
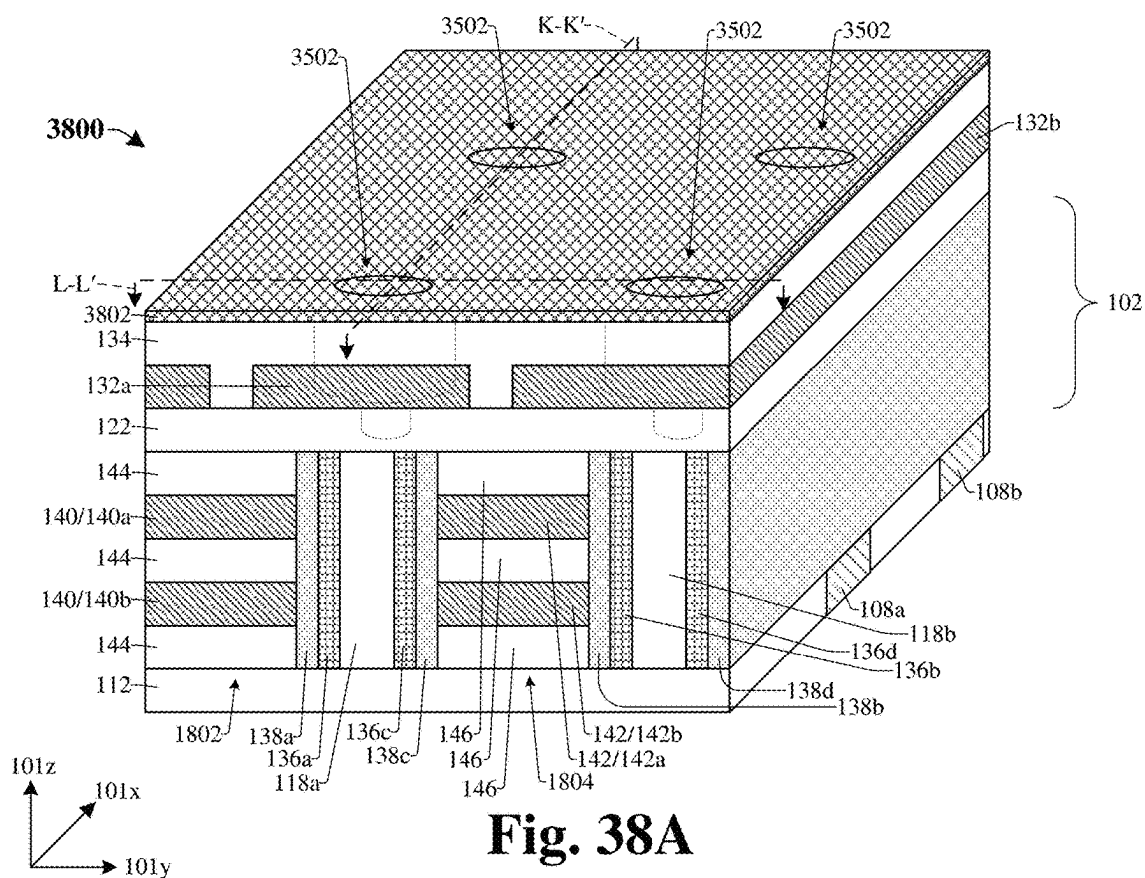
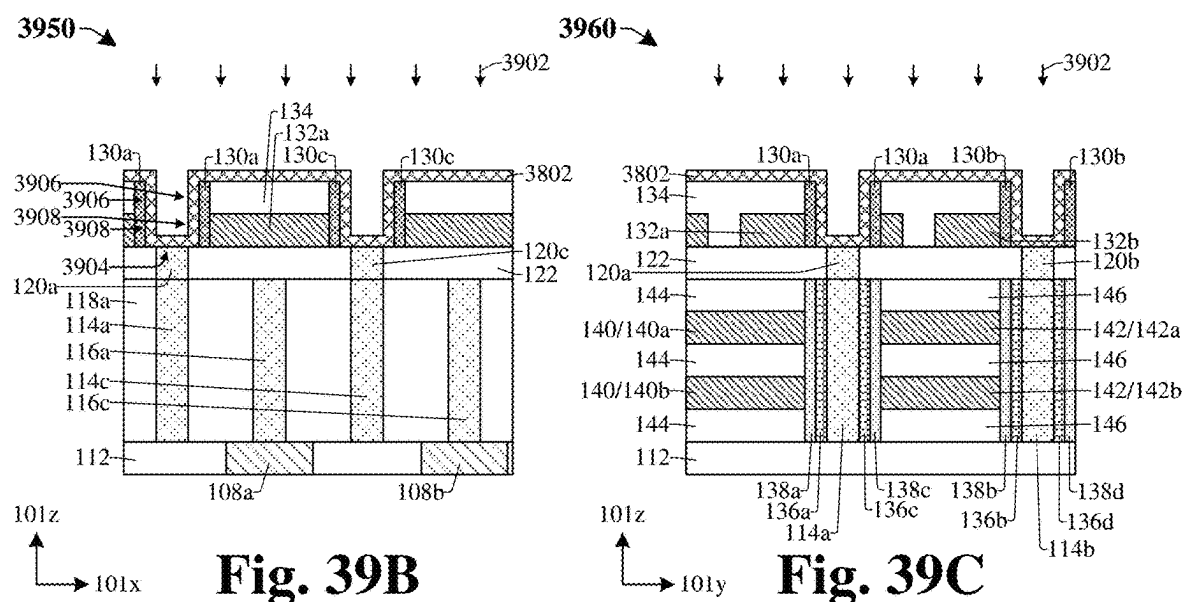
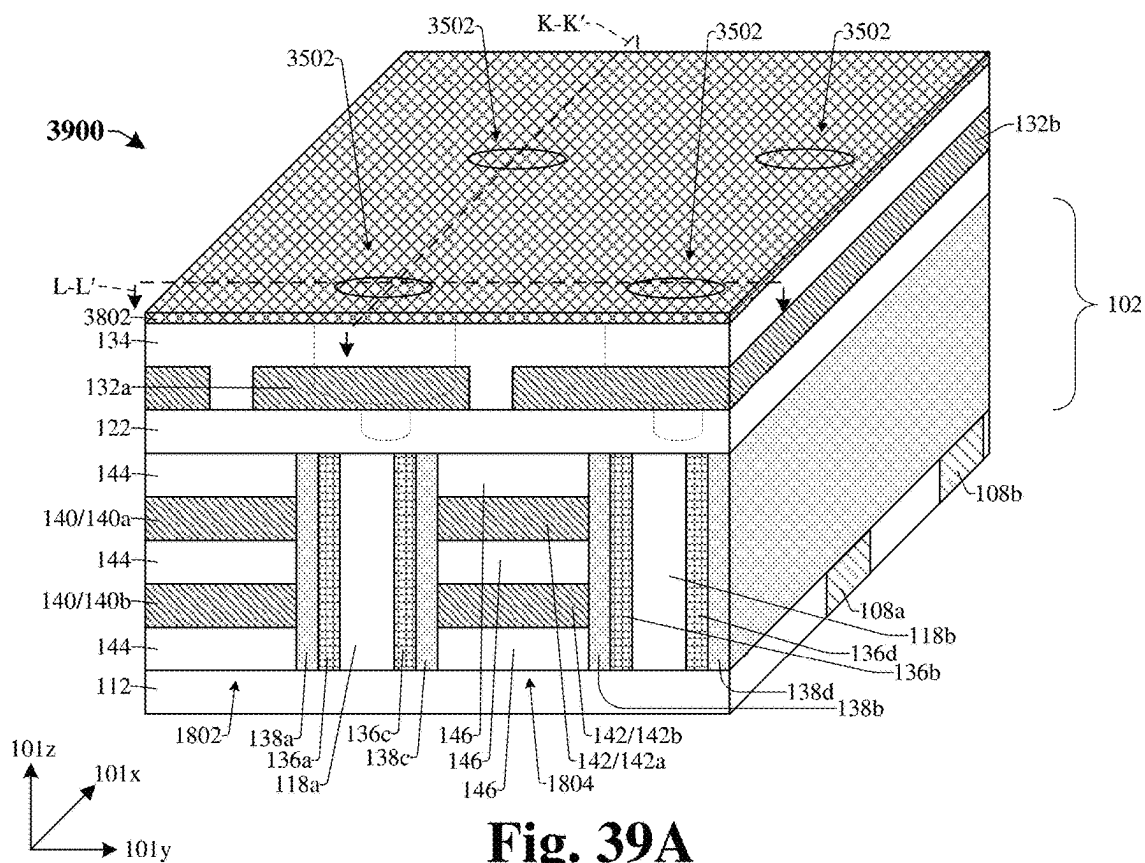
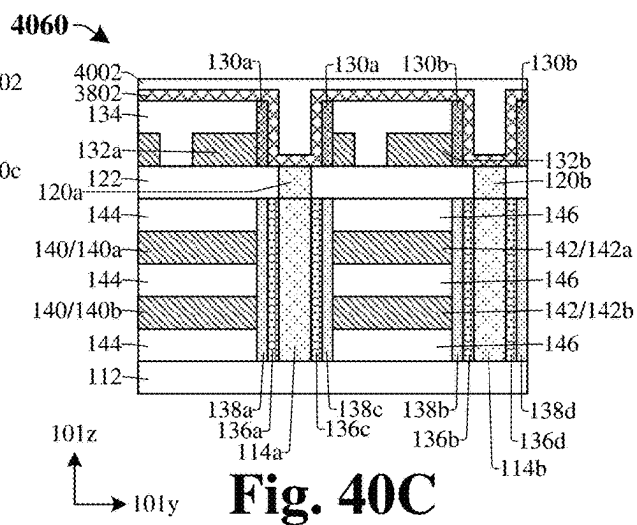
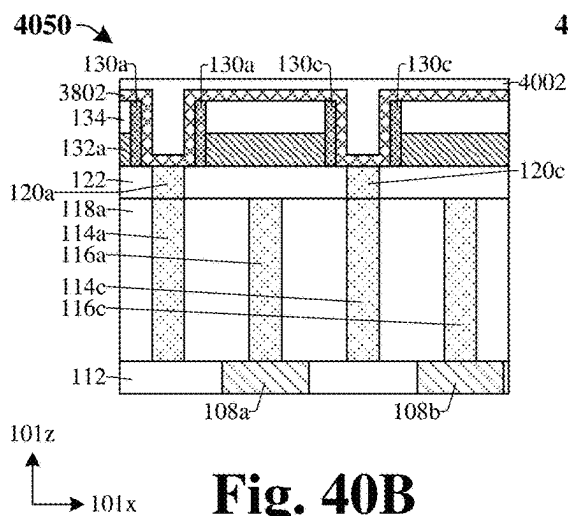
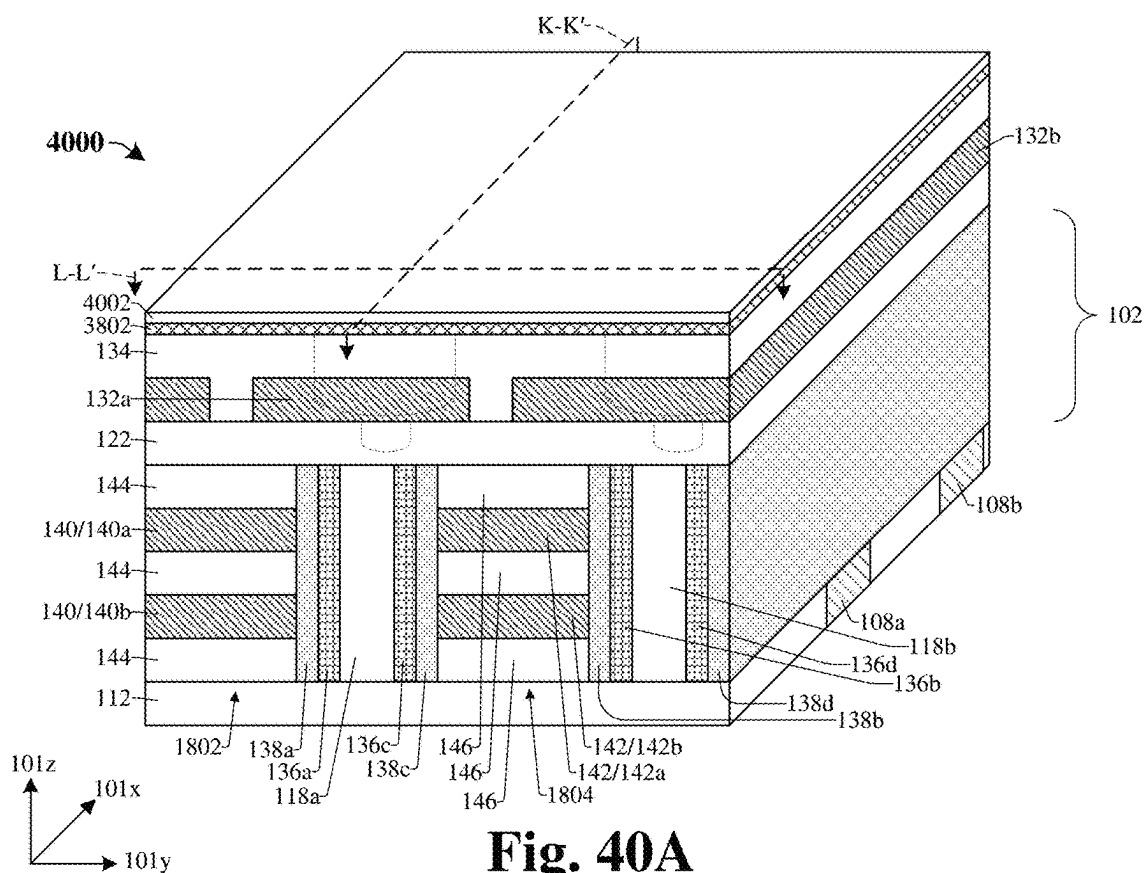
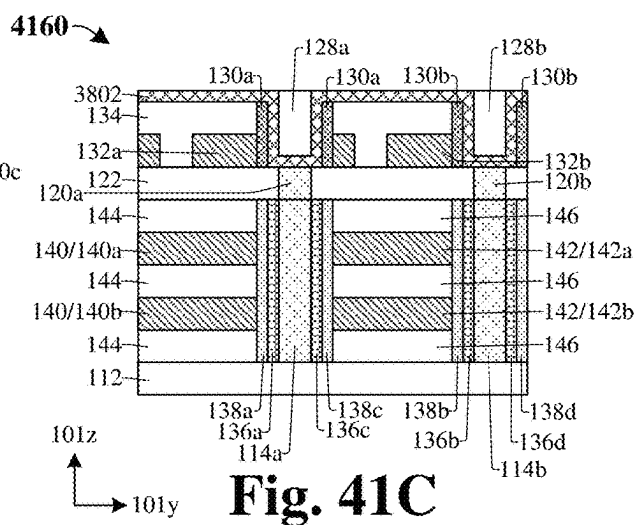
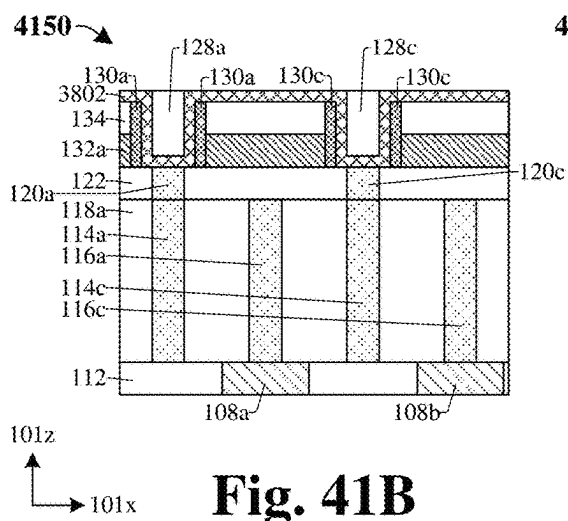
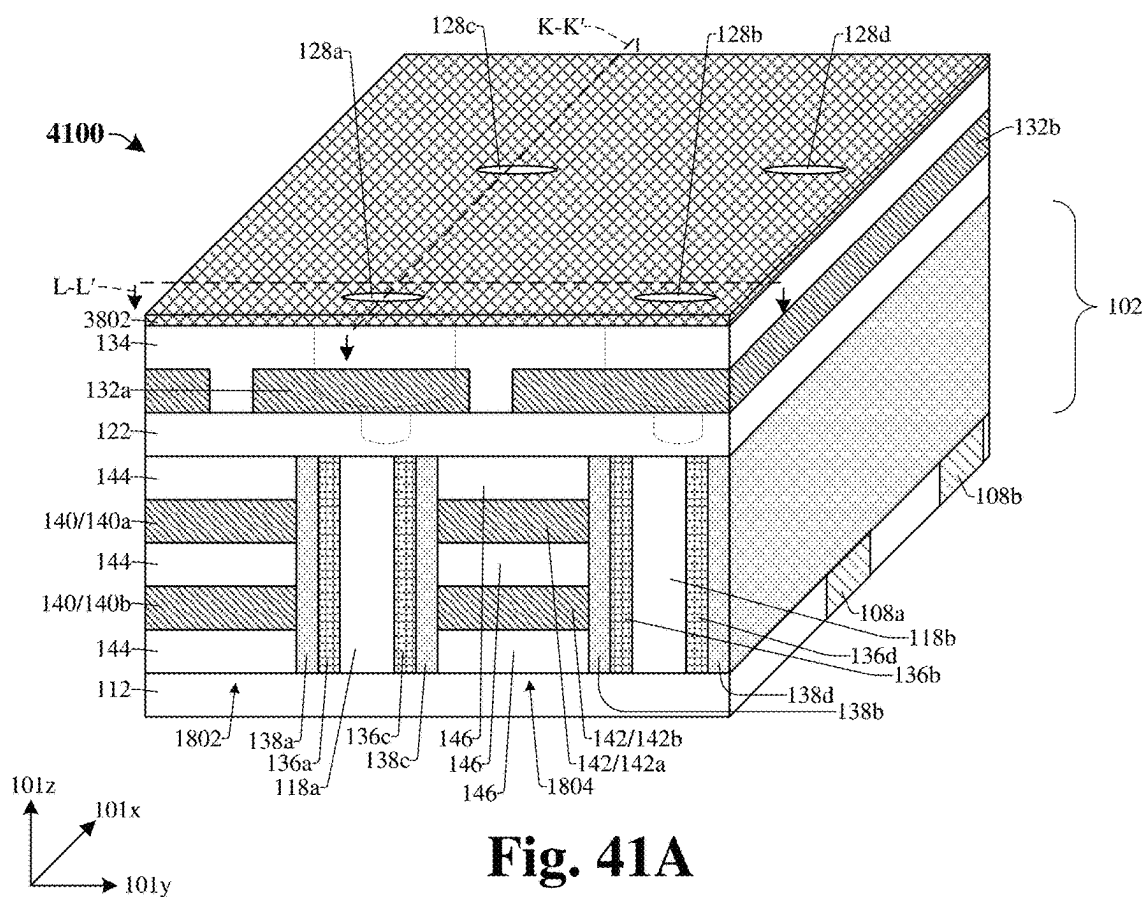


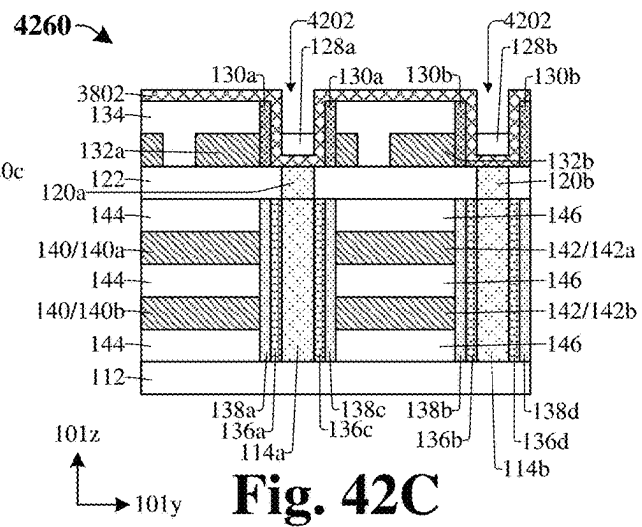
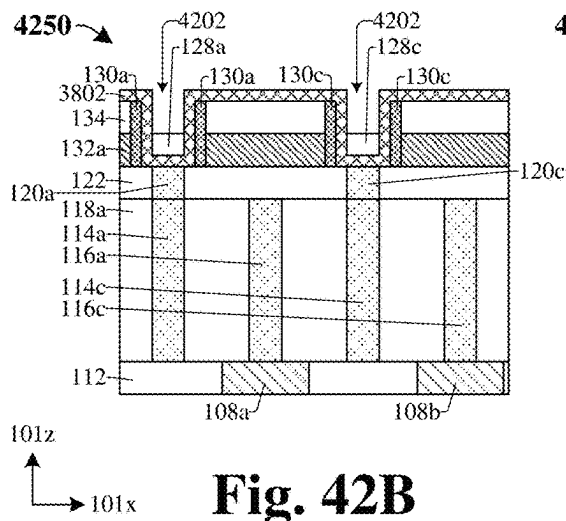
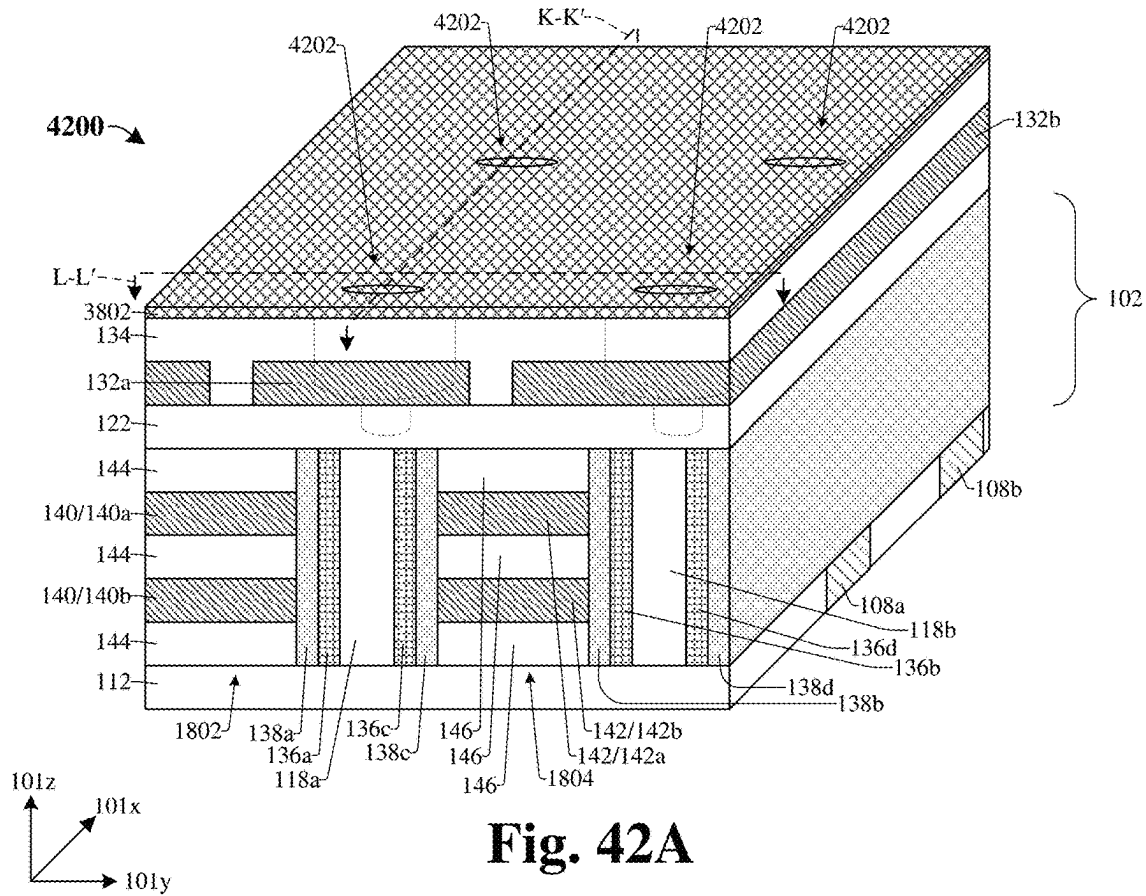
Fig. 37C

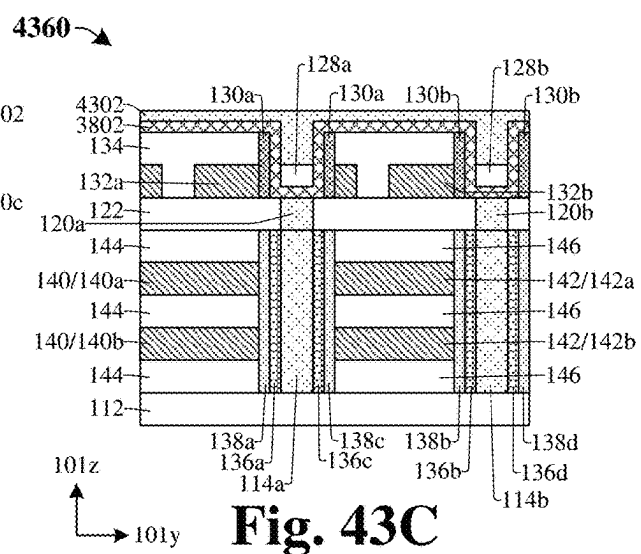
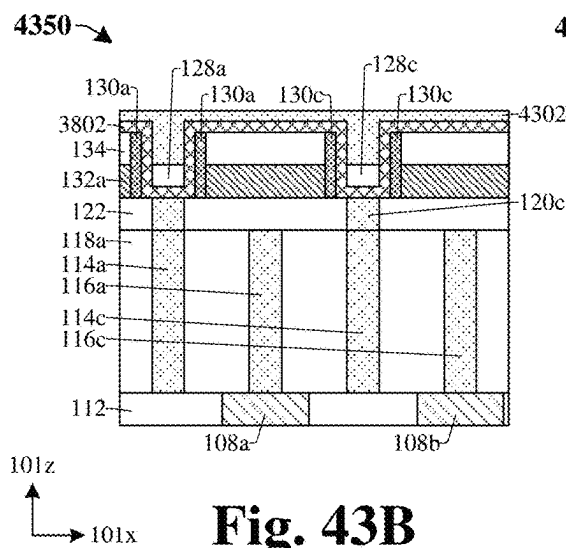
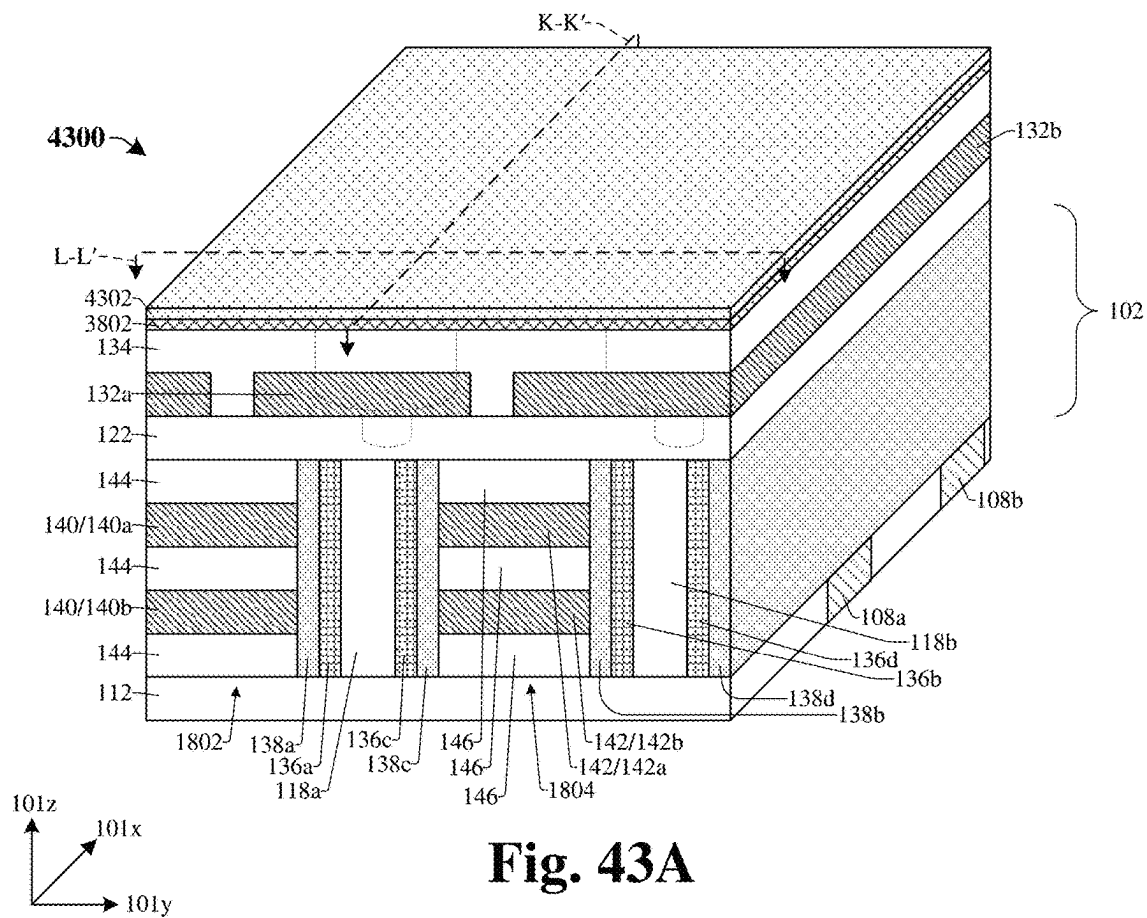


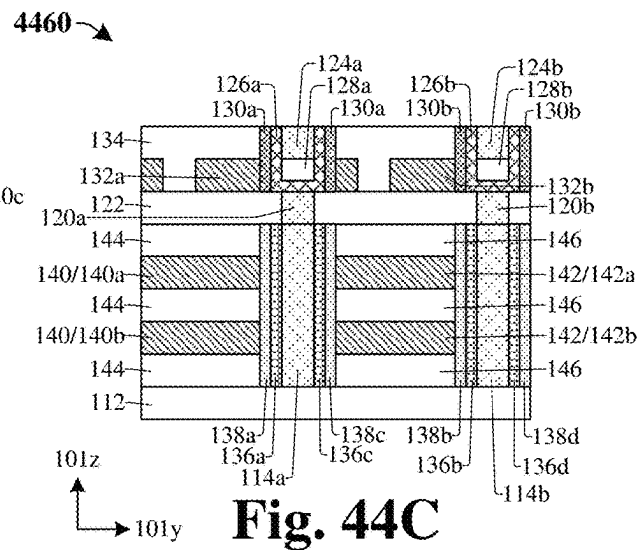
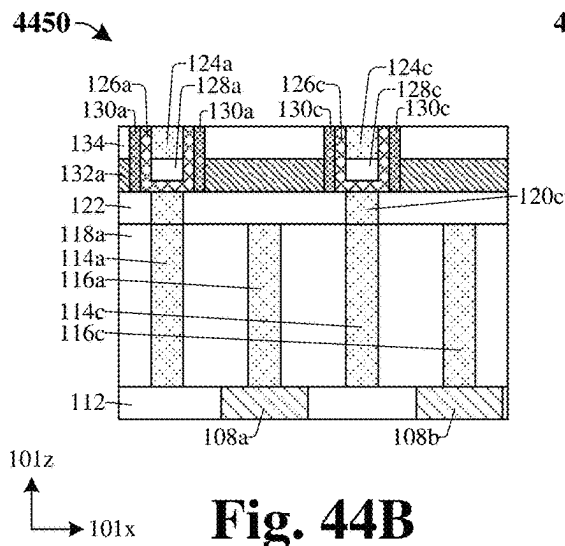
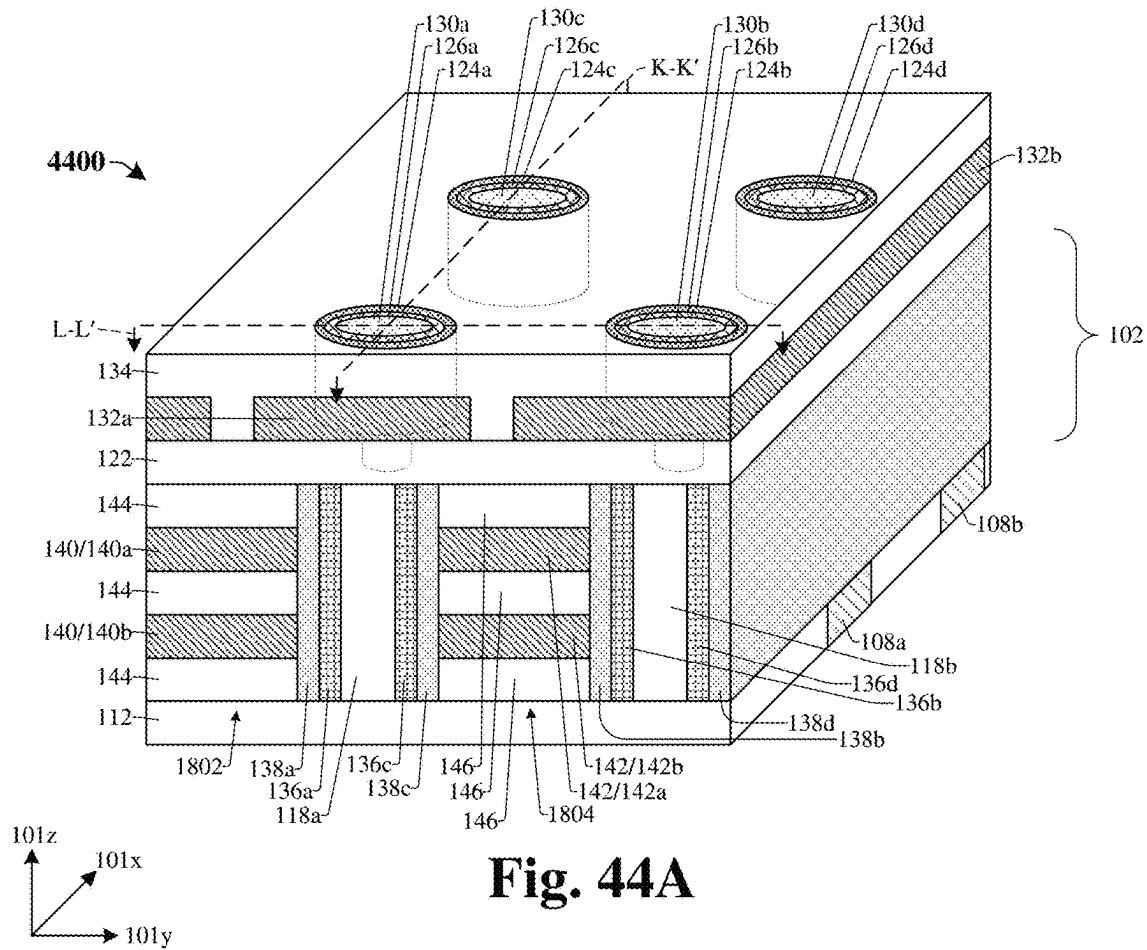


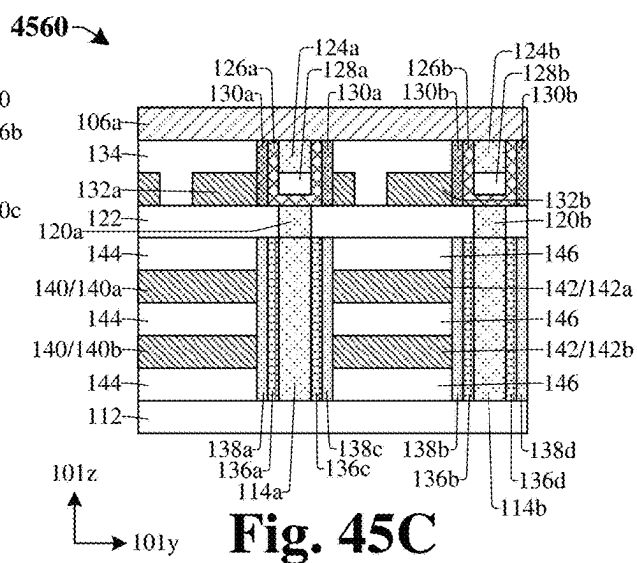
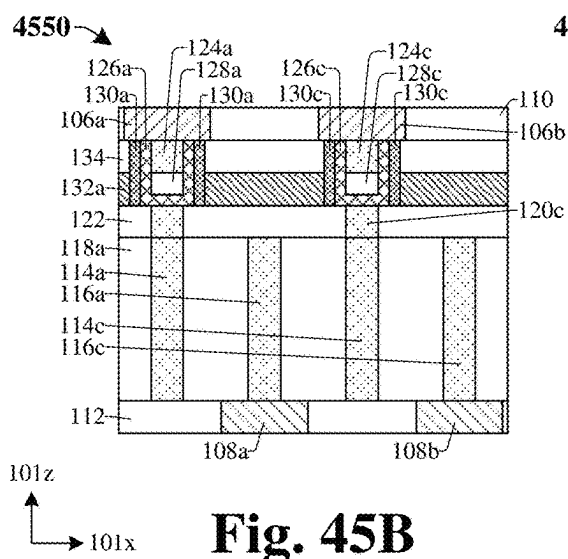
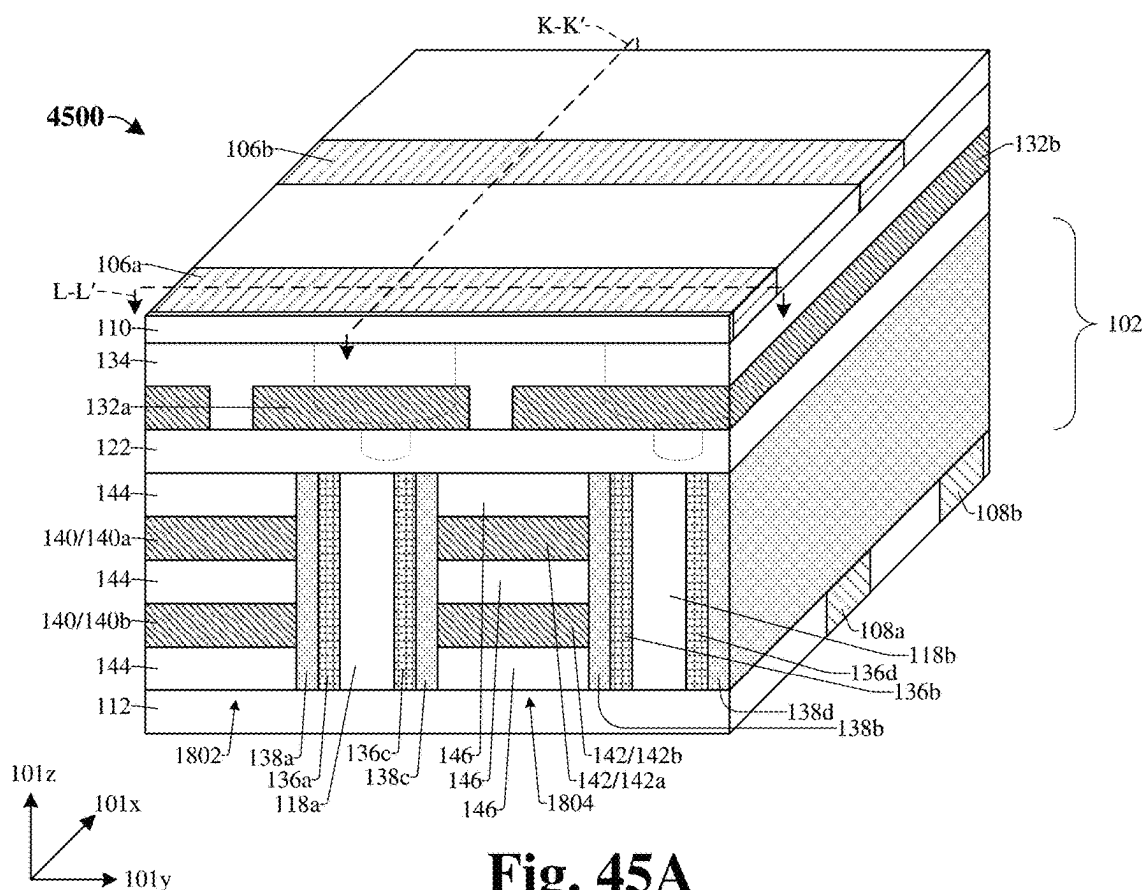




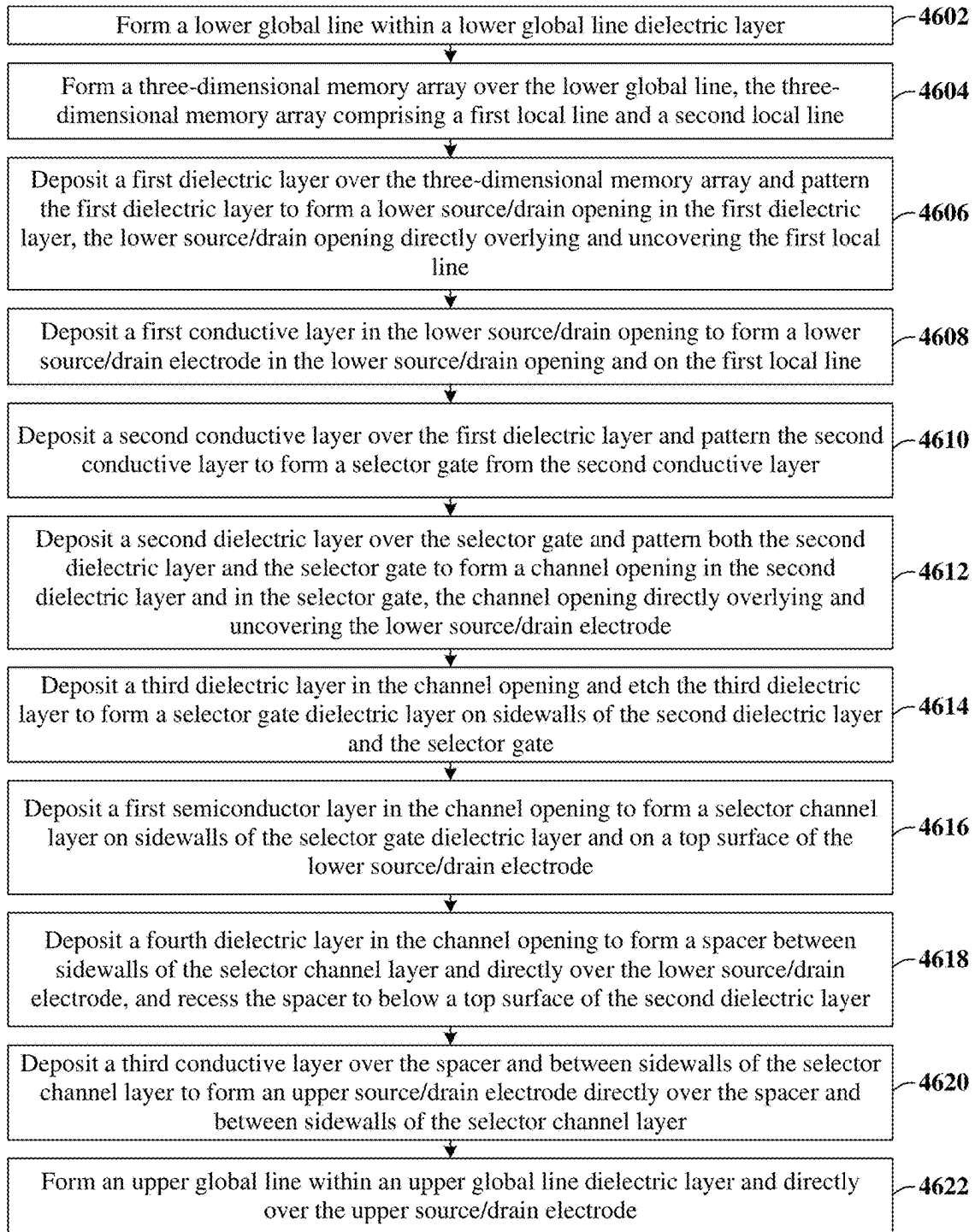








4600 →

**Fig. 46**

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THREE-DIMENSIONAL MEMORY ARRAY WITH LOCAL LINE SELECTOR

REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of U.S. application Ser. No. 18/308,031, filed on Apr. 27, 2023, which is a Divisional of U.S. application Ser. No. 17/168,317, filed on Feb. 5, 2021 (now U.S. Pat. No. 11,672,123, issued on Jun. 6, 2023). The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

Two-dimensional (2D) memory arrays are prevalent in electronic devices and may include, for example, NOR flash memory arrays, NAND flash memory arrays, dynamic random-access memory (DRAM) arrays, and so on. However, 2D memory arrays are reaching scaling limits and are hence reaching limits on memory density. Three-dimensional (3D) memory arrays are a promising candidate for increasing memory density and may include, for example, 3D NAND flash memory arrays, 3D NOR flash memory arrays, and so on.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A illustrates a diagram of some embodiments of an integrated chip comprising a three-dimensional memory array and a first bit line selector that extends between a first local bit line and a first global bit line.

FIG. 1B illustrates a diagram of some embodiments of the integrated chip of FIG. 1A in which a first source line selector extends between a first local source line and a first global source line.

FIG. 2 illustrates a three-dimensional view of some embodiments of an integrated chip comprising a first bit line selector arranged over a first local bit line of a three-dimensional memory array.

FIGS. 3 and 4 illustrate cross-sectional views of some embodiments of the integrated chip of FIG. 2.

FIGS. 5 and 6 illustrate top views of some embodiments of the integrated chip of FIG. 2.

FIG. 7 illustrates a three-dimensional view of some embodiments of the integrated chip of FIG. 2 in which the first bit line selector is arranged under the first local bit line of the three-dimensional memory array.

FIGS. 8 and 9 illustrate cross-sectional views of some embodiments of the integrated chip of FIG. 7.

FIG. 10 illustrates a three-dimensional view of some embodiments of the integrated chip of FIG. 2 in which the first bit line selector is arranged over the first local bit line of the three-dimensional memory array and a first source line selector is arranged over a first local source line of the three-dimensional memory array.

FIGS. 11 and 12 illustrate cross-sectional views of some embodiments of the integrated chip of FIG. 10.

FIG. 13 illustrates a three-dimensional view of some embodiments of the integrated chip of FIG. 2 in which the first bit line selector is arranged under the first local bit line

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of the three-dimensional memory array and the first source line selector is arranged over the first local source line of the three-dimensional memory array.

FIGS. 14 and 15 illustrate cross-sectional views of some embodiments of the integrated chip of FIG. 13.

FIGS. 16A-16D illustrate cross-sectional views of some alternative embodiments of the first bit line selector of FIG. 2.

FIGS. 17A-17C through 45A-45C illustrate views of some embodiments of a method for forming an integrated chip comprising a first bit line selector arranged over a first local bit line of a three-dimensional memory array.

FIG. 46 illustrates a flow diagram of some embodiments of a method for forming an integrated chip comprising a selector arranged over a first local line of a three-dimensional memory array.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Many integrated chips include three-dimensional memory arrays. A three-dimensional memory array may include a plurality of memory cells arranged in a plurality of rows, a plurality of columns, and a plurality of levels. For example, the memory array may include a first stack of memory cell devices arranged across the plurality of levels and coupled in parallel. Memory cells of the first stack of memory cells may extend between a first local bit line and a first local source line. The memory array may also include a second stack of memory cells arranged across the plurality of levels and coupled in parallel. Memory cells of the second stack of memory cells may extend between a second local bit line and a second local source line. The first local bit line and the second local bit line may both be coupled to a global bit line. The first local source line and the second local source line may both be coupled to a global source line.

A challenge with the three-dimensional memory array is that a global bit line and a global source line may be connected to large numbers of local bit lines and local source lines, respectively. As a result, the global bit line and the

global source line may collect capacitance and/or leakage current from a large number of local bit lines and a large number of local source lines, respectively, even when the local bit lines and/or local source lines are not in use (e.g., not being used to perform a read operation, a write operation, or the like). As a result, performance (e.g., speed, power consumption, etc.) of the three-dimensional memory may be reduced.

Various embodiments of the present disclosure are related to an integrated chip comprising a first bit line selector arranged over a first local bit line of a three-dimensional memory array for improving performance of the integrated chip. The three-dimensional memory array comprises a first stack of memory cells arranged across a plurality of levels and coupled in parallel. Memory cells of the first stack of memory cells extend between a first local bit line and a first local source line. The first bit line selector extends vertically between the first local bit line and a first global bit line.

By including the selector in the integrated chip between the first global bit line and the first local bit line, the first local bit line may be selectively decoupled from the first global bit line. Thus, the first bit line selector may prevent capacitance and/or leakage current of the first local bit line from collecting at and affecting the first global bit line when the first local bit line is not in use. For example, a leakage current at the first local bit line may be isolated from the first global bit line when the first local bit line is not in use by setting the first bit line selector to an "OFF" state. As a result, performance of the integrated chip may be improved.

FIG. 1A illustrates a diagram 100 of some embodiments of an integrated chip comprising a three-dimensional memory array and a first bit line selector S_{B1} that extends between a first local bit line LBL_1 and a first global bit line GBL_1 .

The three-dimensional memory array comprises a plurality of memory cells (e.g., C_1 , C_2 , etc.) arranged in rows and columns on a plurality of levels. In other words, the three-dimensional memory array comprises an array of a first plurality of rows and a first plurality of columns on a first level of the plurality of levels, a second plurality of rows and a second plurality of columns on a second level of the plurality of levels, and so on.

For example, a first memory cell C_1 is in a first row and a first column of a first level, and a second memory cell C_2 is in a first row and a first column of a second level. The first memory cell C_1 and the second memory cell C_2 are coupled in parallel between the first local bit line LBL_1 and a first local source line LSL_1 .

A third memory cell C_3 is in a second row and the first column of the first level, and a fourth memory cell C_4 is in a second row and the first column of the second level. The third memory cell C_3 and the fourth memory cell C_4 are coupled in parallel between a second local bit line LBL_2 and a second local source line LSL_2 .

A fifth memory cell C_5 is in the first row and a second column of the first level, and a sixth memory cell C_6 is in the first row and a second column of the second level. The fifth memory cell C_5 and the sixth memory cell C_6 are coupled in parallel between a third local bit line LBL_3 and a third local source line LSL_3 .

A seventh memory cell C_7 is in the second row and the second column of the first level, and an eighth memory cell C_8 is in the second row and the second column of the second level. The seventh memory cell C_7 and the eighth memory cell C_8 are coupled in parallel between a fourth local bit line LBL_4 and a fourth local source line LSL_4 .

A first word line of the first level $WL_{1,1}$ extends along the first row of the first level from a gate of the first cell C_1 to a gate of the fifth cell C_5 . A second word line of the first level $WL_{2,1}$ extends along the second row of the first level from a gate of the third cell C_3 to a gate of the seventh cell C_7 . A first word line of the second level $WL_{1,2}$ extends along the first row of the second level from a gate of the second cell C_2 to a gate of the sixth cell C_6 . A second word line of the second level $WL_{2,2}$ extends along the second row of the second level from a gate of the fourth cell C_4 to a gate of the eighth cell C_8 . In other words, the gate of the first cell C_1 is coupled to the gate of the fifth cell C_5 , the gate of the second cell is coupled to the gate of the sixth cell C_6 , the gate of the third cell C_3 is coupled to the gate of the seventh cell C_7 , and the gate of the fourth cell C_4 is coupled to the gate of the eighth cell C_8 .

The first bit line selector S_{B1} extends between the first local bit line LBL_1 and the first global bit line GBL_1 . A second bit line selector S_{B2} extends between the second local bit line LBL_2 and the first global bit line GBL_1 . A third bit line selector S_{B3} extends between the third local bit line LBL_3 and a second global bit line GBL_2 . A fourth bit line selector S_{B4} extends between the fourth local bit line LBL_4 and the second global bit line GBL_2 .

The bit line selectors (e.g., S_{B1} - S_{B4}) are configured to selectively decouple (i.e., isolate) their respective local bit lines from their respective global bit lines. Thus, the bit line selectors may prevent capacitance and/or leakage current of the local bit lines from collecting at and affecting the global bit lines when the local bit lines are not in use. For example, a leakage current of the first local bit line LBL_1 may be isolated from the first global bit line GBL_1 when the first local bit line LBL_1 is not in use by setting the first bit line selector S_{B1} to an "OFF" state. Likewise, a leakage current of the second local bit line LBL_2 may be isolated from the first global bit line GBL_1 when the second local bit line LBL_2 is not in use by setting the second bit line selector S_{B2} to an "OFF" state, and so on. As a result, performance of the integrated chip may be improved.

In some embodiments, the local source lines (e.g., LSL_1 - LSL_4) are directly coupled to their respective global source lines (e.g., GSL_1 and GSL_2), as illustrated in FIG. 1.

Further, in some embodiments, a gate of the first bit line selector S_{B1} is coupled to a gate of the third bit line selector S_{B3} , and a gate of the second bit line selector S_{B2} is coupled to a gate of the fourth bit line selector S_{B4} .

In some embodiments, the first level is directly over the second level. Although FIG. 1A illustrates the three-dimensional memory array as having two rows and two columns on two levels, it will be appreciated that in some embodiments, some other number of rows, columns, and levels may be feasible. For example, in some embodiments, the array may alternatively comprise eight rows and eight columns on eight levels.

In some embodiments, the three-dimensional memory array may, for example, be or comprise a three-dimensional NOR memory array or the like. In some embodiments, the selectors may, for example, be or comprise gate-all-around transistors or some other suitable type of selector devices.

Although selectors S_{B1} - S_{B4} are referred to as bit line selectors and extend between the local bit lines and the global bit lines, it will be appreciated that in some alternative embodiments, the selectors S_{B1} - S_{B4} may alternatively be source line selectors that extend between the local source lines and the global source lines.

FIG. 1B illustrates a diagram 150 of some embodiments of the integrated chip of FIG. 1A in which a first source line

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selector S_{S1} extends between the first local source line LSL_1 and the first global source line GSL_1 .

In such embodiments, source line selectors (e.g., S_{S1} - S_{S4}) may be arranged between the local source lines (e.g., LSL_1 - LSL_4) and the global source lines (e.g., GSL_1 and GSL_2). For example, the first source line selector S_{S1} extends between the first local source line LSL_1 and the first global source line GSL_1 , a second source line selector S_{S2} extends between the second local source line LSL_2 and the first global source line GSL_1 , a third source line selector S_{S3} extends between the third local source line LSL_3 and the second global source line GSL_2 , and a fourth source line selector S_{S4} extends between the fourth local source line LSL_4 and the second global source line GSL_2 .

The source line selectors (e.g., S_{S1} - S_{S4}) are configured to selectively decouple (i.e., isolate) their respective local source lines from their respective global source lines. Thus, the source line selectors may prevent capacitance and/or leakage current of the local bit lines from collecting at and affecting the global bit lines when the local bit lines are not in use. For example, a leakage current of the first local source line LSL_1 may be isolated from the first global source line GSL_1 when the first local source line LSL_1 is not in use by setting the first source line selector S_{S1} to an "OFF" state. As a result, performance of the integrated chip may be improved.

Referring to FIGS. 2-6, FIG. 2 illustrates a three-dimensional view **200** of some embodiments of an integrated chip comprising a first bit line selector **104a** arranged over a first local bit line **114a** of a three-dimensional memory array **102**, FIGS. 3 and 4 illustrate cross-sectional views **300** and **400** of some embodiments of the integrated chip of FIG. 2, and FIGS. 5 and 6 illustrate top views **500** and **600** of some embodiments of the integrated chip of FIG. 2.

In some embodiments, the cross-sectional view **300** may, for example, be taken across line A-A' of FIGS. 2, 5, and 6. In some embodiments, the cross-sectional view **400** may, for example, be taken across line B-B' of FIGS. 2, 5, and 6. In some embodiments, FIG. 5 illustrates the top view **500** of the integrated chip as seen from line C-C' of FIGS. 3 and 4. In some embodiments, FIG. 6 illustrates the top view **600** of the integrated chip as seen from line D-D' of FIGS. 3 and 4.

A first global source line **108a** and a second global source line **108b** are disposed at a first height and are within a lower global line dielectric layer **112**. The first global source line **108a** and the second global source line **108b** are elongated (i.e., extend) along a y-axis **101y** and are laterally separated by the lower global line dielectric layer **112**.

The memory array **102** includes the first local bit line **114a** and a first local source line **116a** that is laterally separated from the first local bit line **114a** by a first local line dielectric layer **118a**. The first local bit line **114a** and the first local source line **116a** are elongated (i.e., extend) along a z-axis **101z**. A second local bit line **114b** and a second local source line **116b** are disposed within a second local line dielectric layer **118b** and are elongated along the z-axis **101z**. A third local bit line **114c** and a third local source line **116c** are disposed within the first local line dielectric layer **118a** and are elongated along the z-axis **101z**. A fourth local bit line **114d** and a fourth local source line **116d** are disposed within the second local line dielectric layer **118b** and are elongated along the z-axis **101z**. In some embodiments, the first local source line **116a** and the second local source line **116b** are directly over the first global source line **108a** and are coupled to the first global source line **108a**, while the third local source line **116c** and the fourth local source line **116d**

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are directly over the second global source line **108b** and are coupled to the second global source line **108b**.

A first memory channel layer **136a** and a third memory channel layer **136c** continuously extend along an x-axis **101x** from the first local bit line **114a** to the first local source line **116a**. In some embodiments, the first memory channel layer **136a** and the third memory channel layer **136c** also continuously extend to the third local bit line **114c** and further to the third local source line **116c**.

A second memory channel layer **136b** and a fourth memory channel layer **136d** continuously extend along the x-axis **101x** from the second local bit line **114b** to the second local source line **116b**. In some embodiments, the second memory channel layer **136b** and the fourth memory channel layer **136d** also continuously extend to the fourth local bit line **114d** and further to the fourth local source line **116d**.

A first memory film **138a** extends along the x-axis **101x** on a sidewall of the first memory channel layer **136a**. A second memory film **138b** extends along the x-axis **101x** on a sidewall of the second memory channel layer **136b**. A third memory film **138c** extends along the x-axis **101x** on a sidewall of the third memory channel layer **136c**. A fourth memory film **138d** extends along the x-axis **101x** on a sidewall of the fourth memory channel layer **136d**.

A first stack of memory gates extends along the first memory channel layer **136a** and the first memory film **138a**. The first stack of memory gates comprises a first plurality of memory gates **140** (e.g., a first memory gate **140a** and a second memory gate **140b**) that are vertically separated by a first plurality of inter-gate dielectric layers **144**. In other words, the first plurality of memory gates **140** and the first plurality of inter-gate dielectric layers **144** are alternately stacked. For example, the first memory gate **140a** is directly over the second memory gate **140b** and is vertically separated from the second memory gate **140b** by a first inter-gate dielectric layer **144a** of the first plurality of inter-gate dielectric layers **144**.

The first memory gate **140a**, the second memory gate **140b**, and the first inter-gate dielectric layer **144a** are elongated along the x-axis **101x**. The first memory gate **140a**, the second memory gate **140b**, and the first inter-gate dielectric layer **144a** continuously extend along the first memory channel layer **136a** and the first memory film **138a** from alongside the first local bit line **114a** to alongside the first local source line **116a**, and further to alongside the third local bit line **114c** and to alongside the third local source line **116c**. The first memory channel layer **136a** and the first memory film **138a** separate both the first memory gate **140a** and the second memory gate **140b** from the first local bit line **114a**, the first local source line **116a**, the third local bit line **114c**, and the third local source line **116c**.

A second stack of memory gates extends along the second memory channel layer **136b** and the second memory film **138b**. The second stack of memory gates comprises a second plurality of memory gates **142** (e.g., a third memory gate **142a** and a fourth memory gate **142b**) that are vertically separated by a second plurality of inter-gate dielectric layers **146**. In other words, the second plurality of memory gates **142** and the second plurality of inter-gate dielectric layers **146** are alternately stacked. For example, the third memory gate **142a** is directly over the fourth memory gate **142b** and is vertically separated from the fourth memory gate **142b** by a second inter-gate dielectric layer **146a** of the second plurality of inter-gate dielectric layers **146**.

The third memory gate **142a**, the fourth memory gate **142b**, and the second inter-gate dielectric layer **146a** are elongated along the x-axis **101x**. The third memory gate

142a, the fourth memory gate 142b, and the second inter-gate dielectric layer 146a continuously extend along the second memory channel layer 136b and the second memory film 138b from the second local bit line 114b to the second local source line 116b, and further to the fourth local bit line 114d and the fourth local source line 116d. The second memory channel layer 136b and the second memory film 138b separate both the third memory gate 142a and the fourth memory gate 142b from the second local bit line 114b, the second local source line 116b, the fourth local bit line 114d, and the fourth local source line 116d.

In some embodiments, the first memory gate 140a forms a first word line of a first level (e.g., similar to $WL_{1,1}$ of FIGS. 1A and 1B), the second memory gate 140b forms a first word line of a second level (e.g., similar to $WL_{1,2}$ of FIGS. 1A and 1B), the third memory gate 142a forms a second word line of the first level (e.g., similar to $WL_{2,1}$ of FIGS. 1A and 1B), and the fourth memory gate 142b forms a second word line of the second level (e.g., similar to $WL_{2,2}$ of FIGS. 1A and 1B).

In some embodiments, the first local bit line 114a, the first local source line 116a, the first memory channel layer 136a, the first memory film 138a, and the first memory gate 140a form a first memory cell, while the first local bit line 114a, the first local source line 116a, the first memory channel layer 136a, the first memory film 138a, and the second memory gate 140b form a second memory cell that is coupled in parallel with the first memory cell.

In some embodiments, the second local bit line 114b, the second local source line 116b, the second memory channel layer 136b, the second memory film 138b, and the third memory gate 142a form a third memory cell, while the second local bit line 114b, the second local source line 116b, the second memory channel layer 136b, the second memory film 138b, and the fourth memory gate 142b form a fourth memory cell that is coupled in parallel with the third memory cell.

In some embodiments, the third local bit line 114c, the third local source line 116c, the first memory channel layer 136a, the first memory film 138a, and the first memory gate 140a form a fifth memory cell, while the third local bit line 114c, the third local source line 116c, the first memory channel layer 136a, the first memory film 138a, and the second memory gate 140b form a sixth memory cell that is coupled in parallel with the fifth memory cell.

In some embodiments, the fourth local bit line 114d, the fourth local source line 116d, the second memory channel layer 136b, the second memory film 138b, and the third memory gate 142a form a seventh memory cell, while the fourth local bit line 114d, the fourth local source line 116d, the second memory channel layer 136b, the second memory film 138b, and the fourth memory gate 142b form an eighth memory cell that is coupled in parallel with the seventh memory cell.

The first bit line selector 104a is directly over the first local bit line 114a. The first bit line selector 104a comprises a first lower source/drain electrode 120a, a first upper source/drain electrode 124a, a first selector channel layer 126a, a first selector gate dielectric layer 130a, a first selector gate 132a, and a first selector spacer 128a.

The first lower source/drain electrode 120a is within a lower selector dielectric layer 122. The first upper source/drain electrode 124a is directly over, and vertically separated from, the first lower source/drain electrode 120a.

The first selector channel layer 126a extends vertically from a top surface of the first lower source/drain electrode 120a to sidewalls of the first upper source/drain electrode

124a. The first selector channel layer 126a laterally surrounds the first upper source/drain electrode 124a in a closed loop. In some embodiments, a region of the first selector channel layer 126a that laterally surrounds the first upper source/drain electrode 124a (e.g., the portion of the first selector channel layer 126a that is above the top surface of the first selector gate 132a) forms an upper source/drain region (not labeled) of the first selector channel layer 126a, a region of the first selector channel layer 126a that is on the top surface of the first lower source/drain electrode 120a forms a lower source/drain region (not labeled) of the first selector channel layer 126a, and a region of the first selector channel layer 126a that extends along the first selector gate 132a forms a channel region (not labeled) of the first selector channel layer 126a.

The first selector gate dielectric layer 130a is on sidewalls of the first selector channel layer 126a and laterally surrounds the first selector channel layer 126a in a closed loop. In some embodiments, the first selector channel layer 126a and the first selector gate dielectric layer 130a are ring shaped. The first selector gate 132a laterally surrounds the first selector gate dielectric layer 130a in a closed loop. An upper selector dielectric layer 134 is over the first selector gate 132a and laterally surrounds the first selector gate dielectric layer 130a.

A second bit line selector 104b is directly over the second local bit line 114b. The second bit line selector 104b comprises a second lower source/drain electrode 120b, a second upper source/drain electrode 124b, a second selector channel layer 126b, a second selector gate dielectric layer 130b, a second selector gate 132b, and a second selector spacer 128b. The first selector gate 132a and the second selector gate 132b are separated by the upper selector dielectric layer 134.

A third bit line selector 104c is directly over the third local bit line 114c. The third bit line selector 104c comprises a third lower source/drain electrode 120c, a third upper source/drain electrode 124c, a third selector channel layer 126c, a third selector gate dielectric layer 130c, a third selector spacer 128c, and the first selector gate 132a. The first selector gate 132a extends along the x-axis 101x from the first bit line selector 104a to the third bit line selector 104c and the first selector gate 132a surrounds both the first selector gate dielectric layer 130a and the third selector gate dielectric layer 130c. In other words, the first bit line selector 104a and the third bit line selector 104c share (e.g., are commonly coupled to) the first selector gate 132a.

A fourth bit line selector 104d is directly over the fourth local bit line 114d. The fourth bit line selector 104d comprises a fourth lower source/drain electrode 120d, a fourth upper source/drain electrode 124d, a fourth selector channel layer 126d, a fourth selector gate dielectric layer 130d, a fourth selector spacer 128d, and the second selector gate 132b. The second selector gate 132b extends along the x-axis 101x from the second bit line selector 104b to the fourth bit line selector 104d and the second selector gate 132b surrounds both the second selector gate dielectric layer 130b and the fourth selector gate dielectric layer 130d. In other words, the second bit line selector 104b and the fourth bit line selector 104d share (e.g., are commonly coupled to) the second selector gate 132b.

A first global bit line 106a is directly over the first local bit line 114a and the second local bit line 114b. The first global bit line 106a is elongated along the y-axis 101y. The first local bit line 114a and the second local bit line 114b are both selectively coupled to the first global bit line 106a through the first bit line selector 104a and the second bit line

selector **104b**, respectively. A second global bit line **106b** is directly over the third local bit line **114c** and the fourth local bit line **114d**. The second global bit line **106b** is elongated along the y-axis **101y** and is adjacent to the first global bit line **106a**. The third local bit line **114c** and the fourth local bit line **114d** are both selectively coupled to the second global bit line **106b** through the third bit line selector **104c** and the fourth bit line selector **104d**, respectively. The first global bit line **106a** and the second global bit line **106b** are disposed at a second height different from the first height, are within an upper global line dielectric layer **110**, and are separated by the upper global line dielectric layer **110**.

In some embodiments, the inter-gate dielectric layers (e.g., **144** and **146**), the local line dielectric layers (e.g., **118a** and **118b**), the lower selector dielectric layer **122**, the upper selector dielectric layer **134**, the lower global line dielectric layer **112**, the upper global line dielectric layer **110**, and the selector spacers (e.g., **128a-d**) may, for example, comprise silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material.

In some embodiments, the memory gates (e.g., **140** and **142**) and the selector gates (e.g., **132a** and **132b**) may, for example, comprise tungsten or some other suitable material.

In some embodiments, the memory channel layers (e.g., **136a-d**) and the selector channel layers (e.g., **126a-d**) may, for example, comprise silicon, germanium, silicon germanium, gallium arsenide, or some other suitable material. In some embodiments, the upper source/drain regions (not labeled) and the lower source/drain regions (not labeled) of the selector channel layers have higher doping concentrations than the channel regions (not labeled) of the selector channel layers.

In some embodiments, the memory films (e.g., **138a-d**) may, for example, comprise a number of different materials. For example, the memory films may, for example, comprise a gate dielectric layer (e.g., a layer of silicon dioxide, hafnium oxide, aluminum oxide, zirconium oxide, or some other suitable dielectric), a floating gate and one or more dielectric layers (e.g., like that of a flash memory cell), an oxide-nitride-oxide layer, or some other suitable layer(s) and/or material(s).

In some embodiments, the local lines (e.g., **114a-d** and **116a-d**), the lower source/drain electrodes (e.g., **120a-d**), and the upper source/drain electrodes (e.g., **124a-d**) may, for example, comprise tungsten, copper, some other suitable metal, doped silicon, some other suitable doped semiconductor, or some other suitable material.

In some embodiments, the global lines (e.g., **106a**, **106b**, **108a**, and **108d**) may, for example, comprise tungsten, copper, or some other suitable material.

In some embodiments, the selector gate dielectric layers (e.g., **130a-d**) may, for example, comprise silicon dioxide, aluminum oxide, hafnium oxide, or some other suitable material.

In some embodiments, the three-dimensional memory array **102** is a NOR-type three-dimensional memory array or some other suitable memory array. In some embodiments, the bit line selectors (e.g., **104a-d**) are gate-all-around transistor devices or some other suitable devices. In some embodiments, gate all-around transistors may provide added performance enhancement because they may exhibit high currents when "ON" and low leakage currents when "OFF".

It should be noted that the upper global line dielectric layer **110** and the lower global line dielectric layer **112** are not illustrated in FIGS. 2, 7, 10, and 13 for the sake of clarity.

Although items **114a-d** are referred to as local bit lines and items **116a-d** are referred to as local source lines, it will

be appreciated in some embodiments, that items **114a-d** may alternatively be local source lines and items **116a-d** may alternatively be local bit lines. Likewise, it will be appreciated that in some embodiments, items **106a** and **106b** may alternatively be global source lines and items **108a** and **108b** may alternatively be global source bit lines.

Referring to FIGS. 7-9, FIG. 7 illustrates a three-dimensional view **700** of some embodiments of the integrated chip of FIG. 2 in which the first bit line selector **104a** is arranged under the first local bit line **114a** of the three-dimensional memory array **102**, and FIGS. 8 and 9 illustrate cross-sectional views **800** and **900** of some embodiments of the integrated chip of FIG. 7. In some embodiments, the cross-sectional view **800** may, for example, be taken across line E-E' of FIG. 7. In some embodiments, the cross-sectional view **900** may, for example, be taken across line F-F' of FIG. 7.

In such embodiments, the first bit line selector **104a** and the second bit line selector **104b** are directly under the first local bit line **114a** and the second local bit line **114b**, respectively. Further, the first global bit line **106a** is within the lower global line dielectric layer **112** and is directly under both the first bit line selector **104a** and the second bit line selector **104b**. In addition, the third bit line selector **104c** and the fourth bit line selector (not shown) are directly under the third local bit line **114c** and the fourth local bit line (not labeled), respectively. Further, the second global bit line **106b** is within the lower global line dielectric layer **112** and is directly under both the third bit line selector **104c** and the fourth bit line selector (not shown).

The first global source line **108a** is within the upper global line dielectric layer **110** and is directly over both the first local source line **116a** and the second local source line (not labeled). The second global source line **108b** is within the upper global line dielectric layer **110** and is directly over both the third local source line **116c** and the fourth local source line (not labeled).

In some embodiments, the upper source/drain electrodes (e.g., **124a**, **124b**, etc.) are in direct contact with the local bit lines (e.g., **114a**, **114b**, etc.), respectively. For example, the first upper source/drain electrode **124a** may be on a bottom surface of the first local bit line **114a**. In some embodiments, the first lower source/drain electrode **120a** and the second lower source/drain electrode **120b** are both in direct contact with the first global bit line **106a**, while the third lower source/drain electrode **120c** and the fourth lower source/drain electrode (not shown) are both in direct contact with the second global bit line **106b**.

In some embodiments, first selector channel layer **126a** and the third selector channel layer **126c** are in direct contact with the first memory channel layer **136a** and the third memory channel layer **136c**. In some embodiments, the second selector channel layer **126b** and the fourth selector channel layer (not shown) are in direct contact with the second memory channel layer **136b** and the fourth memory channel layer **136d**. In some embodiments, the first selector gate dielectric layer **130a** and the third selector gate dielectric layer **130c** are in direct contact with the first memory film **138a** and the third memory film **138c**. In some embodiments, the second selector gate dielectric layer **130b** and the fourth selector gate dielectric layer (not shown) are in direct contact with the second memory film **138b** and the fourth memory film **138d**.

Referring to FIGS. 10-12, FIG. 10 illustrates a three-dimensional view **1000** of some embodiments of the integrated chip of FIG. 2 in which the first bit line selector **104a** is arranged over the first local bit line **114a** of the three-

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dimensional memory array **102** and a first source line selector **105a** is arranged over the first local source line **116a** of the three-dimensional memory array **102**, and FIGS. **11** and **12** illustrate cross-sectional views **1100** and **1200** of some embodiments of the integrated chip of FIG. **10**. In some embodiments, the cross-sectional view **1100** may, for example, be taken across line G-G' of FIG. **10**. In some embodiments, the cross-sectional view **1200** may, for example, be taken across line H-H' of FIG. **10**.

In such embodiments, the first global bit line **106a**, the second global bit line **106b**, the first global source line **108a**, and the second global source line **108b** are disposed at a same height (e.g., a first height), are within the upper global line dielectric layer **110**, and extend over the three-dimensional memory array **102**.

The first bit line selector **104a**, the second bit line selector **104b**, the third bit line selector **104c**, and the fourth bit line selector (not shown) are directly over the first local bit line **114a**, the second local bit line **114b**, the third local bit line **114c**, and the fourth local bit line (not shown), respectively. The first bit line selector **104a** and the second bit line selector **104b** extend vertically from the first local bit line **114a** and the second local bit line **114b**, respectively, to the first global bit line **106a**. The third bit line selector **104c** and the fourth bit line selector (not shown) extend vertically from the third local bit line **114c** and the fourth local bit line (not shown), respectively, to the second global bit line **106b**.

In addition, the first source line selector **105a**, a second source line selector (not shown), a third source line selector **105c**, and a fourth source line selector (not shown) are directly over the first local source line **116a**, the second local source line (not shown), the third local source line **116c**, and the fourth local source line (not shown), respectively. The first source line selector **105a** and the second source line selector (not shown) extend vertically from the first local source line **116a** and the second local source line (not shown), respectively, to the first global source line **108a**. The third source line selector **105c** and the fourth source line selector (not shown) extend vertically from the third local source line **116c** and the fourth local source line (not shown), respectively, to the second global source line **108b**.

Each of the selectors (e.g., **104a**, **105a**, etc.) comprise a lower source/drain electrode (e.g., **120a**, **120b**, etc.), a selector channel layer (e.g., **126a**, **126b**, etc.), a selector spacer (e.g., **128a**, **128b**, etc.), a selector gate dielectric layer (e.g., **130a**, **130b**, etc.), an upper source/drain electrode (e.g., **124a**, **124b**, etc.), and a selector gate (e.g., **132a** and **132b**). For example, the first source line selector **105a** comprises a fifth lower source/drain electrode **120e**, a fifth selector channel layer **126e**, a fifth selector spacer **128e**, a fifth selector gate dielectric layer **130e**, and a fifth upper source/drain electrode **124e**, and the third source line selector **105c** comprises a seventh lower source/drain electrode **120g**, a seventh selector channel layer **126g**, a seventh selector spacer **128g**, a seventh selector gate dielectric layer **130g**, and a seventh upper source/drain electrode **124g**.

The first bit line selector **104a**, the third bit line selector **104c**, the first source line selector **105a**, and the third source line selector **105c** comprise the first selector gate **132a** while the second bit line selector **104b**, the fourth bit line selector (not shown), the second source line selector (not shown), and the fourth source line selector (not shown) comprise the second selector gate **132b**. In other words, the first bit line selector **104a**, the third bit line selector **104c**, the first source line selector **105a**, and the third source line selector **105c** share the first selector gate **132a** while the second bit line selector **104b**, the fourth bit line selector (not shown), the

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second source line selector (not shown), and the fourth source line selector (not shown) share the second selector gate **132b**.

By additionally including the source line selectors between the local source lines and the global source lines, the global source lines may be selectively isolated from the local source lines. Thus, the source line selectors may prevent capacitance and/or leakage currents of the local source lines from affecting the global source lines when the local source lines are not in use.

Although FIGS. **10-12** illustrate the selectors (e.g., the bit line selectors and the source line selectors) directly over the three-dimensional memory array **102**, it will be appreciated that in some other embodiments, the selectors may alternatively be directly under the three-dimensional memory array **102**.

Referring to FIGS. **13-15**, FIG. **13** illustrates a three-dimensional view **1300** of some embodiments of the integrated chip of FIG. **2** in which the first bit line selector **104a** is arranged under the first local bit line **114a** of the three-dimensional memory array **102** and the first source line selector **105a** is arranged over the first local source line **116a** of the three-dimensional memory array **102**, and FIGS. **14** and **15** illustrate cross-sectional views **1400** and **1500** of some embodiments of the integrated chip of FIG. **13**. In some embodiments, the cross-sectional view **1400** may, for example, be taken across line I-I' of FIG. **13**. In some embodiments, the cross-sectional view **1500** may, for example, be taken across line J-J' of FIG. **13**.

In such embodiments, the first global bit line **106a** and the second global bit line **106b** are within the lower global line dielectric layer **112** while the first global source line **108a** and the second global source line **108b** are within the upper global line dielectric layer **110**.

The first bit line selector **104a**, the second bit line selector **104b**, the third bit line selector **104c**, and the fourth bit line selector (not shown) are directly under the first local bit line **114a**, the second local bit line **114b**, the third local bit line **114c**, and the fourth local bit line (not shown), respectively. The first bit line selector **104a**, the second bit line selector **104b**, the third bit line selector **104c**, and the fourth bit line selector (not shown) are within a first lower selector dielectric layer **122a** and a first upper selector dielectric layer **134a**.

The first source line selector **105a**, the second source line selector (not shown), the third source line selector **105c**, and the fourth source line selector (not shown) are directly over the first local source line **116a**, the second local source line (not shown), the third local source line **116c**, and the fourth local source line (not shown), respectively. The first source line selector **105a** and the third source line selector **105c** comprise a third selector gate **132c**. In other words, the first source line selector **105a** and the third source line selector **105c** share the third selector gate **132c**. The second source line selector (not shown) and the fourth source line selector (not shown) comprise a fourth selector gate **132d**. Further, the first source line selector **105a**, the second source line selector (not shown), the third source line selector **105c**, and the fourth source line selector (not shown) are within a second lower selector dielectric layer **122b** and a second upper selector dielectric layer **134b**.

Although FIGS. **13-15** illustrate the global bit lines (e.g., **106a** and **106b**) under the three-dimensional memory array **102** and the global source lines (e.g., **108a** and **108b**) over the three-dimensional memory array **102**, it will be appreciated that in some other embodiments, the global bit lines may alternatively be over the three-dimensional memory

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array **102** while the global source lines are under the three-dimensional memory array **102**.

FIGS. **16A-D** illustrate cross-sectional views **1600**, **1610**, **1620**, and **1630** of some alternative embodiments of the first bit line selector **104a** of FIG. **2**.

In some embodiments, the first lower source/drain electrode **120a** has a curved lower surface, as illustrated in FIG. **16A**. This may be the result of an etch (see, for example, FIGS. **29A-C**) that forms a lower source/drain opening (e.g., **2902** of FIGS. **29A-C**) extending into a top surface of first local bit line **114a**, thereby removing a portion of the first local bit line **114a**.

In some embodiments, the first selector channel layer **126a** and the first selector gate dielectric layer **130a** may have curved lower surfaces, as illustrated in FIG. **16A**. This may be the result of an etch (see, for example, FIGS. **35A-C**) that forms a channel opening (e.g., **3502** of FIGS. **35A-C**) extending into the first lower source/drain electrode **120a** and/or into the lower selector dielectric layer **122**.

In some embodiments, the first upper source/drain electrode **124a** has a curved lower surface, as illustrated in FIG. **16A**. This may be the result of an etch (see, for example, FIGS. **42A-C**) that recesses the first selector spacer **128a** removing the center of the first selector spacer **128a** at a faster rate than the perimeter of the of the first selector spacer **128a**.

In some embodiments, the first selector channel layer **126a** has a curved lower surface, as illustrated in FIG. **16B**. This may be the result of an etch (see, for example, FIGS. **37A-C**) that is performed on a dielectric layer (e.g., **3602** of FIGS. **36A-C**) extending into the first lower source/drain electrode **120a** and into the lower selector dielectric layer **122**.

In some embodiments, the first selector channel layer **126a** has a curved lower surface, as illustrated in FIG. **16C**. This may be the result of an etch (see, for example, FIGS. **37A-C**) that is performed on a dielectric layer (e.g., **3602** of FIGS. **36A-C**) extending into the lower selector dielectric layer **122** at a substantially faster rate than the etch extends into the first lower source/drain electrode **120a**.

In some embodiments, the first selector channel layer **126a** has a curved lower surface, as illustrated in FIG. **16D**. This may be the result of an etch (see, for example, FIGS. **37A-C**) that is performed on a dielectric layer (e.g., **3602** of FIGS. **36A-C**) extending into the first lower source/drain electrode **120a** at a substantially faster rate than the etch extends into the lower selector dielectric layer **122**.

Although FIGS. **16A-D** illustrate the first bit line selector **104a**, it will be appreciated that any other selector of the integrated chip may exhibit similar features.

FIGS. **17A-C** through FIGS. **45A-C** illustrate views **1700**, **1750**, and **1760** through **4500**, **4550**, and **4560** of some embodiments of a method for forming an integrated chip comprising a first bit line selector **104a** arranged over a first local bit line **114a** of a three-dimensional memory array **102**. Although FIGS. **17A-C** through **45A-C** are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **17A-C** through **45A-C** are not limited to such a method, but instead may stand alone as structures independent of the method.

In some embodiments, cross-sectional view **1750** of FIG. **17B** may, for example, be taken across line K-K' of FIG. **17A**, cross-sectional view **1760** of FIG. **17C** may, for example, be taken across line L-L' of FIG. **17A**, cross-sectional view **1850** of FIG. **18B** may, for example, be taken across line K-K' of FIG. **18A**, cross-sectional view **1860** of

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FIG. **18C** may, for example, be taken across line L-L' of FIG. **18A**, and so on (e.g., through FIGS. **45A-45C**).

As shown in three-dimensional view **1700** of FIG. **17A**, cross-sectional view **1750** of FIG. **17B**, and cross-sectional view **1760** of FIG. **17C**, a plurality of first dielectric layers **1702** and a plurality of first conductive layers **1704** are formed over a first global source line **108a** and a second global source line **108b** in an alternating fashion. Further, the plurality of first dielectric layers **1702** and the plurality of first conductive layers **1704** are formed over a lower global line dielectric layer **112** within which the first and second global source lines **108a**, **108b** are arranged.

In some embodiments, the plurality of first dielectric layers **1702** may, for example, be formed by depositing silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material by a chemical vapor deposition (CVD) process, a physical vapor deposition (PVD) process, an atomic layer deposition (ALD) process, or some other suitable process. In some embodiments, the plurality of first conductive layers **1704** may, for example, be formed by depositing tungsten, copper, or some other suitable material by a sputtering process, an electroless deposition (ELD) process, an electrochemical plating (ECP) process, or some other suitable process.

As shown in three-dimensional view **1800** of FIG. **18A**, cross-sectional view **1850** of FIG. **18B**, and cross-sectional view **1860** of FIG. **18C**, the plurality of first dielectric layers (e.g., **1702** of FIGS. **17A-C**) and the plurality of first conductive layers (e.g., **1704** of FIGS. **17A-C**) are patterned to form a first memory gate stack **1802** and a second memory gate stack **1804** from the plurality of first dielectric layers and the plurality of first conductive layers. The second memory gate stack **1804** is adjacent to the first memory gate stack **1802** and is laterally separated from the first memory gate stack **1802**. In some embodiments, the first memory gate stack **1802** comprises a first plurality of memory gates **140** and a first plurality of inter-gate dielectric layers **144**, and the second memory gate stack **1804** comprises a second plurality of memory gates **142** and a second plurality of inter-gate dielectric layers **146**.

In some embodiments, the patterning may, for example, comprise forming a masking layer (not shown) over the plurality of first dielectric layers and the plurality of first conductive layers, and by subsequently etching the plurality of first dielectric layers and the plurality of first conductive layers according to the masking layer. The masking layer may, for example, be or comprise a photoresist mask, a hard mask, or some other suitable layer. The etching may, for example, comprise a dry etching process (e.g., a reactive ion etching (RIE) process, an ion beam etching (IBE) process, some plasma etching process, or the like) or some other suitable process.

As shown in three-dimensional view **1900** of FIG. **19A**, cross-sectional view **1950** of FIG. **19B**, and cross-sectional view **1960** of FIG. **19C**, a first memory layer **1902** is conformally formed over the first memory gate stack **1802**, over the second memory gate stack **1804**, and on sidewalls of the first and second memory gate stacks **1802**, **1804**.

In some embodiments, the first memory layer **1902** may, for example, be formed by depositing one or more dielectric layers, one or more conductive layers, or any combination of the foregoing by a CVD process, a PVD process, an ALD process, or some other suitable process.

As shown in three-dimensional view **2000** of FIG. **20A**, cross-sectional view **2050** of FIG. **20B**, and cross-sectional view **2060** of FIG. **20C**, the first memory layer (e.g., **1902** of FIGS. **19A-C**) is etched to form a first memory film **138a**,

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a second memory film **138b**, a third memory film **138c**, and a fourth memory film **138d** from the first memory layer. The etching removes the first memory layer from top surfaces of the lower global line dielectric layer **112**, the first global source line **108a**, the second global source line **108b**, the first memory gate stack **1802**, and the second memory gate stack **1804**.

In some embodiments, the etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **2100** of FIG. **21A**, cross-sectional view **2150** of FIG. **21B**, and cross-sectional view **2160** of FIG. **21C**, a first semiconductor layer **2102** is conformally formed on sidewalls of the first memory film **138a**, the second memory film **138b**, the third memory film **138c**, and the fourth memory film **138d**, and also over the first and second memory gate stacks **1802**, **1804**.

In some embodiments, the first semiconductor layer **2102** may, for example, be formed by depositing silicon, germanium, silicon germanium, gallium arsenide, or some other suitable material by a CVD process, a PVD process, an ALD process, an epitaxy process, or some other suitable process.

As shown in three-dimensional view **2200** of FIG. **22A**, cross-sectional view **2250** of FIG. **22B**, and cross-sectional view **2260** of FIG. **22C**, the first semiconductor layer (e.g., **2102** of FIGS. **21A-C**) is etched to form a first memory channel layer **136a**, a second memory channel layer **136b**, a third memory channel layer **136c**, and a fourth memory channel layer **136d** from the first semiconductor layer. The etching removes the first semiconductor layer from top surfaces of the lower global line dielectric layer **112**, the first global source line **108a**, the second global source line **108b**, the first memory gate stack **1802**, and the second memory gate stack **1804**.

In some embodiments, the etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **2300** of FIG. **23A**, cross-sectional view **2350** of FIG. **23B**, and cross-sectional view **2360** of FIG. **23C**, a second dielectric layer **2302** formed over the first and second memory gate stacks, between sidewalls of the first memory channel layer **136a** and the third memory channel layer **136c**, and between sidewalls of the second memory channel layer **136b** and the fourth memory channel layer **136d**.

In some embodiments, the second dielectric layer **2302** may, for example, be formed by depositing silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material by a CVD process, a PVD process, an ALD process, or some other suitable process.

As shown in three-dimensional view **2400** of FIG. **24A**, cross-sectional view **2450** of FIG. **24B**, and cross-sectional view **2460** of FIG. **24C**, the second dielectric layer (e.g., **2302** of FIGS. **23A-C**) is removed from top surfaces of the first memory gate stack **1802**, the second memory gate stack **1804**, the first memory channel layer **136a**, the second memory channel layer **136b**, the third memory channel layer **136c**, the fourth memory channel layer **136d**, the first memory film **138a**, the second memory film **138b**, the third memory film **138c**, and the fourth memory film **138d**. The removal process forms a first local line dielectric layer **118a** and a second local line dielectric layer **118b** from the second dielectric layer.

In some embodiments, removing the second dielectric layer may, for example, comprise a planarization process (e.g., a chemical mechanical planarization (CMP) or the

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like), an etching process (e.g., a dry etching process or the like), or some other suitable process.

As shown in three-dimensional view **2500** of FIG. **25A**, cross-sectional view **2550** of FIG. **25B**, and cross-sectional view **2560** of FIG. **25C**, the first local line dielectric layer **118a** and the second local line dielectric layer **118b** are patterned to form a plurality of local line openings **2502** in the first local line dielectric layer **118a** and the second local line dielectric layer **118b**.

In some embodiments, the patterning may, for example, comprise forming a masking layer (not shown), and subsequently etching the first local line dielectric layer **118a** and the second local line dielectric layer **118b** according to the masking layer. The masking layer may, for example, be or comprise a photoresist mask, a hard mask, or some other suitable layer. The etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **2600** of FIG. **26A**, cross-sectional view **2650** of FIG. **26B**, and cross-sectional view **2660** of FIG. **26C**, a second conductive layer **2602** is formed over and in the plurality of local line openings (e.g., **2502** of FIGS. **25A-C**).

In some embodiments, the second conductive layer **2602** may, for example, be formed by depositing tungsten, copper, or some other suitable metal by a sputtering process, an ELD process, an ECP process, or some other suitable process. In some other embodiments, the second conductive layer **2602** may, for example, be formed by depositing doped silicon or some other doped semiconductor by a CVD process, a PVD process, an ALD process, an epitaxy process, or some other suitable process.

As shown in three-dimensional view **2700** of FIG. **27A**, cross-sectional view **2750** of FIG. **27B**, and cross-sectional view **2760** of FIG. **27C**, the second conductive layer (e.g., **2602** of FIGS. **26A-C**) is removed from top surfaces of the first memory gate stack **1802**, the second memory gate stack **1804**, the first memory channel layer **136a**, the second memory channel layer **136b**, the third memory channel layer **136c**, the fourth memory channel layer **136d**, the first memory film **138a**, the second memory film **138b**, the third memory film **138c**, the fourth memory film **138d**, the first local line dielectric layer **118a**, and second local line dielectric layer **118b**. The removal process forms a plurality of local lines (e.g., a first local bit line **114a**, a first local source line **116a**, a second local bit line **114b**, a second local source line **116b**, a third local bit line **114c**, a third local source line **116c**, a fourth local bit line **114d**, and fourth local source line **116d**) from the second conductive layer.

In some embodiments, removing the second conductive layer may, for example, comprise a planarization process (e.g., a CMP or the like), an etching process (e.g., a dry etching process or the like), or some other suitable process. In some embodiments, this step completes the formation of the three-dimensional memory array **102**.

As shown in three-dimensional view **2800** of FIG. **28A**, cross-sectional view **2850** of FIG. **28B**, and cross-sectional view **2860** of FIG. **28C**, a lower selector dielectric layer **122** is formed over the three-dimensional memory array **102**.

In some embodiments, the lower selector dielectric layer **122** may, for example, be formed by depositing silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material by a CVD process, a PVD process, an ALD process, or some other suitable process.

As shown in three-dimensional view **2900** of FIG. **29A**, cross-sectional view **2950** of FIG. **29B**, and cross-sectional view **2960** of FIG. **29C**, the lower selector dielectric layer **122** is patterned to form a plurality of lower source/drain

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openings **2902** in the lower selector dielectric layer **122**. The lower source/drain openings **2902** are directly over the local bit lines (e.g., **114a**, **114b**, etc.) of the three-dimensional memory array **102**.

In some embodiments, the patterning may, for example, comprise forming a masking layer (not shown) over the lower selector dielectric layer **122**, and subsequently etching the lower selector dielectric layer **122** according to the masking layer. The masking layer may, for example, be or comprise a photoresist mask, a hard mask, or some other suitable layer. The etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **3000** of FIG. **30A**, cross-sectional view **3050** of FIG. **30B**, and cross-sectional view **3060** of FIG. **30C**, a third conductive layer **3002** is formed over the lower selector dielectric layer **122** and in the plurality of lower source/drain openings **2902**.

In some embodiments, the third conductive layer **3002** may, for example, be formed by depositing tungsten, copper, or some other suitable metal by a sputtering process, an ELD process, an ECP process, or some other suitable process. In some other embodiments, the third conductive layer **3002** may, for example, be formed by depositing doped silicon or some other doped semiconductor by a CVD process, a PVD process, an ALD process, an epitaxy process, or some other suitable process.

As shown in three-dimensional view **3100** of FIG. **31A**, cross-sectional view **3150** of FIG. **31B**, and cross-sectional view **3160** of FIG. **31C**, the third conductive layer (e.g., **3002** of FIGS. **30A-C**) is removed from a top surface of the lower selector dielectric layer **122**. The removal process forms a plurality of lower source/drain electrodes (e.g., a first lower source/drain electrode **120a**, a second lower source/drain electrode **120b**, a third lower source/drain electrode **120c**, and a fourth lower source/drain electrode **120d**) from the third conductive layer.

In some embodiments, removing the third conductive layer may, for example, comprise a planarization process (e.g., a CMP or the like), an etching process (e.g., a dry etching process or the like), or some other suitable process.

As shown in three-dimensional view **3200** of FIG. **32A**, cross-sectional view **3250** of FIG. **32B**, and cross-sectional view **3260** of FIG. **32C**, a fourth conductive layer **3202** is formed over the lower selector dielectric layer **122** and over the plurality of lower source/drain electrodes.

In some embodiments, the fourth conductive layer **3202** may, for example, be formed by depositing tungsten, copper, or some other suitable material by a sputtering process, an ELD process, an ECP process, or some other suitable process.

As shown in three-dimensional view **3300** of FIG. **33A**, cross-sectional view **3350** of FIG. **33B**, and cross-sectional view **3360** of FIG. **33C**, the fourth conductive layer (e.g., **3202** of FIGS. **32A-C**) is patterned to form a first selector gate **132a** and a second selector gate **132b** from the fourth conductive layer.

In some embodiments, the patterning may, for example, comprise forming a masking layer (not shown) over the fourth conductive layer, and subsequently etching the fourth conductive layer according to the masking layer. The masking layer may, for example, be or comprise a photoresist mask, a hard mask, or some other suitable layer. The etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **3400** of FIG. **34A**, cross-sectional view **3450** of FIG. **34B**, and cross-sectional view **3460** of FIG. **34C**, an upper selector dielectric layer

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134 is formed over the first selector gate **132a**, over the second selector gate **132b**, and between the selector gates.

In some embodiments, the upper selector dielectric layer **134** may, for example, be formed by depositing the upper selector dielectric layer **134** and subsequently planarizing a top surface of the upper selector dielectric layer **134**. The upper selector dielectric layer **134** may be deposited as silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material and/or may be deposited by a CVD process, a PVD process, an ALD process, or some other suitable process. The planarization may, for example, be performed by a CMP or some other suitable process.

As shown in three-dimensional view **3500** of FIG. **35A**, cross-sectional view **3550** of FIG. **35B**, and cross-sectional view **3560** of FIG. **35C**, the upper selector dielectric layer **134** and the selector gates (e.g., the first selector gate **132a** and the second selector gate **132b**) are patterned to form a plurality of channel openings **3502** in the upper selector dielectric layer **134** and in the selector gates. The channel openings **3502** extend through the upper selector dielectric layer **134** and through the selector gates to top surfaces of the lower source/drain electrodes (e.g., **120a-120d**).

In some embodiments, the patterning may, for example, comprise forming a masking layer (not shown) over the upper selector dielectric layer **134**, and subsequently etching the upper selector dielectric layer **134** according to the masking layer. The masking layer may, for example, be or comprise a photoresist mask, a hard mask, or some other suitable layer. The etching may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **3600** of FIG. **36A**, cross-sectional view **3650** of FIG. **36B**, and cross-sectional view **3660** of FIG. **36C**, a third dielectric layer **3602** is conformally formed over the upper selector dielectric layer **134** and in first portions of the plurality of channel openings **3502**. In other words, the third dielectric layer **3602** is formed on sidewalls of the selector gates (e.g., **132a** and **132b**) and on sidewalls of the upper selector dielectric layer **134**.

In some embodiments, the third dielectric layer **3602** may, for example, be formed by depositing silicon dioxide, aluminum oxide, hafnium oxide, or some other suitable material by a CVD process, a PVD process, an ALD process, or some other suitable process.

As shown in three-dimensional view **3700** of FIG. **37A**, cross-sectional view **3750** of FIG. **37B**, and cross-sectional view **3760** of FIG. **37C**, the third dielectric layer (e.g., **3602** of FIGS. **36A-C**) is removed from a top surface of the upper selector dielectric layer **134** and from top surfaces of the lower source/drain electrodes (e.g., **120a-d**). The removal process forms a plurality of selector gate dielectric layers (e.g., a first selector gate dielectric layer **130a**, a second selector gate dielectric layer **130b**, a third selector gate dielectric layer **130c**, and a fourth selector gate dielectric layer **130d**) from the third dielectric layer.

In some embodiments, the removal process may, for example, comprise a dry etching process or some other suitable process.

As shown in three-dimensional view **3800** of FIG. **38A**, cross-sectional view **3850** of FIG. **38B**, and cross-sectional view **3860** of FIG. **38C**, a second semiconductor layer **3802** is conformally formed over the upper selector dielectric layer **134** and in second portions of the plurality of channel openings **3502**. In other words, the second semiconductor layer **3802** is formed on sidewalls of the selector gate dielectric layers (e.g., **130a-d**) and on top surfaces of the lower source/drain electrodes (e.g., **120a-d**).

In some embodiments, the second semiconductor layer **3802** may, for example, be formed by depositing silicon, germanium, silicon germanium, gallium arsenide, or some other suitable material by a CVD process, a PVD process, an ALD process, an epitaxy process, or some other suitable process.

As shown in three-dimensional view **3900** of FIG. **39A**, cross-sectional view **3950** of FIG. **39B**, and cross-sectional view **3960** of FIG. **39C**, an implantation process (e.g., illustrated by arrows **3902**) is performed on the second semiconductor layer **3802** to dope the second semiconductor layer **3802**. For example, in some embodiments, the implantation process dopes a first region (e.g., **3904** of FIG. **39B**) of the second semiconductor layer **3802** and a second region (e.g., **3906** of FIG. **39B**) of the second semiconductor layer **3802** more highly than a third region (e.g., **3908**) of the second semiconductor layer **3802** to form a first lower source/drain region (e.g., from the first region), a first upper source/drain region (e.g., from the second region), and a channel region (e.g., from the third region) in the second semiconductor layer **3802**.

In some embodiments, the implantation process may, for example, be or comprise an ion implantation process or some other suitable process.

As shown in three-dimensional view **4000** of FIG. **40A**, cross-sectional view **4050** of FIG. **40B**, and cross-sectional view **4060** of FIG. **40C**, a fourth dielectric layer **4002** is formed over the second semiconductor layer **3802** and in remaining portions of the plurality of channel openings (e.g., **3502** of FIGS. **39A-D**). In other words, the fourth dielectric layer **4002** is formed between sidewalls of the second semiconductor layer **3802** and on top surfaces of the second semiconductor layer **3802**.

In some embodiments, the fourth dielectric layer **4002** may, for example, be formed by depositing silicon dioxide, silicon nitride, silicon oxynitride, or some other suitable material by a CVD process, a PVD process, an ALD process, or some other suitable process.

As shown in three-dimensional view **4100** of FIG. **41A**, cross-sectional view **4150** of FIG. **41B**, and cross-sectional view **4160** of FIG. **41C**, the fourth dielectric layer (e.g., **4002** of FIGS. **40A-C**) is removed from a top surface of the second semiconductor layer **3802**. The removal process forms a plurality of selector spacers (e.g., a first selector spacer **128a**, a second selector spacer **128b**, a third selector spacer **128c**, and a fourth selector spacer **128d**) from the fourth dielectric layer.

In some embodiments, removing the fourth dielectric layer may, for example, comprise a wet etching process, a dry etching process, or some other suitable process.

As shown in three-dimensional view **4200** of FIG. **42A**, cross-sectional view **4250** of FIG. **42B**, and cross-sectional view **4260** of FIG. **42C**, the plurality of selector spacers (e.g., **128a-128d**) are recessed such that top surfaces of the selector spacers are below top surfaces of the upper selector dielectric layer **134**. In some embodiments, the top surfaces of the selector spacers are approximately disposed at a same height as top surfaces of the selector gates (e.g., **132a** and **132b**) as a result of the recessing. The recessing forms a plurality of upper source/drain openings **4202** directly over the selector spacers and between sidewalls of the second semiconductor layer **3802**.

In some embodiments, recessing the plurality of selector spacers may, for example, comprise a wet etching process, a dry etching process, or some other suitable process. In some embodiments, the recessing process illustrated in FIGS. **42A-C** and the removal process illustrated in FIGS.

41A-C may be achieved using a single etching process. In some other embodiments, the recessing process illustrated in FIGS. **42A-C** and the removal process illustrated in FIGS. **41A-C** may be achieved using separate etching processes.

As shown in three-dimensional view **4300** of FIG. **43A**, cross-sectional view **4350** of FIG. **43B**, and cross-sectional view **4360** of FIG. **43C**, a fifth conductive layer **4302** is formed over the second semiconductor layer **3802** and in the plurality of upper source/drain openings (e.g., **4202** of FIGS. **42A-C**). In other words, the fifth conductive layer **4302** is formed between sidewalls of the second semiconductor layer **3802** and on top surfaces of the selector spacers (e.g., **128a-128d**).

In some embodiments, the fifth conductive layer **4302** may, for example, be formed by depositing tungsten, copper, or some other suitable metal by a sputtering process, an ELD process, an ECP process, or some other suitable process. In some other embodiments, the fifth conductive layer **4302** may, for example, be formed by depositing doped silicon or some other doped semiconductor by a CVD process, a PVD process, an ALD process, an epitaxy process, or some other suitable process.

As shown in three-dimensional view **4400** of FIG. **44A**, cross-sectional view **4450** of FIG. **44B**, and cross-sectional view **4460** of FIG. **44C**, the second semiconductor layer (e.g., **3802** of FIGS. **43A-C**) and the fifth conductive layer (e.g., **4302** of FIGS. **43A-C**) are removed from over the upper selector dielectric layer **134**. The removal process forms a plurality of selector channel layers (e.g., a first selector channel layer **126a**, a second selector channel layer **126b**, a third selector channel layer **126c**, and a fourth selector channel layer **126d**) from the second semiconductor layer, and forms a plurality of upper source/drain electrodes (e.g., a first upper source/drain electrode **124a**, a second upper source/drain electrode **124b**, a third upper source/drain electrode **124c**, and a fourth upper source/drain electrode **124d**) from the fifth conductive layer.

In some embodiments, the removal process may, for example, comprise a planarization process (e.g., a CMP or the like) or some other suitable process.

As shown in three-dimensional view **4500** of FIG. **45A**, cross-sectional view **4550** of FIG. **45B**, and cross-sectional view **4560** of FIG. **45C**, an upper global line dielectric layer **110** is formed over the upper selector dielectric layer **134** and a first global bit line **106a** and a second global bit line **106b** are formed within the upper global line dielectric layer **110**.

In some embodiments, the upper global line dielectric layer **110** may, for example, be formed by depositing silicon dioxide or some other suitable material by a CVD process, a PVD process, an ALD process, or some other suitable process. In some embodiments, the global bit lines (e.g., **106a** and **106b**) may, for example, be formed by patterning the upper global line dielectric layer **110** to form openings in the upper global line dielectric layer **110**, depositing a conductive layer (e.g., copper or the like) in the openings, and planarizing (e.g., via a CMP or the like) the conductive layer to form the global bit lines.

FIG. **46** illustrates a flow diagram of some embodiments of a method **4600** for forming an integrated chip comprising a selector arranged over a first local line of a three-dimensional memory array. While method **4600** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those

illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **4602**, a lower global line is formed within a lower global line dielectric layer. FIGS. **17A-C** illustrate views **1700**, **1750**, and **1760** of some embodiments corresponding to act **4602**.

At **4604**, a three-dimensional memory array is formed over the lower global line, the three-dimensional memory array comprising a first local line and a second local line. FIGS. **17A-C** through FIGS. **27A-C** illustrate views **1700**, **1750**, and **1760** through **2700**, **2750**, and **2760** of some embodiments corresponding to act **4604**.

At **4606**, a first dielectric layer is deposited over the three-dimensional memory array and the first dielectric layer is patterned to form a lower source/drain opening in the first dielectric layer, the lower source/drain opening directly overlying and uncovering the first local line. FIGS. **28A-C** and FIGS. **29A-C** illustrate views **2800**, **2850**, **2860**, **2900**, **2950**, and **2960** of some embodiments corresponding to act **4606**.

At **4608**, a first conductive layer is deposited in the lower source/drain opening to form a lower source/drain electrode in the lower source/drain opening and on the first local line. FIGS. **30A-C** and FIGS. **31A-C** illustrate views **3000**, **3050**, **3060**, **3100**, **3150**, and **3160** of some embodiments corresponding to act **4608**.

At **4610**, a second conductive layer is deposited over the first dielectric layer and the second conductive layer is patterned to form a selector gate from the second conductive layer. FIGS. **32A-C** and FIGS. **33A-C** illustrate views **3200**, **3250**, **3260**, **3300**, **3350**, and **3360** of some embodiments corresponding to act **4610**.

At **4612**, a second dielectric layer is deposited over the selector gate and both the second dielectric layer and the selector gate are patterned to form a channel opening in the second dielectric layer and in the selector gate, the channel opening directly overlying and uncovering the lower source/drain electrode. FIGS. **34A-C** and FIGS. **35A-C** illustrate views **3400**, **3450**, **3460**, **3500**, **3550**, and **3560** of some embodiments corresponding to act **4612**.

At **4614**, a third dielectric layer is deposited in the channel opening and the third dielectric layer is etched to form a selector gate dielectric layer on sidewalls of the second dielectric layer and the selector gate. FIGS. **36A-C** and FIGS. **37A-C** illustrate views **3600**, **3650**, **3660**, **3700**, **3750**, and **3760** of some embodiments corresponding to act **4614**.

At **4616**, a first semiconductor layer is deposited in the channel opening to form a selector channel layer on sidewalls of the selector gate dielectric layer and on a top surface of the lower source/drain electrode. FIGS. **38A-C** and FIGS. **39A-C** illustrate views **3800**, **3850**, **3860**, **3900**, **3950**, and **3960** of some embodiments corresponding to act **4616**.

At **4618**, a fourth dielectric layer is deposited in the channel opening to form a spacer between sidewalls of the selector channel layer and directly over the lower source/drain electrode, and the spacer is recessed to below a top surface of the second dielectric layer. FIGS. **40A-C** through FIGS. **42A-C** illustrate views **4000**, **4050**, and **4060** through **4200**, **4250**, and **4260** of some embodiments corresponding to act **4618**.

At **4620**, a third conductive layer is deposited over the spacer and between sidewalls of the selector channel layer to form an upper source/drain electrode directly over the spacer and between sidewalls of the selector channel layer. FIGS.

43A-C and FIGS. **44A-C** illustrate views **4300**, **4350**, **4360**, **4400**, **4450**, and **4460** of some embodiments corresponding to act **4620**.

At **4622**, an upper global line is formed within an upper global line dielectric layer and directly over the upper source/drain electrode. FIGS. **45A-C** illustrate views **4500**, **4550**, and **4560** of some embodiments corresponding to act **4622**.

Thus, the present disclosure relates to an integrated chip comprising a first bit line selector arranged over a first local bit line of a three-dimensional memory array for improving a performance of the integrated chip.

Accordingly, in some embodiments, the present disclosure relates to an integrated chip comprising a three-dimensional memory array. The three-dimensional memory array comprises a first local line and a second local line that are elongated vertically, a first memory cell extending between the first local line and the second local line, and a second memory cell directly under the first memory cell, extending between the first local line and the second local line, and coupled in parallel with the first memory cell. The first memory cell is coupled to a first word line and the second memory cell is coupled to a second word line. The first word line and the second word line are elongated horizontally. A first global line is disposed at a first height and is elongated horizontally. A first selector extends vertically from the first local line to the first global line.

In other embodiments, the present disclosure relates to an integrated chip comprising a first local line and a second local line elongated vertically. The second local line is adjacent to the first local line and is separated from the first local line. A first memory channel layer extends horizontally from the first local line to the second local line. A first memory gate is directly over, and vertically separated from, a second memory gate. The first memory gate and the second memory gate extend along the first memory channel layer from alongside the first local line to alongside the second local line. A first global line is elongated horizontally. The first global line is separated from the first local line. A first selector extends vertically between the first local line and the first global line.

In yet other embodiments, the present disclosure relates to a method for forming an integrated chip. The method comprises forming a three-dimensional NOR memory array. The three-dimensional NOR memory array comprises a first local line and a second local line. A lower source/drain electrode is formed in a first dielectric layer and directly over the first local line. A selector gate is formed over the lower source/drain electrode. A second dielectric layer is deposited over the selector gate. Both the second dielectric layer and the selector gate are patterned to form a channel opening in the second dielectric layer and in the selector gate. The channel opening directly overlies and uncovers the lower source/drain electrode. A selector gate dielectric layer is formed in the channel opening on sidewalls of the second dielectric layer and on sidewalls of the selector gate. A first semiconductor layer is deposited in the channel opening to form a selector channel layer on sidewalls of the selector gate dielectric layer and on a top surface of the lower source/drain electrode. A spacer is formed in the channel opening between sidewalls of the selector channel layer and directly over the lower source/drain electrode. An upper source/drain electrode is formed directly over the spacer and between the sidewalls of the selector channel layer. An upper global line is formed directly over the upper source/drain electrode.

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The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated chip comprising:
 - a memory array comprising a first local line and a second local line elongated vertically, a first memory cell extending laterally between the first local line and the second local line and coupled to a first word line, and a second memory cell under the first memory cell, extending laterally between the first local line and the second local line, and coupled to a second word line; a first global line vertically spaced from the first local line; and
 - a first selector between the first local line and the first global line, the first selector comprising a first channel and a first gate laterally surrounding the first channel.
2. The integrated chip of claim 1, wherein the first selector selectively couples the first global line to the first local line.
3. The integrated chip of claim 1, the first selector further comprising a first source/drain and second source/drain on opposite sides of the first channel and a gate dielectric layer between the first gate and the first channel.
4. The integrated chip of claim 3, wherein the gate dielectric layer laterally surrounds the first channel in a first closed loop, and wherein the first gate laterally surrounds the gate dielectric layer in a second closed loop.
5. The integrated chip of claim 4, the first selector further comprising a spacer between the first source/drain and the second source/drain, wherein the first channel laterally surrounds the spacer.
6. The integrated chip of claim 1, wherein the memory array is a three-dimensional memory array.
7. The integrated chip of claim 1, further comprising:
 - a second global line vertically aligned with the second local line; and
 - a second selector between the second local line and the second global line, the second selector comprising a second channel and a second gate laterally surrounding the second channel.
8. An integrated chip comprising:
 - a lower source/drain;
 - an upper source/drain spaced over the lower source/drain, wherein the lower source/drain and the upper source/drain are between a first local line and a first global line of a memory array;
 - a selector channel layer between the lower source/drain and the upper source/drain;
 - a selector gate laterally surrounding the selector channel layer; and
 - a selector gate dielectric layer between the selector gate and the selector channel layer.
9. The integrated chip of claim 8, further comprising:
 - a second local line laterally spaced from the first local line; and

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- a first memory cell extending laterally between the first local line and the second local line.
10. The integrated chip of claim 8, further comprising:
 - a spacer over the lower source/drain and under the upper source/drain, wherein the selector channel layer laterally surrounds the spacer.
11. The integrated chip of claim 10, wherein the upper source/drain is on a top surface of the spacer, and wherein the selector channel layer extends directly between a bottom surface of the spacer and a top surface of the lower source/drain.
12. The integrated chip of claim 8, wherein the selector channel layer and the selector gate dielectric layer extend above the selector gate.
13. The integrated chip of claim 8, wherein the selector channel layer extends from a top surface of the lower source/drain to sidewalls of the upper source/drain, and wherein the selector channel layer laterally surrounds the upper source/drain.
14. The integrated chip of claim 8, wherein the lower source/drain is on the first local line, wherein the first global line is on the upper source/drain and the selector channel layer, and wherein the first global line is vertically spaced from the selector gate.
15. An integrated chip comprising:
 - a memory array comprising a first local line;
 - a first global line spaced from the first local line;
 - a lower source/drain and an upper source/drain between the first local line and the first global line;
 - a spacer between the lower source/drain and the upper source/drain;
 - a selector channel layer between the lower source/drain and the upper source/drain and laterally surrounding the spacer;
 - a selector gate dielectric layer laterally surrounding the selector channel layer; and
 - a selector gate laterally surrounding the selector gate dielectric layer.
16. The integrated chip of claim 15, wherein the selector channel layer laterally surrounds the upper source/drain and extends between the spacer and the lower source/drain.
17. The integrated chip of claim 15, further comprising:
 - a first dielectric layer under the selector gate and laterally surrounding the lower source/drain; and
 - a second dielectric layer over the selector gate and laterally surrounding the upper source/drain.
18. The integrated chip of claim 17, wherein the selector channel layer and the selector gate dielectric layer are on a top surface of the first dielectric layer, and wherein the second dielectric layer laterally surrounds the selector channel layer and the selector gate dielectric layer.
19. The integrated chip of claim 15, further comprising:
 - a second local line spaced from the first local line;
 - a first memory channel layer extending from the first local line to the second local line;
 - a first memory gate extending along the first memory channel layer; and
 - a first memory film between the first memory channel layer and the first memory gate.
20. The integrated chip of claim 1, wherein the first selector comprises a spacer, and wherein the first channel laterally surrounds the spacer.